Survey Research 2019-08-03

2019 K-Innovation ODA Program with Uganda

Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (Ⅱ) — Policy Capacity-building Program

STEPI SCIENCE AND TECHNOLOGY POLICY INSTITUTE

UNCST

2019 K-Innovation ODA Program with Uganda

Title	Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (Ⅱ) – Policy Capacity-building Program
Institute	Science and Technology Policy Institute (STEPI)
Partner Country	Government of Republic of Uganda
In Cooperation with	Ministry of Science, Technology, and Innovation of the Republic of Uganda, Uganda National Council for Science and Technology (UNCST)
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Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks(II) – Policy Capacity-building Program

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Executive Summary

The Science and Technology Policy Institute (STEPI) of the Republic of Korea signed a Memorandum of Understanding (MOU) with the Uganda National Council for Science and Technology (UNCST) on 30 May 2016 during the Korean President's visit to Uganda. The purpose of the MOU is to establish a framework of practical cooperation between both parties to propel further collaboration in the fields of Science, Technology, and Innovation (STI). Within the framework of the agreement, the National Research Foundation of Korea (NRF) and STEPI serve as practical development practitioners in S&T Official Development Assistance (ODA) for Uganda. Specifically, STEPI, NRF, and UNCST agreed to host joint events such as seminars on STI, capacity-building workshop, joint research projects, and any other activities jointly decided upon by the parties.

As part of the operationalization of the MOU, UNCST submitted to STEPI two project proposals -- (i) "Capacity Building to Enhance the Development of Science, Technology, and Innovation in Uganda"; and (ii) "Strengthening Uganda's National STI System and Establishment of STI Parks" -- on 17 March 2017. The project was approved to run for a two-year period from February 2018 to December 2019.

The partnership has since resulted into the successful Implementation of the 2018 K-Innovation ODA program aimed at strengthening Uganda's Science, Technology, and Innovation system through capacity building. The purpose was: (i) to identify specific areas of capacity building and (ii) discuss the scope of the 2018 program. As a result, the following thematic areas were identified as strategic for capacity building: (i) New and emerging technologies (focused on ICT); (ii) Technology transfer; (iii) Science communication; and (iv) Research, innovation, and commercialization/Roadmap for the development of science parks.

Under the title "Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (II) – Policy Capacity-building Program," the 2019 K-

Innovation ODA Program with Uganda ("the project") was launched with the aims of developing and strengthening policy capacity on Uganda's national Science, Technology, and Innovation (STI) system and establishing STI parks. Thus, the project will support Uganda's policymakers to build STI capacity, which will be a key driver for Uganda's sustainable social and economic development.

Five priority themes for the 2019 K-Innovation Program with Uganda were selected based on the demand survey with MoSTI and UNCST: 1) Industry-University Cooperation; 2) Incubation and Commercialization of R&D Outputs; 3) Research Evaluation and Assessment of Research Quality; 4) STI Parks; and 5) Emerging Technologies Focused on ICT.

The 2019 project consists of five phases to support the social and economic development of Uganda by strengthening policy capacity on the S&T innovation system and S&T innovation.

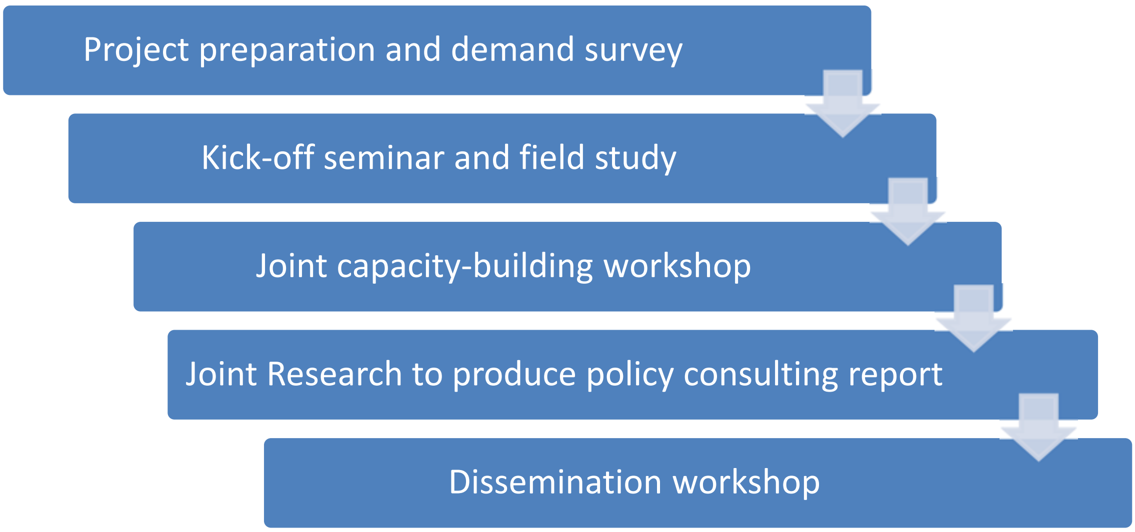
In the first phase, the demand survey to select themes for the 2019 project was conducted with Ugandan experts. The Ugandan team shared rationales for the priority themes to get policy consulting from Korea. STEPI created a Korean research team by inviting external experts according to the themes and their expertise.

In support of MoSTI and UNCST, STEPI organized a kick–off seminar and a field study in Kampala, Uganda. In addition to the kick-off seminar for the 2019 project, Korean experts collected local data and conducted interviews with local experts as part of the field study.

The capacity-building workshop is the key element of the 2019 K-Innovation ODA Program with Uganda. The program for capacity-building workshop was designed for Ugandan and Indonesian experts. Participants welcomed the type of joint workshop as they can learn from Korea's experiences as shared by Korean experts and from colleagues from other developing countries as they can exchange ideas and discuss their findings.

A policy consulting paper was prepared based on the joint research by Korean experts and Ugandan experts. Policy implications on priority themes were drawn not only from the literature review but also through various discussions with stakeholders during the kick-off seminar, field study, and capacity-building workshop. The policy consulting report was produced through online discussions as well as the dissemination workshop.

<Project framework>



Sharing the outcomes of the project among Ugandan stakeholders from ministries, research institutes, academe, and industries is important to get support in ongoing efforts to improve the National Innovation System and prepare for STI Parks in Uganda.

Key policy implications on Industry-University Cooperation consist of four parts, and first is the Entrepreneurial Leadership and Industry-University Collaboration (UIC) Plan. University entrepreneurial leadership from the administration and faculty is very important for the ecosystem. The leadership of administration and faculty provides a top-down, bottom-up approach to campus-wide entrepreneurship and UIC. Entrepreneurial leadership creates a culture of industry involvement and administration support for cooperative programs. The faculty engages in programs, course creation, and charges of thought leadership and skills training. A faculty champion paves a road to UIC education and supports the UIC plan (Korean UIC Plan for 5 years).

Second is “Student Focus.” The student is a core asset of UIC, and the university has to develop and provide a UIC program to students at an adequate level for their undergraduate and graduate degrees. Universities should offer UIC education opportunities and extracurricular activities (visits, competitions). The industry should

collaborate to create corporate entrepreneurship programs and host visits by university students. It should also establish hiring and promotion criteria for UIC and develop job descriptions for UIC so that students can see a new window of opportunities.
Third is “Faculty Champion.” University faculty are often hesitant to become involved in University-Industry Collaboration because of their interests as well as time constraints, evaluation criteria, and sometimes ignorance of the industry. The faculty identify and support University-Industry Collaboration for students and corporate businesses such as INC (Idea-Needs-Capability) model. Universities provide time and leave policies to participate in the industry and establish IUC-friendly evaluation criteria. They require course development support and need to provide the faculty with University-Industry Collaboration funding. For instance, the Korean government provides a “faculty leave system” for UIC. Faculty should have time to develop new courses on University-Industry Collaboration.

Fourth is “Community Engagement and Cluster.” Community engagement is the process by which community interests and individuals form a lasting, permanent relationship to apply the collective vision for community benefit. Communities create joint programs such as speakers, games, and competitions with the university and university extension services for corporate entrepreneurship and innovative cluster. They contribute to regional economic development and establish an innovation cluster.

Policy recommendations on the Technology Commercialization of R&D Outputs suggested by a Korean expert consist of four parts. First, in addition to relying on Makerere University, government agencies such as the UGANDA Management Training and Advisory Center (MTAC) and Uganda Industrial Research Institute (URI) should develop new technologies themselves and act as technology providers. Deficient technologies can be received through technology transfers with the support of advanced countries and international organizations such as the United Nations and should be enhanced in terms of their capacity through international trade, technology licensing, attraction of multinational companies, and human resources training. In particular, it is necessary to attract transnational companies actively through various tax benefits, given that they are accompanied by capital and technology. Second, capability building should be strengthened. In order for the aforementioned technology transfer to be successful, the amount of technology absorption to materialize the technology transferred from developed countries should be increased.

Based on this, the government should develop detailed technologies and support existing companies or push for successful technological commercialization by fostering startups. Third, R&BD strategies as the first mover should be developed and implemented in order to secure the same semi-heat or better technological capability as that of developed countries and occupy the market as the next step. The role of startup companies is paramount in the process of leading technology development as the last gateway to becoming an advanced country and which requires creativity, since the process of more flexible and quick acting in the process of discovering and verifying new ideas is more advantageous to startups than to existing ones. Lastly, the fact that global investors (VCs) are finding good technologies and startups worldwide can also serve as a good opportunity. A startup company based on good technology, if it makes inroads into the global market from the start, will be able to attract investment from investors around the world.

At the dissemination workshop, key stakeholders of Uganda's STI Policy particularly with MoSTI and UNCST reviewed policy implications from the policy consulting paper. The STEPI Team and Ugandan Team had a close discussion on the implications and areas for further cooperation. Korean and Ugandan experts explored the linkage of outcomes of the project with the existing national and international projects or development of pilot projects.

The 2019 K-Innovation ODA Program with Uganda greatly contributed to enhancing the STI policy capacity of Ugandan policymakers by providing policy consulting on STI policies under five themes -- Industry-University cooperation, Technology Commercialization, Research Quality Evaluation, STI Parks, and Emerging Technologies (focusing on ICT) -- and by organizing a capacity-building workshop. Since Uganda is a key partner country, this project contributes to facilitating bilateral cooperation on STI between Korea and Uganda. The outcomes of STI policy consulting will support sustainable development for the people of Uganda.

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Project Overview

Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (Ⅱ) – Policy Capacity-building Program

Chapter 1. Project Overview

1. Introduction

The Science and Technology Policy Institute (STEPI) of the Republic of Korea signed a Memorandum of Understanding (MOU) with the Uganda National Council for Science and Technology (UNCST) on 30 May 2016 during the Korean President's visit to Uganda. The purpose of the MOU is to establish a framework of practical cooperation between both parties to propel further collaboration in the fields of Science, Technology, and Innovation (STI). Within the framework of the agreement, the National Research Foundation of Korea (NRF) and STEPI serve as practical development practitioners in S&T Official Development Assistance (ODA) for Uganda. Specifically, STEPI, NRF, and UNCST agreed to host joint events such as seminars on STI, capacity-building workshop, joint research projects, and any other activities that may be jointly decided upon by the parties.

As part of the operationalization of the MOU, UNCST submitted to STEPI two project proposals -- (i) "Capacity Building to Enhance the Development of Science, Technology, and Innovation in Uganda" and (ii) "Strengthening Uganda's National STI System and Establishment of STI Parks" -- on 17 March 2017. The project proposals were subsequently approved to run for a two-year period from February 2018 to December 2019.

The partnership has since resulted in the successful implementation of the 2018 K-Innovation ODA program aimed at strengthening Uganda's Science, Technology, and Innovation system through capacity building. The purpose was (i) to identify specific areas of capacity building and (ii) discuss the scope of the 2018 program. As a result, the following thematic areas were identified as strategic for capacity building: (i) New and emerging technologies (focused on ICT); (ii) Technology transfer; (iii) Science communication; and (iv) Research, innovation, and commercialization/ Roadmap for the development of science parks.

2. Objectives

Under the title "Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (II) – Policy Capacity-building Program," the 2019 K-Innovation ODA Program with Uganda ("the project") was aimed at developing and strengthening policy capacity on Uganda's national Science, Technology, and Innovation (STI) system and establishing STI parks. Thus, this project will support Uganda's policymakers in building STI capacity, which will be a key driver for Uganda's sustainable social and economic development.

Five priority themes for the 2019 K-Innovation Program with Uganda were selected based on the demand survey with MoSTI and UNCST:

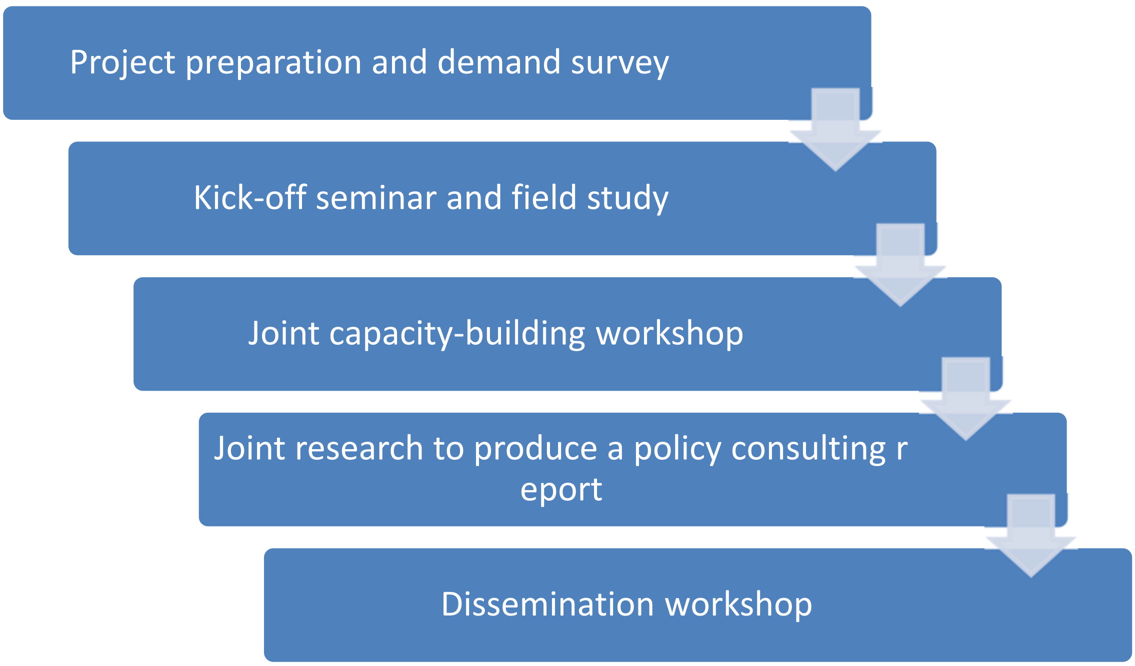
1) Industry-University Cooperation
2) Incubation and Commercialization of R&D Outputs
3) Research Evaluation and Assessment of Research Quality
4) STI Parks
5) Emerging Technologies Focused on ICT

Among the five themes, the project put more focus on Industry-University Cooperation and Incubation and Commercialization of R&D Outputs. With full expertise on the themes, Korean experts provided policy consulting by participating in the kick-off seminar and capacity-building workshop and conducting joint research. In addition to policy consulting, additional guidelines were given, i.e., Ugandan experts were expected to draft policy guidelines on Research Evaluation and Assessment of Research Quality and STI Parks and Emerging Technologies focused on ICT, which were covered in the 2018 project.

3. Project Framework

The 2019 project consisted of five components to support the social and economic development of Uganda by strengthening policy capacity on the S&T innovation system and S&T innovation.

<Project framework>



In the first phase, the demand survey to select themes for the 2019 project was conducted with Ugandan experts. The Ugandan team shared rationales for the priority themes to get policy consulting from Korea. STEPI invited external experts to create a Korean research team according to the themes and their expertise. Literature review on the theme of Ugandan science and technology were conducted to identify demands for S&T innovation policy in Uganda.

In support of MoSTI and UNCST, STEPI organized a kick–off seminar and a field study in Kampala, Uganda. In addition to the kick-off seminar of the 2019 project, Korean experts collected local data and conducted interviews with local experts as part of the field study.

Capacity-building workshop is the key element of the 2019 K-Innovation ODA Program with Uganda. The program for capacity-building workshop was designed for Ugandan and Indonesian experts. Participants welcomed the type of joint workshop as they can learn from Korea’s experiences as shared by Korean experts and from colleagues from similar developing countries as they can exchange ideas and discuss their findings.

A policy consulting paper was produced based on joint research of Korean experts and Ugandan experts. Policy implications on priority themes were drawn not only from the literature review but also through various discussions with stakeholders during the kick-off seminar, field study, and capacity-building workshop. The policy consulting report is finalized through online discussion as well as the dissemination workshop.

Sharing the outcomes of the project among Ugandan stakeholders from ministries, research institutes, academe, and industries is important to get support in the ongoing efforts to improve the National Innovation System and prepare for STI Parks in Uganda.

4. Project Team

4.1 Korean Research Team

The research team of the project consists of STEPI researchers and external specialists in the fields of Industry-University cooperation and Technology Commercialization of R&D Outputs and Emerging Technologies.

[Table 1-1] Composition of Research Team

Name	Organization	Position	Remarks
Dr. Wandong Kim	STEPI	Senior Research Fellow	Program Manager for the 2019 K-Innovation ODA Program
Dr. Deok Soon Yim		Senior Research Fellow	Project Manager
Ms. Elly Hyanghee Lee		Researcher	Project Coordinator
Prof. Jong-in Choi	Hanbat National University	Professor	Policy Consulting and Joint Research on Industry-University Cooperation
Dr. Jong Heung Park	Electronics and Telecommunications Research Institute (ETRI)	Vice President	Policy Consulting and Joint Research on Technology Commercialization of R&D Outputs
Prof. Hansung Yoon	Hansung University	Professor	Providing Advice on Emerging Technologies

4.2 Ugandan Research Team

The Ministry of Science, Technology, and Innovation (MoSTI) and The Uganda National Council for Science and Technology of the Republic of Uganda are responsible for strengthening the Ugandan National STI system. In partnership with the Korean research team, the Ugandan research team representing three officers at the director level from MoSTI and two researchers from UNCST conducted joint research on the five themes covered by the 2019 K-Innovation ODA Program with Uganda.

[Table 1-2] Ugandan Research Team

Name	Organization	Position	Contact Information
Dr. Maxwell Otim-Onapa	MoSTI	Director for Science, Research, and Innovation	maxwell.otim@mosti.go.ug +(256) 41478882066
Mr. Basil Ajer	MoSTI	Director for Technopreneurship	abasil@ugandainvest.go.ug +256-772687774
Ms. Monica Rose Nyakaisiki	MoSTI	Assistant Commissioner, Science, Technology, and Innovation Infrastructure Development	monica.nyakaisiki@gmail.com +256-782951393
Mr. Bashir Rajab Kagere	UNCST	Head, Policy Analysis and Development	bkagere@gmail.com +256 782 86 94 89 +256 702 775 460
Mr. Geoffrey Sempiri	UNCST	Head, Science and Technology Infrastructure Development	sempiri.geoffrey@gmail.com g.sempiri@uncst.go.ug +256 773 690 740

5. Project Activities

5.1 Activity 1: Project preparation and demand survey

The Korean team held online consultations with the Ugandan team and decided the details of the 2019 K-Innovation ODA project including themes, project activities, terms of consultant, and project timeline. The priority themes of the Ugandan research team, which were decided based on Uganda's demands, include: 1) Industry-University linkage; 2) Incubation and commercialization of R&D outputs; Research evaluation and assessment of research quality; 4) New and emerging technologies including ICT; and 5) STI parks.

5.1.1 Themes

■ **Industry-University Cooperation**

There is currently a huge divide between knowledge generation at the university and its utilization in the industry. The mismatch between knowledge creation and industry demands is partly responsible for the high levels of unemployment, low capacity utilization, poor absorption of talent, and low levels of innovation. Therefore, it is imperative that we build an industrial sector that provides conduits for technological learning and linkage creation between industry, academe, and research institutes. Specifically, there is a need to identify the principal technology constraints facing the industrial sector, ways through which firms draw on the relevant research conducted by universities and public research institutes, most severe human resource constraints, and size and proportion of S&T personnel, among others.

■ **Incubation and commercialization of R&D Outputs**

Incubation is a unique, highly flexible combination of business development processes, infrastructure, and people designed to support entrepreneurs and nurture and grow new and small businesses, products, and innovations through the early stages of development or change. For the industrialization process to be sustainable, it requires a set of core competences and resources such as skilled human resource, technological innovation, and enhanced knowledge capacity, access to affordable finance and infrastructure, comprehensive and coherent policy processes, and culture of entrepreneurship and competitiveness.

In Uganda, the industrial sector predominantly consists of MSMEs, which account for 95% of the entire sector and employ more than 2.5 million people. These MSMEs constitute 90 per cent of the private sector and contribute 20% of GDP, and they are very dynamic with 80 per cent located in urban and peri-urban areas countrywide and are largely involved in trade, agro-processing, and small-scale manufacturing. Due to globalization, internal challenges, and international competition, MSMEs have faced global challenges; like the rest of the world, less than 30% of the startups are able to see their first birthday in Uganda (Hatega, 2006). Still, the rest of the world that have tried business incubation have claimed a success rate of over 85%. Business incubation programs were considered a remedy for the disadvantages encountered by small and new firms by providing useful business support services.

UNCST plans to set up professional business incubation centers that will offer support services and resources for nurturing startup enterprises with the goal of developing them into financially viable businesses equipped with the tools for long-term survival.

■ **Research evaluation and assessment of research quality**

A fundamental question that can be posed within any field of research is, “What constitutes good- or high-quality *research* (*or scientific*) *practice*?” This question is relevant for research in both university context and organizational or innovation context for research and development activities. Evaluation of the quality of research practice is a truly important issue in most scientific domains and at many levels. More or less elaborate efforts have been made in recent years to evaluate the quality of research practice in a host of different settings. These efforts affect resource allocation and scientific activity as well as the very lives of researchers across the globe. Quality is the focus for several different reasons and is examined in various contexts such as in the evaluation of the following: research grant applications; research manuscripts and publications; specific research topics; research groups and constellations; institutions; and national systems for producing science and innovation. Regarding the issue of measuring the quality of research in the wider scientific community, it is important for us to find the right balance and learn lessons from countries that have excelled in this line, such as Korea.

5.1.2 Terms of consultant

After the priority themes for 2019 were finalized, STEPI drafted the terms of consultant for the Ugandan research team to help them understand what roles and cooperation are expected of them.

■ **Background**

This project was developed based on the MOU signed in Kampala, Uganda in 2016 between UNCST and STEPI to strengthen cooperation and provide policy advice and education for the strengthening of Uganda's S&T innovation policy. Uganda is also one of Korea's priority partner countries for development cooperation. STEPI has been operating the STI Policy Program (STIP) through the 2018-2019 K-Innovation ODA Program with Uganda in partnership with UNCST.

As part of the 2018-2019 K-Innovation ODA Program for Uganda in partnership with UNCST and Uganda's Ministry of Science, Technology, and Innovation (MSTI), policy consulting in 2018 was focused on Science Communication, STI Parks, and Emerging Technologies as requested by UNCST. Following the successful implementation of the 2018 policy consulting, the K-Innovation Project for Uganda 2019 will focus on three themes: Industry-University linkage; Incubation and commercialization of R&D Outputs; and Research evaluation and assessment of research quality.

The research report contains useful and detailed information for the themes faced by Uganda regarding the background and status as well as in which areas policy consulting is most needed. This report will be presented to the Korean government as a major outcome of the K-Innovation Project for Uganda 2019.

■ Duties, responsibilities, and output expectations

a) Terms of Reference

The consultant will be contracted by STEPI to conduct research for the K-Innovation Project for Uganda 2019 on the theme assigned by STEPI. He/she shall write a report on the selected theme with minimum length of 30 pages. The research activity will mainly involve desk study and virtual interactions with STEPI. The consultant will undertake the following tasks:

- Conduct literature reviews pertaining to the theme such as background, status, and stakeholders as well as where Uganda needs policy consulting the most;
- Participate in the kick-off meeting for the K-Innovation Project for Uganda 2019 (scheduled in Kampala, Uganda on 29-30 May 2019) and give a presentation on the theme assigned to him/her;
- Write a full report on the theme assigned to him/her;
- Interact with various stakeholders on the theme to reflect on different perspectives; and
- Perform other necessary tasks as advised by STEPI.

b) Output Expectations

- Report on literature review by 15 August
- Presentation materials for the kick-off meeting
- Draft research report by 1 September
- Final research report by 30 September

5.2 Activity 2: Kick-off seminar and field study

The Korean research team held a kick-off seminar and conducted a field study in Kampala, Uganda from 26 May to 1 June 2019. During this mission, the Korean and Ugandan research teams discussed the work plan for the K-Innovation Program for Uganda 2019 such as themes, project activities, timeline (particularly on the capacity-building workshop and workshop for sharing outcomes), local consultants, and role allocation. A bilateral meeting with UNCST was held to hear its specific demands and expectations on policy consulting for the K-Innovation Program for Uganda 2019.

At the kick-off seminar for the 2019 K-Innovation ODA Program with Uganda, Korean experts shared Korea's experiences particularly in Industry-University Cooperation and Technology Commercialization of R&D Outputs. Local consultants gave a presentation on the themes. Experts from ministries, academe, and SMEs participated in rigorous discussion after the presentations.

In addition to the kick-off seminar, the Korean research team had bilateral meetings with Mr. Obong O. David, Permanent Secretary of MoSTI, and the Ugandan Research Team. The Korean research team consulted with Ambassador Byungku Ha from the Embassy of the Republic of Korea in Uganda and KOICA's Ugandan Office to share the outcomes and progress of the K-Innovation Program for Uganda 2018-2019 and to seek support for the K-Innovation Program for Uganda 2020-2022.

5.3 Activity 3: Joint capacity-building workshop

As part of the 2019 K-Innovation Program with Uganda and Indonesia, experts from Uganda and Indonesia in the areas of STI policy participated in the joint capacity-building workshop. Through the workshop, the participants were able to deepen their understanding of the themes covered by the K-Innovation Program 2019 with Uganda and Indonesia. Major themes for the workshop include:

- Structure and history of the Korean S&T policy in general
- STI policy and necessary policy measures
- Industry-University linkage, incubation and commercialization of R&D Outputs

The four-day workshop was composed of two elements: 1) lectures and group discussions and 2) field visits. Participants actively participated in discussions and presented an action plan for developing innovative science and technology policies that could be applicable to Indonesia and Uganda based on the Korean experiences.

The theoretical and practical intensive program was designed for the Indonesian and Ugandan government officials and practitioners looking to broaden their knowledge through group discussions. Field visits to public research institutes and other related agencies were arranged.

On the last day of the capacity-building workshop, representatives from each country gave a presentation on what they learnt from the workshop as well as the way forward on STI policy.

According to the satisfaction survey, all the participants said that the program was helpful, and 92 per cent of the participants said that they would recommend the program to others.

5.4 Activity 4: Joint research to produce a policy consulting report

Local consultants and Korean experts conducted a joint research on the five themes to produce a policy consulting report as part of the 2019 K-Innovation ODA Program. The local consultants consisted of officers from MoSTI and researchers from UNCST who are responsible for formulating Uganda's STI policy and measures.

Among the five themes, two Korean experts focused on two themes -- industry-university cooperation policy and technology commercialization of R&D outputs – and conducted joint research with Ugandan policymakers. In addition to sharing Korea's experiences, they diagnosed Uganda's relevant policy and suggested policy recommendations. The STEPI team provided guidelines for joint research and reviewed the policy consulting report drafted by Ugandan and Korean experts.

5.5 Activity 5: Workshop to disseminate the project outcomes

As the last activity of the 2019 project, STEPI, MoSTI, and UNCST are co-organizing the workshop to disseminate the outcomes of policy consulting in Kampala, Uganda on 22 November 2019. At the workshop, STEPI presents the outcomes of the 2019 project along with policy consulting on the two focused themes. Ugandan consultants will share the outcomes of their research including the policy implications drawn through bilateral consultations with Korean experts and future actions. This workshop gives a great opportunity for various stakeholders related to STI to publicize Korea's ongoing efforts to support Uganda's capacity building on STI through the 2019 K-Innovation ODA Project.

6. Project Schedule

The STEPI team proposed the project schedule, and it was finalized through consultations with the Ugandan government. Note, however, that the schedule was adjusted according to Ugandan circumstances.

[Table 1-3] Project Schedule

Activities	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
Project Preparation / Theme selection												
Local consultant identification and contracting												
Literature review												
Joint research								Draft report		Final Report		
Kick-off seminar and field study (Uganda)												
Capacity-building workshop (Korea)												
Mid-term report												
Dissemination workshop (Uganda)												
Final report										Draft	Final	

7. Project Outputs

STEPI provided the following deliverables:

- **Deliverable 1**: Workplan (including the background of the themes' selection, ToR for consultant)
- **Deliverable 2**: Presentation materials for the kick-off seminar on 29 May 2019
- **Deliverable 3**: Presentations and training materials from the Joint Capacity-building Workshop on 8 ~ 11 July 2019
- **Deliverable 4**: Lessons learnt and way forward from the Joint Capacity-building Workshop
- **Deliverable 5**: Research papers through the Korea-Uganda joint research on five themes
- **Deliverable 6**: Policy Consulting Report

CHAPTER 2

Current Status of STI System and STI Parks in Uganda

Policy Consulting for Strengthening Uganda's National
STI System and Establishment of STI Parks (Ⅱ)
– Policy Capacity-building Program

Chapter 2. Current Status of STI System and STI Parks in Uganda

Part I. Industry-University Cooperation in Uganda

Dr. Maxwell Otim Onapa, Director for Science, Research, and Innovation,
Ministry of Science, Technology, and Innovation, Uganda

1. Introduction

1.1 Background

Uganda lies astride the equator, bordered by Tanzania and Rwanda in the south, the Democratic Republic of Congo in the west, South Sudan in the north, and Kenya in the east. Lake Victoria, the third largest freshwater lake in the world, also forms part of the southern border. The country is mainly covered by woody hills, fertile plateau, swampy lowlands, and semi-arid environment in the northeast.

Uganda's economic growth continues to be positive with real GDP growth at 4.8% in 2017 compared to 5.7% in 2015 and 2.3% for 2016 (Figure 2-1). Uganda's economic growth over the last few years has mainly been attributed to strong performance in the industry and service sectors. Investments in priority public infrastructure such as road and energy investments have equally contributed to the strong economic performance. These large infrastructure projects are mainly funded through concessional loans, playing a major role in boosting manufacturing as well as services, notably tourism.

Credit expansion, which has been increasing over the years, continues to boost consumption. Investment in the energy sector and roads has also boosted growth, although the fluctuation in oil prices has also impacted the rate of growth. Oil exploration in the greater Albertan region is playing a role in increasing the much-needed foreign direct investment.

Employing 72% of the population, agriculture remains one of the most important sectors of the economy (Figure 2-2). Uganda’s export market suffered a major setback following the outbreak of armed conflict in South Sudan. This has largely recovered lately thanks to record-high coffee harvests -- which account for 16% of exports -- and increase in gold exports, which constitute 10% of exports.

Uganda's industrial sector is relatively and highly dependent on imported inputs such as refined oil and heavy equipment. Productivity is constrained by a number of factors including weak infrastructure, inadequate modern technology in agriculture, and corruption.

Uganda’s industrial sector is highly reliant on imported inputs. These include refined petroleum products as well as heavy industrial equipment. Note, however, that Uganda’s full productivity potential is constrained by numerous challenges such as corruption and low adoption and lack of modern technology in agriculture coupled with weak infrastructure.

Uganda’s long-term growth is largely driven by donor support in the agriculture, health, and education sectors. [1]

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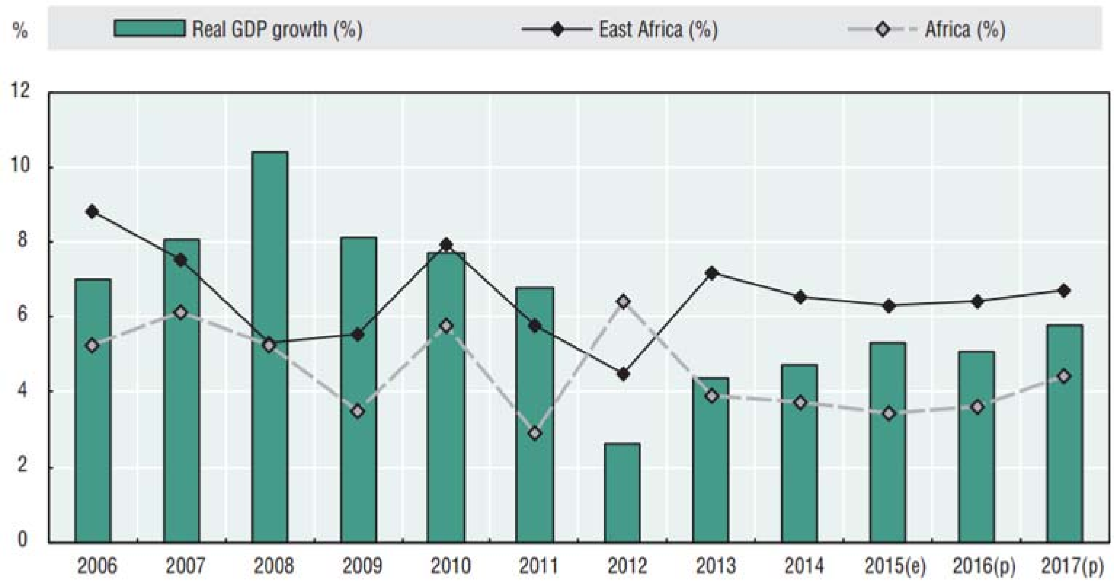
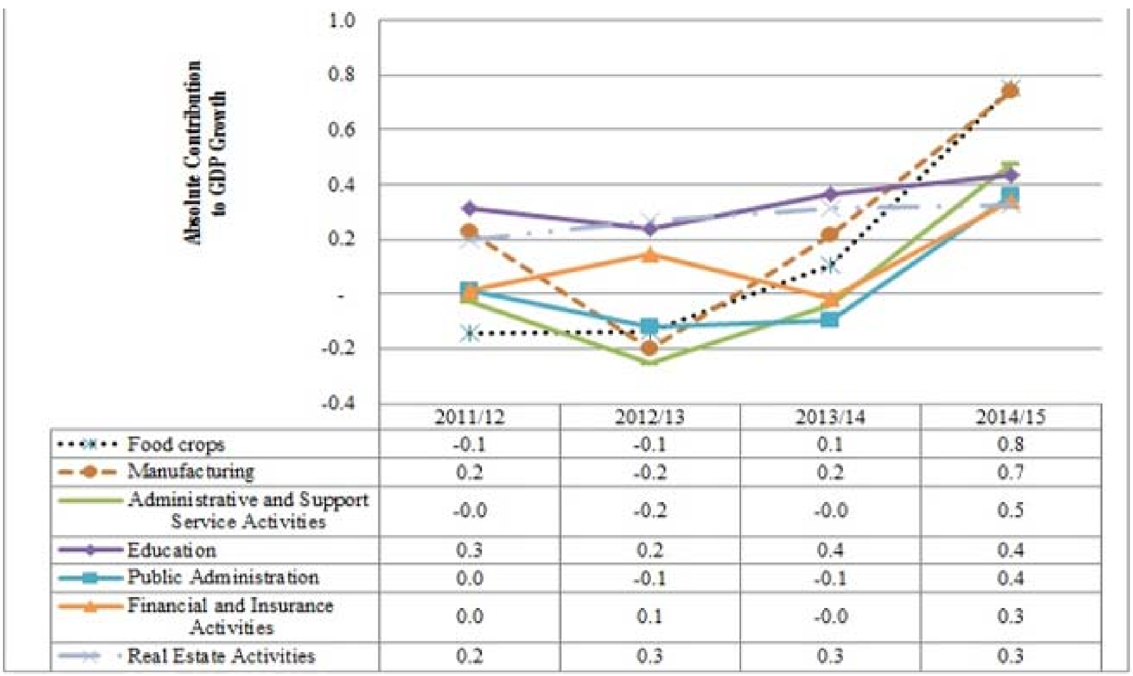
[1] https://theodora.com/wfbcurrent/uganda/uganda_economy.html.

[Figure 2-1] Real GDP Growth



Source: AfDB, Statistics Department AEO, Estimates€; projections (p).

[Figure 2-2] Sources of Growth (2011/2012 - 2014/2015)



	2011/12	2012/13	2013/14	2014/15
Food crops	-0.1	-0.1	0.1	0.8
Manufacturing	0.2	-0.2	0.2	0.7
Administrative and Support Service Activities	-0.0	-0.2	-0.0	0.5
Education	0.3	0.2	0.4	0.4
Public Administration	0.0	-0.1	-0.1	0.4
Financial and Insurance Activities	0.0	0.1	-0.0	0.3
Real Estate Activities	0.2	0.3	0.3	0.3

Source: BTTB (MoFPED, 2015).

Regarding gender inclusiveness, Uganda has registered remarkable progress in its performance in empowering women and attaining gender equality. Such progress notwithstanding, the discrimination of women especially in the areas of economic opportunities and the quest for ownership of assets persists.

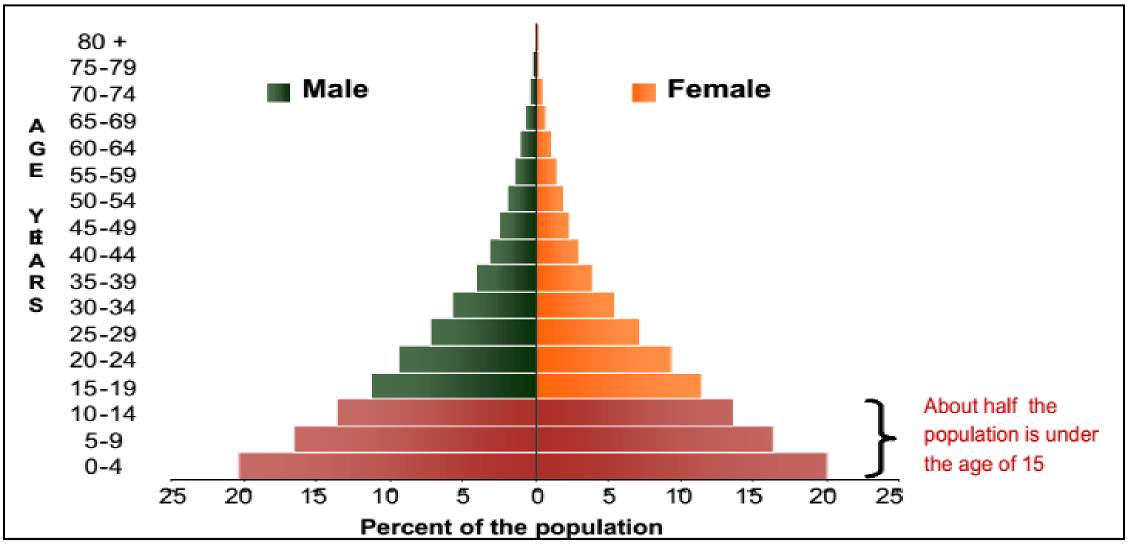
The Human Development Report (HDR) 2019 ranks Uganda at 159th out of the 189 countries that were assessed. With a Human Development Index value of 0.528 in 2018, Uganda falls under the low human development category. Nonetheless, the country has made marginal progress, improving from HDI of 0.516 and ranking of 182nd out of 189 countries in 2019.

It is also noteworthy that, between 1990 and 2018, Uganda’s Human Development Index (HDI) value increased from 0.312 to 0.528, representing an increase of 69.1 per cent. In the 2019 HDR, Uganda’s human inequality coefficient stands at 26.7%. Life expectancy, education, and income inequality stand at 27.2%, 27.9%, and 24.9%, respectively.

1.2 Demographics

Contemporary statistics indicate that 77% of the Ugandan population is below 30 years old, making the country one of the youngest in the world. Estimates indicate that about 7.4 million youth in Uganda fall within the age bracket of 15–24 years (UBOS, 2017). With an average of 5.8 children per woman, Uganda’s total fertility rate is among the highest in the world. The actual fertility is estimated to exceed a woman’s desired fertility by one or two children; thus suggesting the widespread unmet need for contraception and possible weak support for family planning as well as cultural preference for large families. The high number of births, short birth intervals, and early age of childbearing contribute to Uganda's high maternal mortality rate. High birth frequencies, short spacing of births, and childbearing at an early age are some of the factors that contribute to Uganda’s poor maternal-child health outcomes and high maternal mortality rate. Figure 3 shows the population pyramid of Uganda.

[Figure 2-3] Uganda's Population Pyramid, 2009



Source: Uganda Bureau of Statistics, 2017

The World Economic Forum on Africa (2016) has touted Africa's young population as a demographic dividend – a major advantage for the continent that could realize 11-15 per cent GDP growth between 2011 and 2030. In order to harness the expected demographic dividend, however, the youth must be provided with opportunities and access to good education and health services so as to avert the negative implications such as frustration, social unrest, or unmanaged migration that this increasing working-age population might suffer from or bring (UNFPA, 2013).

High population growth unmatched by a concomitant economic growth creates strains on the economy, natural resources, infrastructures, and services; thus making it difficult for the economy to grow, especially since an overwhelming majority (70%) of Ugandans still dwell in rural settings and predominantly depend on agriculture for their livelihood. This is most likely to entrap the country in poverty and instability (Daumerie and Madsen, 2010). Therefore, the real point is that Uganda has a population challenge to deal with (Serunjogi and Musisi, 2013). While the population grows, rapid urbanization is also occurring in the country, creating a strain on services and compromising sustainability.

1.3 Uganda's Development Framework

Uganda's development aspiration is articulated in Vision 2040 (Ubos, 2013), the blueprint articulating the development pathways and strategies through which Uganda's Vision statement of "*A Transformed Ugandan Society from a Peasant to a Modern, Prosperous Country within 30 years*" is being operationalized. The framework seeks to transform Uganda from a predominantly peasant population and a low-income country to one that is competitive with upper middle-income population. The approach seeks to builds on the progress made so far in tackling strategic bottlenecks that have stifled national development following the country's independence, including ideological disorientation, weak private sector, underdeveloped human resources, inadequate infrastructure, small market, lack of industrialization, underdeveloped service sector, underdeveloped agriculture, and poor democracy, among others.

Uganda's Vision 2040 is premised on reinforcing the fundamentals of the economy so that the country can tap into the abundant opportunities that it possesses (NPA…ORG). Such opportunities include abundant labor force accruing from the demographic dividend, oil and gas, ICT business, tourism, abundant minerals, geographic location and trade, water resources, agriculture, and industrialization, among others, that are considerably under-exploited to date. Therefore, achieving the transformational goal will depend on the country's ability to leverage the fundamentals above and others that include: infrastructure energy, human resource transport, and Science, Technology, Engineering, and Innovation (STEI); land use and management; and urban development as well as peace, security, and defense.

The key core projects identified by Uganda Vision 2040 are: Hi-tech ICT city and associated ICT infrastructure; internationally competitive human capital development centers; and renewal energy sources such as solar energy, hydro, and nuclear power plants as well science, technology, and innovation parks in each regional city. The Vision is implemented within the context of a comprehensive National Development Planning Framework wherein interventions are sequenced and detailed in the 5-year national development plan and annual budgets. Financing of the National Development Plan is mainly through the government, CSOs, development partners, and private sector. The government is responsible for mobilizing resources using conventional and innovative unconventional means of financing, which include revenue from oil and gas, public-private partnerships, concessional loans and grants from international sources, and borrowings from both domestic and international markets. The revenue from oil and gas has already been invested to kick-start major infrastructure development projects such as roads and airport. These in turn are expected to enhance the country's competitiveness.

1.4 Second National Development Plan (NDP II)

In her aspirations toward Vision 2040, Uganda has so far implemented two national development plans. This current National Development Plan (NDP II), which lapses in 2020/2021, is the second in a series of six five-year Plans aimed at achieving Uganda Vision 2040.

The Plan underscores the need to strengthen and harness the natural resources endowment including land, water, oil, and gas as well as ICT, STEI, and infrastructure including energy and transport, if it is to achieve its transformational goal. Its goal was originally to propel Uganda toward middle-income status by 2020 by strengthening the country's ability to create wealth and employment competitively and sustainably so that it can place itself on a path of inclusive growth. The science, technology, and innovation field is one of the key sectors expected to contribute to the attainment of this aspiration.

Thus, the Plan sets the following four key objectives to be attained during the five-year period: enhancing sustainable production and productivity and promoting value addition in key growth opportunities; increasing the stock and quality of strategic infrastructure to accelerate the country's competitiveness; enhancing human capital development; and strengthening the quality, effectiveness, and efficiency of service delivery.

In order to meet the objectives above, the government has been pursuing a number of development strategies including:

(i) Ensuring macroeconomic stability and increasing the level of financial resources committed to infrastructure investments;
(ii) Expanding the industrial base and investments in value addition through selected heavy and light manufacturing; agro processing and mineral beneficiation to support export-oriented growth
(iii) Creating employment by fast-tracking skills development and harnessing the demographic dividend
(iv) Strong Public/Private Partnerships (PPPs) model for sustainable development
(v) Mixed economy led by the private sector and quasi-market approach
(vi) Strengthening governance mechanisms and structures

This Plan prioritizes investment in key growth opportunities. These are:

Agriculture

Agricultural research, which has been performing well over the last decades, is being strengthened, with the single spine agricultural extension system and farm level technology adaptation being promoted. Emphasis is also being placed on increasing access to quality and effective use of key farm inputs. The challenge of land fragmentation is being addressed by promoting sustainable land use and improved soil management. The critical challenge of lack of agricultural credit is being handled by increasing access to agricultural finance with focus on women farmers. All these interventions are being buttressed by agricultural institutions, which are being strengthened to support effective coordination and service delivery.

Tourism

In the tourism sector, focus is on value addition where emphasis is placed on improvement and diversification as well as exploitation of new tourism products. Aggressive marketing and investment in infrastructure that facilitate tourism, such as transport, water, energy, and ICT, is being promoted. This is being supported by the development of the relevant skills and improvement in related services such as the quantity and quality of hotels and accommodation facilities, provision of security to tourists, and protection of tourist attraction sites. Combating poaching is being pursued as a measure for conserving tourism products including flora and fauna. The problem of wildlife dispersal is being addressed, thereby allowing their concentration in specific locations for easy viewing and interaction. These efforts in turn help ensure the maximum exploitation of tourist attractions and amenities and promote capacity building and tourism management.

Minerals, Oil and Gas

Minerals that have been earmarked for exploitation and value include: iron ore, copper, cobalt, limestone/marble, precious stones, phosphates, and uranium. In addition, the exploitation of an estimated 6.5 billion barrels of oil with recoverable potential of 1.4 billion barrels is prioritized.

In addition, the extraction of an estimated 6.5 billion barrels of crude oil with approximately 2.2 billion barrels of recoverable quantity has been prioritized. Key investments in this area include: improvement of geological surveys; investment in further survey and exploration; expedited acquisition of land; construction of 3 pipelines to carry crude oil to the Indian Ocean coast; refined products to Kampala, Eldoret, and Kigali and Liquefied Petroleum Gas (LPG) to Kampala and Gulu; construction of a refinery for oil and gas; and increased prospecting and processing of and adding value to the selected minerals.

Infrastructure

Regarding transport infrastructure-strategic roads to facilitate the exploitation of oil and gas, minerals, tourism, and decongestion of traffic within the city and surrounding areas have been targeted. In addition, efforts are being made toward increasing the volume of both passenger and cargo traffic via marine transport. With respect to energy infrastructure, investments are being promoted by targeting the exploitation of abundant renewable energy sources including the following: hydro power and geothermal; expansion of the national electricity power grid network; and promotion of energy efficiency and use of alternative sources of energy.

Regarding other supportive infrastructure, the priority in the ICT sector is the extension of the National Backbone Infrastructure (NBI) to cover underserved parts of the country and establishment of ICT incubation hubs/ centers and ICT parks to boost innovations in ICTs. As part of the strategy to boost commercial agriculture and industrial activities, provision and harvesting of water for production are being promoted.

Human Capital

Enhancing human capital development is being addressed by increasing the quality and number of skilled and healthy workforce that is expected to contribute to and accelerate the realization of the demographic dividend accruing from Uganda's young population.

Within the education component, focus is on the following: strengthening Early Childhood Development (ECD); increasing retention at primary and secondary school levels with specific emphasis on the girl child as well as increasing primary-to-secondary transition; and increasing investment in school inspection.

The skills development component is also focusing on the review and reform of the education curriculum at all levels. This is expected to facilitate the production and expansion of skills relevant to the labor market demands. Expanding skills development includes both formal and informal training and establishment of centers of excellence for skills development in prioritized areas.

In the health sector, emphasis is on the mass management of malaria and HIV through prevention and treatment. Other approaches include the provision of universal national health coverage through the national health insurance scheme as well as access to family planning services. Efforts are being made to addresses maternal and child health by reducing maternal, neonatal, and child morbidity and mortality. Other areas of focus include developing health infrastructure and centers of excellence for handling and managing various ailments including cancer treatment and related services.

1.5 National STI Policy

The government put in place a national STI policy in 2009 (NSTIP, 2009). NSTIP -- which aims at calibrating the contribution and role of STI in the national development process as highlighted in Vision 2040 and National Development Plan -- is anchored on the Constitution and National Vision 2040. It underlines the contribution of innovation and research in transforming Uganda into a knowledge-based economy and articulates the government's intent to foster research and development that build the human capital required by Uganda for a knowledge-based economy (Brar, et al, 2010).

Therefore, at the strategic and policy level, STI is mainstreamed into and adequately addressed through the National STI Policy 2009, STI Plan (2013/14-2017/18), and National Development Plan.

The goal of the STI policy is to strengthen national capability to generate, transfer, and apply technologies and to ensure sustainable utilization of natural resources for the realization of Uganda's development objectives (MoFED, 2009)[2].

The policy is anchored around 4 key objectives: creating an enabling environment to facilitate the growth and deployment of STI and enhancing their contribution to national development processes; building STI sector capacity to generate and transfer technology; establishing and strengthening the policy and legal and regulatory frameworks to promote ethics and safety in the development and application STI; and strengthening the STI coordination framework to enhance sector performance.

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[2] Ministry of Finance Planning and Economic Development. (2009), National Science, Technology, and Innovation Policy.

The STI policy is in line with the national development objectives of poverty eradication, industrialization, agricultural modernization, increased productivity, employment creation, value addition, and gender mainstreaming.

The purpose of NSTIP is to facilitate the attainment of Uganda's development aspirations as articulated in Vision 2040 through inter-alia, getting the population out of absolute poverty by meeting basic needs, transforming the current agrarian-based economy to an industrial, knowledge-based one, and ensuring that Uganda participates in the global economy and development processes. NSTIP builds on the existing initiatives in the various sectors to guide Uganda's STI development path toward achieving the national vision for STI (MoFPED, 2012)[3].

Essentially, NSTIP contributes to the implementation and results framework that translates the national STI policy into measurable actions within a five-year dispensation.

The framework is driven by the expected results per STI Policy objectives. The policy envisages harmonized, integrated related policies and regulatory regimes within the national development framework for greater consolidated impact on national development and poverty eradication. The policy provides a framework for the support of local innovation and scientific excellence with regard to national research priorities and places Uganda on the path to active participation and contribution to the STI activities within the East African Community while seeking to guide the STI sector in implementing the country's international obligations in international agreements.

Safety and Health are clearly articulated in the policy within the context of promoting research ethics and safety for the protection of the environment, humans, and animal health.

[3] Ministry of Finance Planning and Economic Development. (MoFPED) (2012), *National Science, Technology, and Innovation Plan*, pp. 1–87.

The policy addresses gender and equity by striving to promote the active participation of all women and men, youth, children, elderly, and other disadvantaged or special needs/marginalized groups and leverages private sector participation and investment in science and technology. In doing so, it provides a framework that is expected to promote product development, quality assurance, and standardization of goods and services in order to improve Uganda's international competitiveness.

The policy seeks to create an enabling environment to foster STI and augment its contribution to national development by undertaking technology forecasting, assessment and transfer; industrial development, intellectual property management, traditional, conventional, and emerging technologies, and gender and equity are being addressed.

By directing efforts toward finance and investment, human capital development and retention, STI infrastructure, research, and technology incubation, the policy seeks to build STI sector capacity to generate and transfer technology.

The policy has strengthened STI safety regulations, ethics standards, and quality assurance that ensure ethical practice and safety in STI development and application.

Through the action of addressing public awareness, enhancing appreciation of STI, and building an information management system and sector coordination and partnerships, the policy seeks to strengthen the STI coordination framework, which in turn is expected to enhance the sectors' performance.

2. University-Industry linkage in the policy

2.1 Global and African contexts

Universities and higher institutions of learning have long been recognized as sources of knowledge creation. They have also served as sources of skills and human resources for industry, developed innovation, and promoted technological advancements. Universities and higher institutions of learning the world over have long been recognized as centers of knowledge generation, innovation, and technological advances[4]. Globally, both developed and emerging economies are leveraging universities and positioning them as strategic assets for innovation and economic competitiveness. Note, however, that universities in Africa have played a very marginal role in innovation and or technology transfer.

Contemporary challenges such as fiscal limitations, graduate unemployment, need to demonstrate relevance and contribution to addressing societal challenges, response to national development imperatives and need for them to be more open and accountable to society have made the discourse on university-industry linkages increasingly prominent in the debate on the state of higher education in Africa.[5] University-industry linkage can play a critical role in mobilizing additional resources for research, innovation, and technology transfer. This in turn would support the skilling of graduates and ensure that they acquire the knowledge required to contribute to the workforce and the labor market effectively.

Universities in Africa face considerable constraints. These challenges in turn affect their research and innovation capacity, skilling capabilities, and economies as well as political environments. Even in the face of these constraints, many universities in Africa are

[4] www.adeanet.org.

[5] www.aau.org.

making efforts to accelerate interventions to strengthen their institutional capacity to support linkages with the industry as well as the broader productive sector (Tiyambe, 2004; Ginies and Mazurelle, 2010). Optimal structures for the interface, appropriate policies, incentives, and funding avenues are necessary for effective university-industry linkages.

The paucity of data prevents the detailed analysis of outlook as to what steps have already been taken by African universities and what is needed to provide a strengthened, more comprehensive platform for promoting, building, and managing synergistic partnerships with the productive sector (Munyoki, et al, 2011).

2.2 Universities - Industry nexus

The emergence of knowledge economy wherein knowledge is widely recognized as a factor of production and is consequently deemed to have a bearing on the economy and competitiveness has given universities - industries fresh impetus (Martin, 2011). University-industry-government linkages embodied in the triple helix within the national innovation systems can take various forms and involve different scopes of engagement.

Forms of linkages include industrial placement of students to pursue internship and staff exchanges wherein university and industry staff participate in activities in the university and industry. More complex partnerships take the form of business and technology incubator as well industry-sponsored projects. Linkage also takes the form of research and development (R&D) partnership, training and joint curriculum development, and consultancy. In other instances, the industry may commission a university to undertake specific research projects. The industry may also support the establishment of a university in a specific area of interest. University-industry linkage promotes technology transfer through prototype development and technology incubation. The linkage is an avenue for the creation of startup companies and spin-off, which are avenues for commercialization of innovations, licensing, and generation of royalty fees.

2.3 Forms of academe-industry linkages

Formal linkage

Formal linkage between university and industry is established through liaison offices and technology transfer offices. In some cases, this can be realized through the establishment of science, technology, and innovation infrastructure such as science parks, which can be co-located on or near the university campus. This approach facilitates the interaction between university and industry (Lundvall, 2009).

In other instances, the industry and the academe can collaboratively set up leadership and dedicated posts and draw up clear strategic direction and policies for managing the effective governance of the linkages (Kruss, 2008).

Informal linkage

Common informal approaches to university-industry linkage include providing guest lectures at the university, holding joint stakeholder meetings, and conducting a collaborative review of the university curriculum to make it responsive to the labor market demands. Synergistic linkages between universities and industry are quite instrumental in mobilizing and influencing additional resources for higher education. The linkages also promote innovation and technology transfer and ensure that graduates have the skills and knowledge required to contribute to the workforce effectively (Ssebuwufu, et al, 2012).

The organizational process of the university-industry linkage can be viewed from the following models:

National Systems of Innovation (NIS) Framework

Developed by Etzkowtitz and Leydesdorff in 1997, this model views innovation as a product of interaction between three main actors: academe, industry, and government. In this model, universities focus on establishing institutional interface structures including industry liaison/technology transfer offices and business and technology incubators and fostering entrepreneurialism through various policies and incentives (Etzkowtitz, 2008).

Triple Helix Model

This model, which was developed by Etzkowtitz and Leydesdorff in 1997, views innovation as a product of interaction between three principal actors: academe, industry, and government (Etzkowtitz and Leydesdorff, 1997). Here, the university focuses on establishing institutional interface structures -- including industry liaison/technology transfer offices and business and technology incubators – and fostering entrepreneurialism through various policies and incentives (Etzkowtitz, 2008).

Mode II Knowledge Production

This was developed by Michael Gibbons in 1994 a model wherein innovation is viewed as context-driven, problem-focused, and interdisciplinary. Knowledge is produced in the context of application, quality control, social accountability reflexivity, and heterogeneity organizational diversity (Goransson and Brundenius, 2011).

3. Status of Industry-University linkages in Uganda

Although universities in Uganda have embraced the approach to institutionalizing linkages with the industry/private sector through endeavors such as creation of dedicated posts and offices, they still face challenges such as low budget and weak expertise in intellectual property management and marketing. Moreover, various capacity gaps in terms of the necessary skills, research and innovation infrastructure, and funding still pose serious challenges to the universities. Universities require appropriate structures, policies, positions, incentives, and funding avenues to enable the establishment of effective linkages with the industry.

A number of Ugandan universities have flagged industry linkages in their strategic plans. Nonetheless, many of them still lack complementary and supportive mechanisms and policies such as institutional intellectual property rights policies for governing interactions with the private sector and industry. Furthermore, research and innovation outputs are also limited by the low percentage of academic faculty members who hold doctoral degrees. There is also limited awareness of intellectual property rights.

Formal structures that help universities develop, nurture, and showcase novel technology and solutions include science parks and technology incubators. Although there are plans to develop such infrastructure to foster university-industry linkage, there are no such facilities in Uganda at the moment. The Ministry of Science, Technology, and Innovation has prioritized the establishment of STI infrastructure as key agenda. Sectors wherein a number of Ugandan universities and industries are collaboratively engaged include: agriculture and agri-business; computing; and ICTs and environmental management.

Within the framework of the linkage, the industry also offers internship attachment / placement opportunities for undergraduate students, whereas universities frequently engage industry players as guest lecturers/speakers to provide business and entrepreneurial advice to the university community.

In order to respond to the labor market demands and to enhance graduate employability, universities are seeking to improve their curricula, skills development, and internship placement by involving the industry in developing and revising their curricula. There is some relationship between the university and the industry through commissioned research projects, investments in research and innovation infrastructure such as laboratories and equipment, provision of scholarships to students, and funding for graduate research, although this is still weak.

Priority areas of intervention for which support from the government, other external partners, and stakeholders is most desired and valued by the university include: assistance in training academic staff in entrepreneurship skills; assistance in developing institutional strategic plans that emphasize engagement with the productive sector of the economy; support for the establishment of technology and innovation incubators and science parks; and provision of opportunities for peer learning among institutions where those with weaker capacity can learn from those with a history of successful engagement with the industry and the private sector (www.heart-resources.org). These priorities are a direct reflection of the lack of entrepreneurial expertise and the lack of science parks and technology incubators highlighted in the university-industry ties. Support is also desirable in the area of intellectual property management and business and project management capacity, including other types of skills required for the effective governance of such structures given the low level of awareness of intellectual property.

3.1 Government efforts toward promoting university-industry linkage

In recognition of the role of Science, Technology, and Innovation (ST&I) as fundamental drivers of socioeconomic growth and transformation, the government of Uganda established a Ministry of Science, Technology, and Innovation (MoSTI) in 2016. The mandate of the ministry is to provide overall policy guidance and support and coordinate the entire chain for scientific research, development, and entire National Innovation System in Uganda. The ministry coordinates the implementation of research, development, and innovation and provides leadership in developing and implementing guidelines for the management of the research and innovation fund, policies, plans, and programs for research promotion and development. The architecture of the ministry has been structured to provide seamless juxtaposition between the different phases of research, innovation, technology development, and transfer value chain.

The ministry is composed of three directorates:

Directorate of science, research, and innovation

This directorate provides strategic leadership in science, research development, and innovation and supports the implementation of all research, development, and innovation programs in the country. The directorate guides research and innovation and technology development and supports the establishment of STI Infrastructure Development. It is also responsible for intellectual property management.

Directorate of technopreneurship

The directorate is responsible for coordinating product testing and commercialization of products and services for startups and Small and Medium Enterprises (SMEs). It is directly responsible for guiding linkages and partnerships between researchers and industrialists and business enterprises, thereby ensuring that products from the directorate of science, research, and innovation reach the market. It plays a very important interlocutory role between the researchers/university and the industry.

Directorate of science, technology, and innovation regulation

This directorate is responsible for providing guidance on the development and implementation of STI Policies, regulations, plans, priorities, standards, and guidelines.
In pursuit of its mandate, the ministry has developed a National Research and Innovation Program (NRIP). The Framework provides for a national research and innovation

program to coordinate and streamline government initiatives and funding in the Science, Technology, and Innovation (ST&I) sector in the country. The program aimed at encouraging creativity and supporting innovations in Uganda. The main goal of the National Research and Innovation program is to promote Research and Development, Technology Incubation, and Technology Commercialization activities. The program seeks to do this by promoting the implementation of innovative projects, with the aim of facilitating the realization of new or improved products, services, or processes designed to raise economic efficiency, improve the innovation potential and enhance the technological level of enterprises, and increase private investments and enhance the dynamics of innovation processes in Uganda.

In an effort to use STI to increase the productivity, inclusiveness, and wellbeing of its population, Uganda has set forth *Innovation, Technology Development, and Transfer* as a development program within its NDPIII, which will run from 2020/2021 -2025/2026. This has been motivated by the challenge of absence of a well-coordinated innovation ecosystem to support innovators coupled with the low uptake of innovation and technology. The program seeks to create requisite systems for identifying and nurturing novel innovations and technologies and increase the development, adoption, transfer, and commercialization of technology and innovation.

Different challenges influence innovation, technology development, and transfer value chain at various stages, consequently affecting university - industry linkages. These include:

i. Low science culture and negative mindset with regard to STI: A significant segment of the population does not seem to appreciate science and technology.
ii. Inadequate education curriculum, wherein the mode of delivery is based on imparting knowledge as opposed to building of research culture and knowledge generation.
iii. Supportive STI Infrastructure is lacking. The current research and innovation infrastructure such as laboratories, technology and innovation hubs, and incubation facilities are grossly inadequate.
iv. Limited linkages between the sectors of the economy

v. Unfavorable policy environment to foster innovation and technology development: The current STI policy formulated in 2009 does not adequately support the spectrum spanning from idea to the market. The policy does not provide incentives to private actors to engage in and/or support research and development.

Enabling policies such as intellectual property policies or business management policies are also necessary for guiding the collaborative work of scientists and innovators, especially the private sector and international partners. Among the many universities in Uganda, only Makerere University has an IP policy approved in 2008. Therefore, considerable effort is needed to build the capacity of local research organizations and universities to design or formulate and implement the relevant institutional strategies and policies supporting R&I and IP governance.

vi. Limited human capacity (both qualitative and quantitative)

Uganda is one of the countries hampered with development challenges due to the low levels of adaptation, transfer, and dissemination of Technology and Innovation. Uganda still suffers from low human capital base in STI. A 2003 study by the World Bank showed that 800 researchers were employed in research and development, mostly in public universities and R&D laboratories. The study also found out that only six universities in Uganda offer science and engineering courses, although this has expanded since then.

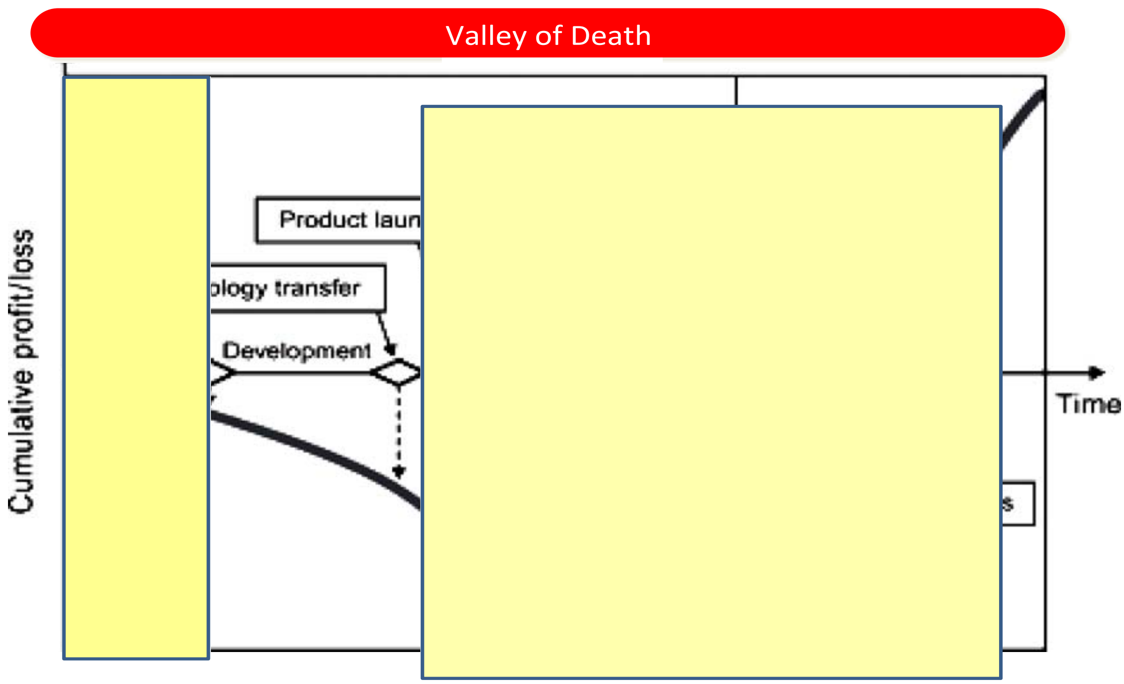
Enrollment in basic science was found to account for below 1% of the overall secondary school enrollment, with less than 10 new PhD degrees awarded annually in science and engineering disciplines (www.researchgate.net).

The national innovation system is weak. This, among other things is manifested by the absence of a platform for facilitating the engagement of the triple helix: government, academe, and private sector.

vii. Low funding of research and technology development

The amount of resources for research and innovation allocated by the government and industry is limited. Uganda's expenditure on research and development accounts for only 0.43 per cent of GDP compared to the 1% recommended by the African Union. About 66% of the research and development expenditure by the government comes from foreign sources (UNCST, 2014a). Research and development funding by the private sector is small. Most of the private sector businesses are small -- operating as informal establishment -- with very few of them doing any form of research.

[Figure 2-4] Valley of Death/Innovation Chasm/Innovation Paradox



Source: Forbes

Business financing

Financial institutions do not fully appreciate their role and contribution as innovation system actors in Uganda. The failure of research and innovation undertakings by universities and other establishments in Uganda to overcome the "innovation paradoxes, "valley of death," or "innovation chasm is largely attributed to the lack of funding to support research and innovation (Figure 2-4).

Note, however, that a few of them such as Uganda Development Bank and Stanbic Bank have started to support business incubation. The national social security fund is a pension fund that runs a program supporting innovators in taking their ideas to the market. Nonetheless, Uganda still lacks venture capital, which could support investment in new science-based and innovation-based business enterprises.

Support for university innovations by financial institution: case of UDB

The i-Growth Accelerator by Uganda Development Bank is aimed at facilitating the sustainable growth of the economy by supporting the commercialization of local startups. More specifically, the UDB i-Growth Accelerator aims to:

- Identify, recognize, and accelerate outstanding innovative startups and support them in commercialization.
- Inspire, stimulate, and create a culture of innovation and entrepreneurship by nurturing creativity from an early age.
- Link successful ventures to investment opportunities that will support their scaling.

For a long time, society and business entities have largely regarded Universities and higher education institutions as the most important source of highly skilled human resources, centers of generation of new knowledge, and repositories of new research findings. Now, however, they are considered the “new frontiers of innovation.” This view paved the way for the first specific science, technology, and innovation policy in 2009 wherein issues of innovation were dealt with. The most common traditional form of relationship between the University and the private sector has been in the form of university-industry ties. Note, however, that emerging science, technology, and innovation policies are underscoring engagement with society and industry through technology transfer and innovation. Contemporary policies address issues of science and technology parks and business incubation, establishment of startups, and spin-offs and technology transfer offices (TTOs) as a means of enhancing the commercialization of innovations.

All these are structured within the framework of the national innovation system embodied by the *Triple Helix.* The concept views innovation activities in a broader perspective. It focuses not only on outputs (innovation) but also on the processes leading to and the various actors in generating those outputs (Ecuru, 2013). These encompass R&D efforts by universities, firms/industry, public policies, learning processes, and incentive schemes. The interaction among the different actors in the innovation space demonstrates the fact that innovation processes are non-linear, iterative processes that involve multidisciplinary groups with common interest in co-creation (Ecuru, 2013).

Ugandan universities still play a very marginal role in technology transfer or innovation. In his study report, Tveit and Webjørnsen (2011) analyzed the university-industry-government linkage. They recommended that universities prioritize the applied research customized to the local needs. Responding to the challenge of the mismatch between the competency of university graduates and the labor market, they emphasized the need to overhaul the content of the curriculum and pedagogical approach to ensure that university graduates have the required skills relevant to the labor market.

In a bid to improve entrepreneurial culture in the university, they encouraged university management to make applied research a key strategic focus. Management was encouraged to provide the professors with requisite incentives.

Regarding industry, they emphasized the need to put more pressure on the universities in order to realize increased linkages. They noted the benefits that accrued for both University and *Umeme* -- electricity power distributor as a result of the model linkage initiated between *Umeme* and the university, and industry-initiated collaboration like what *Umeme* has done. In other words, there are observed benefits for both actors.

Tveit and Webjørnsen (2011) recommended that the government make policies, specifically those that encourage competition between universities to increase motivation for University-Industry linkage. They stressed that increased communication between university and industry is important with regard to result-based salaries systems in the universities.

3.2 Earlier efforts to promote University–Industry linkage

The Millennium Science Initiative (MSI)

The development objective of the initiative was *for universities and research organizations to train more and better-qualified scientists and engineers, to conduct high-quality, relevant research, and for firms and industry to utilize the scientists, engineers, and research output to increase their profitability* (World Bank, 2007).

The Millennium Science Initiative (MSI) operated within three windows. One of the windows (Window C) provided a platform for the private sector and researchers to work collaboratively to provide solutions to problems of direct interest to the industry. This was done through technology platforms and formal internship placements for students pursuing science, engineering, and business courses.

The window aimed at creating linkages between the academe and industry to promote greater access to technology and skills. Technology platforms were to bring together firms and partners from the academe or public research institutes to identify areas where technologies existed outside Uganda and they could be transferred within the country or identify solutions to problems faced by firms. The goal was to adopt or adapt technology to increase the company's productivity. Feasibility studies to understand commercial viability were permitted. Internships were also used to create university-industry linkages and provide practical training for students. Grantees under this window could receive up to $50,000; thus, technology creation was not the aim.

Makerere University – Kakira Sugar Works Limited

Under the research-academe-industry linkage of the MSI, researchers from the College of Agriculture of Makerere University partnered with Kakira Sugar Works Limited to address the challenge of low milk production in dairy herds around the sugar factory. The partnership was able to produce a feed supplement called milk booster, a new product from a project conducted under a research project titled "*Participatory research for technology development on the use of Molasses Urea Blocks (MUB) and local feedstuff for improved dairy cattle production in Uganda.*"

The research was a collaboration among a team of researchers from Makerere University, Kakira Sugar Works Ltd. (industry), and Dairy Development Authority (a government agency). It exploited the locally available resources such as sugar cane industrial waste of molasses from Kakira Sugar Works Limited as well as other ingredients including maize bran, lint seed cake, urea, mineral salts, and lime.

The innovation is a feed that has the effect of boosting milk in dairy cattle. The commercialized innovation is contributing to increased productivity and enhancing growth in the dairy industry through improved milk production.

Case of Consortium for enhancing University Responsiveness to Agribusiness Development Limited (CURAD)

The Consortium for enhancing University Responsiveness to Agribusiness Development Limited (CURAD) is a public-private partnership initiative promoted by Makerere University, National Union of Coffee Agribusinesses and Farm Enterprises Limited (NUCAFE), and National Agricultural Research Organization (NARO), University of Copenhagen (UC), and NIRAS International. CURAD is one of the six agribusiness incubators in Africa supported by the Forum for Agricultural Research in Africa under the UniBRAIN facility with funding from DANIDA.

CURAD is a public-private partnership initiative with the aim of producing innovative young entrepreneurs and agribusiness leaders to champion productivity and profitability of agricultural enterprises that can spin off new enterprises. This is an agribusiness innovation incubator geared toward the creation of jobs and boosting of income within the agricultural sector in Uganda, starting with the coffee value chain in the first four years. As a non-profit company limited by guarantee (certificate no. 144130), CURAD was established to support profit-oriented agribusinesses.

4. Conclusion

Based on the discussions and analysis above, it is quite evident that research- and knowledge-intensive innovation in Uganda is mainly supported through international collaborations and by the government, with the latter bearing the biggest responsibility of financing research and innovation. The current gross expenditure on research and development stands at 0.25%, which is way below the 1% recommended by the African Union.

Low science culture and negative mindset with regard to STI, outdated and rigid curriculum, poor *STI* Infrastructure, limited interface between the sectors of the economy, unfavorable policy environment, and limited human capacity are some of the binding constraints that limit university–industry linkages in Uganda.

Unlike in developed economies like OECD, the private sector in Uganda is still too weak to make any substantial investments in R&D:

- There are no venture capitalists to support the commercialization of research results.
- Financial institutions do not appreciate their role in supporting the commercialization of university research results!
- Frameworks such as IP policy support linkages to support collaboration in universities and public organizations are either still weak or non-existent.
- Universities need to be supported to develop this

Clear policy supporting U-I linkage is necessary. The current STI policy produced in 2009 does not adequately support the spectrum spanning from idea up to the market. The policy does not provide private actors with incentives to engage. Note, however, that there are success stories of university-industry linkages as presented earlier.

Part II. Incubation and Commercialization of Research and Development Outputs in Uganda

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1. Introduction

In Uganda, the industrial sector predominantly consists of Small and Medium Enterprises (SMEs), which account for 95% of the entire sector and employ more than 2.5 million people; thus making them one of the largest employers in the country ((MFPED, 2011, Financial Sector Deepening Uganda, 2015)). In fact, SMEs have important contributions to the economy -- for instance, the jobs they create absorb the most vulnerable in society, thereby contributing to poverty reduction. The SME sector also contributes fundamentally to the development of skilled and semi-skilled labor force, thereby supporting future industrial growth. The sector also creates opportunities to cultivate and nurture entrepreneurial and managerial skills across the country and reduce the rural–urban drift as smaller companies allow people to remain in the rural areas.

Despite the sector's enormous size and contribution to the economy, small-scale entrepreneurs still face a myriad of challenges that affect the competitiveness of the sector and the economy in general. Consequently, SMEs are predominantly informal and young. Majority of enterprises in Uganda are aged between 1 and 5 years. For every business created, nearly another is closed; hence the high mortality rate, with less than 10% of the enterprises having operated for more than 20 years (Financial Sector Deepening Uganda, 2015). As the Ugandan population continues to grow above 3 per cent annually and with 78 per cent aged below 35, the high number of young people demands stronger development focus particularly with regard to business and employment opportunities. This, however, cannot be made possible unless appropriate research & development (R&D) is conducted to ensure the success of business ideas. R&D provides the foundation for creativity and innovation that helps make business ideas flourish.

Many countries have increasingly been engaged in establishing business incubators and have claimed different success rates. Incubators are also widely acknowledged to be a technology transfer mechanism and a means of promoting entrepreneurship and commercialization of new knowledge and innovations (Phillips, 2002). Business incubation programs have been considered a remedy for the disadvantages faced by small and new firms by providing useful business support services (Sherman and Chappell, 1998). They are useful in fostering technological innovation, entrepreneurship, commercialization, and industrial renewal (Hackett and Dilts, 2004).

Business incubation is largely at its infancy in many Sub-Saharan African countries; hence the still not so well-developed opportunities for innovation and entrepreneurial networking. For example, the Ugandan manufacturing sector is not technology-intensive or innovation-led but dominated by production activities that are standardized and which require low technology by global standards.

In the National Industrial Policy (2008), the government of Uganda advocates the establishment of technology-based incubators. The driving force behind this has been to foster innovation and enterprise creation. Universities in Uganda and other institutions have developed business incubators to provide training opportunities for students and serve as commercial outlets for faculty research. As a traditional incubator run by the government, the Uganda Industrial Research Institute (UIRI) has made a significant impact by setting up value-added processing systems from its Kampala headquarters at farmer communities. While the model lacks the necessary innovation development, UIRI offers SME clients in these regions the opportunity to expand their personal income and their existing businesses through local market development and value-added food processing. At the same time, UIRI's traditional incubator has been challenged to graduate incubatees who do not have the financial resources to stand on their own.

1.1 The Context

While the world's acceptable standard ratio for R&D expenditure to GDP is 2 %, Uganda, according to the United Nations Educational, Scientific, and Cultural Organization (UNESCO) Institute for Statistics, spent a mere 0.17 % in 2014, down from 0.48 in 2010 and compared to Kenya's 0.79 in 2010. According to the National Survey on Research and Development Report 2012, in 2010, government expenditure on research and development (GOVERD) -- expressed as a percentage of GDP -- stood at 0.19%, with 66.1% and 31.4% financed by sources from abroad and government, respectively. The total government sector headcount (HC) and full-time equivalent (FTE) R&D personnel were 744 and 545.9, respectively; it is of little wonder that fewer R&D outputs are being commercialized as the government commits only over 30% of the R&D annual budget.

Research and Development (R&D) comprise the creative work undertaken on a systematic basis in order to increase the stock of knowledge (including knowledge of man, culture, and society) and the use of this knowledge to devise new applications.

The main aggregate used for international comparisons is gross domestic expenditure on R&D (GERD). This consists of the total expenditure (current and capital) on R&D carried out by all resident companies, research institutes, university and government laboratories, etc. It includes R&D funded from abroad but excludes domestic funds for R&D performed outside the domestic economy. GERD here is expressed in constant 2005 dollars (adjusted for purchasing power parity) and as a share of GDP (R&D intensity) (OECD, 2012).

The Government of Uganda (GOU) undertakes R&D directly through its ministries, departments, and agencies (MDAs), at the same time providing competitive and noncompetitive research grants in strategic areas of national interest. Uganda's science and technology infrastructure currently consists of 53 Universities, 6 of which offer science and engineering courses; 33 are science-related vocational and technical institutes (NCHE 2018), 20 are active R&D institutes, 2 are national museums, 1 is a functional public library, and 5 are private laboratories (UNCST, 2008).

1.2 Commercialization

Commercialization driven by market and profit motivation is the utilization of research results or new findings (transfer of knowledge to end users) wherein firms seek to gain an economic return of investment from research, product development, and marketing.

Commercialization is a mechanism for transforming the knowledge into products, services, and institutes by having competitive advantage to achieve regional economic growth (Mueller, 2005). Other scholars argue that commercialization became more popular for its participation in economic growth through the university platform, considered the backbone of a knowledge-based economy. Universities have evolved as an “entrepreneurial university” to support the commercialization of research and knowledge for a sustainable, progressive ecosystem (Audretsch, 2014). Makerere University in Uganda is taking the lead in R&D. Thus, commercialization and knowledge transfer to society should become the third mission of universities apart from the conventional teaching and research.

1.3 Incubation

Business Incubation is a unique, highly flexible combination of business development processes, infrastructure, and people; it is designed to support entrepreneurs and nurture and grow new and small businesses, products, and innovations through the early stages of development and/or change (Philips, 2002). An incubator is as a place where incubation activities are carried out and where the would-be entrepreneurs can find a suitable place in terms of facilities and expertise to address their needs and develop business ideas and transform them into sustainable realities (Xavier, Martins & Lima, 2008). Note, however, that there are an increasing number of virtual incubators -- that is, incubators without walls – that use e-platforms of online services.

The benefits of a well-managed business incubator can be manifold for different stakeholders: for incubatees, it provides a holistic business development service that enhances the chances of success, raises credibility, helps improve skills, creates synergy among client firms, and facilitates access to mentors, information, networks, and seed capital; for government, the incubator helps promote entrepreneurship and overcome market failures, promotes regional development, generates jobs, income, and taxes, and becomes a demonstration of the political commitment to small businesses; for research institutes and universities, BIC helps turn science and innovations into business for the benefit of researchers and institutions alike, strengthens interactions between university, research, and industry, promotes research commercialization and gives opportunities for faculty/graduate students to utilize these capabilities better and become job creators and not job seekers, provides a conduit for the masses -- especially farming and agribusiness who need them -- to avail themselves of new technologies.

For business, BIC can develop opportunities for acquiring innovations and new technologies, novel and/or reliable suppliers especially in agribusiness value chains, supply chain management, and spin-offs and help them fulfill their social responsibilities. For local communities, BIC creates jobs and wealth, self-esteem, and entrepreneurial culture together with local income as majority of graduating businesses stay within the area. For the international community, BIC generates opportunities for trade and technology transfer between client companies and their host incubators, understanding of business culture, and facilitated exchanges of experience through associations and alliances.

Some of the establishments offering incubation services in Uganda include:

- Uganda Management Training and Advisory Center (MTAC)
- Uganda Gatsby Trust (UGT)
- Uganda Industrial Research Institute (UIRI)
- Textile Development Agency (TEXDA)
- Presidential Initiative on Banana Industrial Development (PIBID)
- Makerere University: (i) Makerere Innovation and Incubation Center (MIIC) and (ii) Food Technology and Business Incubator
- Business Incubation Skilling Center (BISC)-Nakyesesa–Kabanyolo

- Science and Technology Incubation Center (STIC) - Kyambogo

Ugandan Incubation Centers still have many limitations including weak business support services, low-level infrastructures, limited capacity for R&D activities, and insufficient funding.

2. Incubation and Commercialization of R&D Outputs

The success of any incubation is dependent on the strong marriage between innovation and commercialization. It also requires cooperation between universities, government, and private industry. Although these three partners work in different reward systems and often with different interests and expectations, business incubation centers provide common ground between the research economy and the commercial economy.

Therefore, making R&D work for development requires its alignment with the national development goals and priority sectors, which should then be followed by technology needs assessment and then finally undertaking of research projects. While the direct product of R&D is knowledge -- which can be in the form of new technology, new product, new process, and improvement of existing product, process, or technology – the utilization of these products consequently leads to economic development, industrialization, job creation, and poverty reduction. Thus, it is fair to say that it is only through the transfer of knowledge that an R&D Institution can become relevant to society.

To achieve high-impact, result-oriented research, however, the government needs to come up with Policy incentives that may catalyze R&D. These may include:

- Institutional Implementation Partner Policy that should clarify ownership, benefit sharing, privately sponsored research, and issues of national interest.
- Other Policies like Technology Transfer policy, University-Industry Partnerships, and Funding for innovation and creativity should be adequately addressed.
- Support Structures should be in place including the Technology Transfer Office, Companies, Business incubation, Science parks, and Spun-out/off companies.

The success of commercialization depends on the involvement of multidimensional parties having different missions and objectives such as government, academicians, business, and community (Markman, et al, 2008). Thus, commercialization has various implementations for academicians, industry, government, students, and researchers.

Scientific, technological, and innovation advancements are a critical precursor for industrialization and socioeconomic development. They enable the creation of new firms and/or industries that act as avenues for translating Research and Development (R&D) outputs into commercialized outcomes consisting of new products, processes, and enterprises.

As Uganda aspires for faster, sustainable, and inclusive growth as expressed in Vision 2040 and NDP II, the Ugandan Science Technology Engineering and Innovation (STEI) ecosystem – having the advantages of a large demographic dividend and a huge talent pool -- will need to play a defining role in achieving the national Vision and Goals. The national Science, Technology Engineering, and Innovation (STEI) enterprises must become central to national development.

New structural mechanisms and models may be needed to address the following pressing challenges:

- Access to clean, low unit-cost energy
- Climate change mitigation, adaptation, and resilience
- Food, nutrition, and sustainable agriculture
- Access to quality health care for all at all ages
- Affordable shelter
- Employment generation
- Wealth creation

3. Challenges for Research and Commercialization Ecosystem

- The country needs appropriate technology, strong skills base especially technical skills, and affordable financing to move from a subsistence economy to industrialization.
- For so many years, the government has been emphasizing research and innovation, but this is not reflected to the national budget allocation as meagre resources are allocated for the same.
- Lack of an enabling policy framework to revitalize and spur R&D in the country.
- Research institutes and universities are detached from the enterprises; hence the low commercialization of R&D outputs.
- Inadequate funding regimes remain a major challenge to the development of more vibrant R&D infrastructure in the country. Most public universities and research institutes predominantly rely on unpredictable international donor support, which, in most cases, is not aligned with the national research agenda.
- Insufficient business support services coupled with inadequate physical and operational infrastructure exacerbate the problem of inadequate capabilities to exploit opportunities in the emerging sectors such as ICT, biosciences, material science, and Nano technologies.
- Low level of private sector participation in R&D activities.

4. Recommendations

First, policymakers are required to have greater awareness and capacity building to ensure that national STI policies and programs capture the national development priorities, and that they are internally and externally consistent in order to promote policy complementarity, coherence, and effectiveness. They should support policies and provide resources to strengthen business linkages between R&D institutions and society.

Second, there is a need to formulate STI policies and strategies that respond to the development needs of the country. Policy and support structures are also required.

Third, in keeping with global trends, there is a need to strengthen the small business environment by implementing and promoting more explicit links between business incubation and a broader portfolio of business growth; investment strategies will realize more benefits.

Fourth is the improvement of business incubation infrastructure to promote the transfer of knowledge, creation of synergies, entrepreneurship development, and ultimately industrial growth.

Fifth, business incubation environments should be close to knowledge-intensive areas surrounded by universities and research institutes or located in science and technology industrial parks.

Lastly, public-private partnerships should be strengthened so that new forms of financing can be promoted. Measures for capacity building need to be improved as well.

Part III. Research Evaluation and Research Quality Management n in Uganda

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1. Introduction

The Uganda National Council for Science and Technology (UNCST) is an Ugandan government agency established in 1990 by the Act of Parliament (CAP 209). The establishment of UNCST was largely inspired by the Lagos Plan of Action (LPA) conference held in Lagos, Nigeria in 1980, which stipulated that each country should create an institution that would "help the country in determining the origins and effects of alleviating the technological dependence and in achieving technological self-reliance by striking a socioeconomically favorable balance between foreign inputs and those inputs generated by the indigenous science and technology system and utilized by the national sectors of production and services." Such institution would be entrusted with the mandate of formulating national science and technology policies and strategies and coordinating all national Research and Development (R&D) programs.

LPA had also been inspired by the 1979 World Bank "Berg report,[6]" which advocates economic liberalization, increased participation of the private sector, development of local solutions to local problems, and self-sufficiency of African states. The overarching aim of LPA was to ensure that member countries attain self-sufficiency and become technologically independent. The UNCST Act sets forth the functions of UNCST as an institution, and the mandate for the management of research in Uganda is derived from the following provisions: Article 4b(iii) mandates UNCST to carry out scientific and technological research and Development (R&D); Article 4(d) mandates UNCST to clear all

[6] Elliot Berg (1981), "Accelerated Development in Sub-Saharan Africa: A Plan for Action," World Bank

information and materials on research and experimental development prior to being undertaken in scientific institutions or centers and any other enterprises; and Article 4(e) allows UNCST to create or set up specialized committees, councils, and institutions and undertake experimental and development activities or other scientific and technological services. As the apex body for research management in Uganda, UNCST registers, approves, and conducts ethical reviews for all researches undertaken in the country.

2. Research Management and Regulation

2.1 Research Management

Management of research in Uganda has undergone profound reforms especially in the 1980s and 1990s. In 1970, the National Research Council (NRC) was established by a cabinet decision to coordinate research activities in the country. NRC was mainly tasked with research oversight, national research policy, and government research grants. The council had six committees including medical, veterinary, and agricultural, among others. Prior to these reforms, research was integrated in most line ministries, e.g., agricultural research in the Ministry of Agriculture and animal industry, health research in the Ministry of Health, among others. Specialized research departments were created within each of the ministries with the NRC coordinating research and determining priorities. During the reform process, however, specialized institutions (academic and research institutions) were either maintained or set up to undertake research while government line ministries were left with the mandate of policy and regulation.

In 1990, the Uganda National Council for Science and Technology (UNCST) replaced NRC as the apex body for research oversight in the country. Based on the recommendations of LPA, UNCST was then mandated to develop science and technology (S&T) policies and strategies and ensure their subsequent integration into the national development agenda and advise the government of Uganda on the strategic pathways essential for S&T advancement in Uganda. Subsequently, the National Agricultural Research Organization (NARO) was established in 1992 to promote and streamline research on agriculture, livestock, fisheries, and forestry. To avoid any conflicts, Section 11 of the NARO Act 1992 stipulates NARO’s relationship with UNCST: NARO shall operate in cooperation with UNCST and shall, in relation to UNCST, assume responsibility for research in the agricultural sector of the economy of Uganda. In addition, NARO was mandated to develop efficient research management systems with particular emphasis on the decentralization of research operations and implementation of research institutes and to review research priorities in agriculture in accordance with the constitution, government policies, and guidelines provided by UNCST, among others.

Prior to these initiatives, there were a number of research institutes particularly in the medical field. For instance, the Uganda Virus Research Institute (UVRI) had been created in 1945 to undertake research on communicable diseases in man and animals with specific emphasis on viral, transmitted infections. The Trypanosomiasis Research Organization in the Fort portal district was established to deal with the sleeping sickness epidemic; this in turn led to the creation of the trypanosome cryobank in 1956 to support the research process by securing the requisite materials. In 1990, the Joint Clinical Research Center (JCRC) was created to address the HIV/AIDS issue, which was at its peak at the time. Later, in 1992, Parliament created the Uganda Trypanosomiasis Control Council (UTCC) in order to rationalize tsetse and trypanosomiasis research and control in Uganda. To date, several other specialized research institutions have been established. Nonetheless, the overall management of Research and Development in the country remains with UNCST.

2.2 Research Regulation

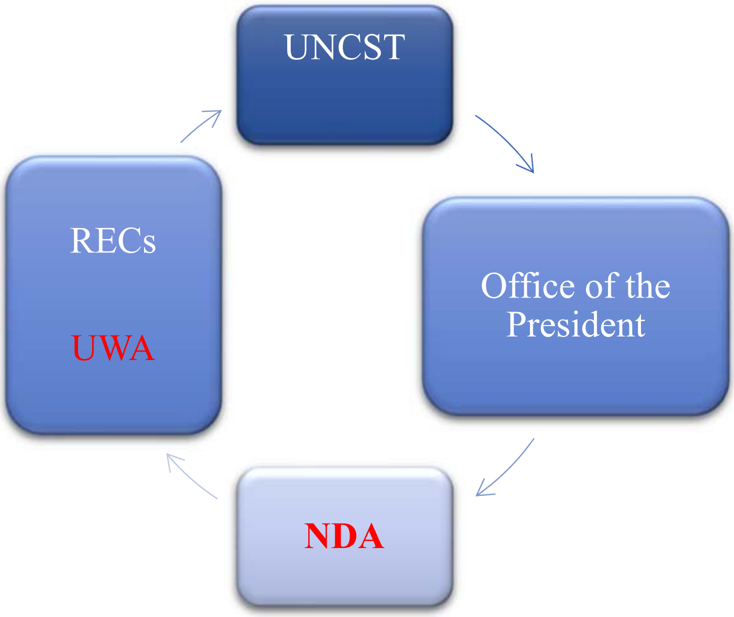
UNCST is mandated to clear all information and materials on research and experimental development prior to being undertaken in scientific institutions or centers and any other enterprises. The research clearing process also takes keen interest in the potential application of the research results. Thus, UNCST has developed regulations and guidelines to protect the rights, wellbeing, and safety of the research participants and the surrounding environment. Both humans and animals as research participants as well as aquatic resources are considered research participants. Therefore, any persons or institution (both local and international) intending to undertake research in Uganda is expected to register their research projects with UNCST, and every proposal submitted is evaluated for its scientific integrity and ethical appropriateness. Some of the regulations and frameworks developed include: the National Guidelines for Research Involving Humans as Research Subjects (2014); Research Registration and Clearance Policy and Guidelines (2007); Patent Amendment Act (2002); and Access to Genetic Resources and

Benefit-sharing Regulations (2008). These regulations warranted the creation of Research Ethics Committees (RECs) within research institutions.

Currently, there are 24 institutional review committees across the country, 7 of which have been accredited by UNCST. The institutional review committees have the responsibility of reviewing both the science and ethics of research protocols that have been submitted to them before such protocols are registered with UNCST. For clinical trials, however, there is an additional requirement for investigators to obtain a clearance certificate from the National Drug Authority for each drug trial or research protocol to be tested; approval must also be obtained from the Uganda Wildlife Authority and National Forestry Authority (where applicable) for research on game parks, forests, and other protected areas.

Research is conducted by universities, research organizations, private companies, local and international NGOs, foreign scholars, and nationals. UNCST is responsible for securing a conducive environment for research undertakings through policy and strategy development, coordination of all S&T activities, research evaluation and quality management, upholding of ethical standards, and intellectual property management in the country. In this regard, and in addition to the research guidelines, UNCST incorporated the national research policy into the National Science, Technology, and Innovation Policy (2009); it has helped establish institutional review boards at scientific institutions to aid in the development of research proposals, reduce plagiarism, undertake scientific peer review, and monitor the implementation research projects. For academic research, faculty research committees with varied specialties are established to vet students’ research proposals. The students can then proceed with and register their research protocol with UNCST and obtain clearance to undertake the study; a viva voce can then be organized at the faculty for the students to present and defend their results. This process is partly regulated by the National Council for Higher Education (NCHE) whose role is to set minimum standards of practice for all academic institutions (public and private) in the country. In addition, academic institutions should adhere to the regulations and frameworks for research undertaking in the country as set by UNCST.

[Figure 2-5] Research Regulation and Management Framework



Oversight by RECs

Research Ethics Committees (RECs) are established under Section 5e of the UNCST Act (Cap 209**),** 1990: UNCST has powers to "establish specialized committees, research councils, organizations, and experimental and development activities or other scientific and technological services." With the support of UNCST, RECs are established and instituted in organizations whose mandate allows them to undertake scientific research and experimental development. In instances wherein a research institution is unable to set up its own REC, it can use an accredited REC of another institution to review the research project independently with the aim of upholding ethics and preserving or protecting the rights of humans and animal subjects as participants in the research process. RECs can approve or reject a research protocol or request researchers to amend and resubmit their protocol.

UNCST has established the Accreditation Committee for RECs in Uganda (ACRECU), a five-member committee appointed by the UNCST Executive Secretary on a three-year renewable term limit. ACRECU was established mainly to help review REC submissions, accredit them, and uphold the ethical standards practice as set forth by UNCST in the research guidelines. The REC accreditation process has two stages. First, REC should undertake self-assessment to identify its strengths, achievements, and areas that need improvement. REC should submit a report of the self-assessment to UNCST along with an application for accreditation. At the second stage, ACRECU will review REC's application and will inspect REC's host institution. ACRECU also assesses REC's operation facilities such as office space, documentation, and storage facilities and standard operating procedures. REC may then be granted accreditation for a period of three (3) years. Upon expiry of the operation period, REC has to re-apply for renewal of accreditation through the same stages.

[Table 2-1] Accredited Research Ethics Committees in Uganda

S/N	Research Ethics Committee (REC)	Institution of Affiliation
1.	National HIV/AIDS Research Committee	Uganda National Council for Science and Technology (UNCST)
2.	Joint Clinical Research Center	Joint Clinical Research Center (JCRC)
3.	Uganda Virus Research Institute	Uganda Virus Research Institute (UVRI)
4.	School of Biomedical Sciences REC	Makerere University's College of Health Sciences (CHS)
5.	Mbarara University of Science and Technology REC	Mbarara University's Science and Technology (MUST)
6.	School of Medicine REC	Makerere University's College of Health Sciences (MUCHS)
7.	TASO REC	TASO
8.	School of Public Health REC	Makerere University's College of Health Sciences (MUCHS)
9.	Mengo Hospital REC	Mengo Hospital
10.	School of Health Sciences REC	Makerere University's College of Health Sciences (MHCH)

S/N	Research Ethics Committee (REC)	Institution of Affiliation
11.	Mbale Referral Hospital REC	Mbale Referral Hospital
12.	Hospice Africa Uganda REC	Hospice Africa Uganda
13.	Mild May Uganda Research and Ethics Committee (MUREC)	Mild May Uganda Hospital
14.	Mulago Hospital Research and Ethics Committee (MHREC)	Mulago National Referral Hospital
15.	International Health Sciences University REC	International Health Sciences University
16.	St Mary's Hospital Lacor Hospital REC	St Mary's Hospital Lacor
17.	Kampala International University REC	Kampala International University
18.	Makerere University School of Social Sciences REC	Makerere University's College of Humanities
19.	Uganda Cancer Institute REC	Uganda Cancer Institute
20.	St Francis Hospital Nsambya REC	St Francis Hospital, Nsambya
21.	Gulu University REC	Gulu University
22.	Vector Control Division REC	Vector Control Division
23.	THETA Uganda REC	THETA UGANDA
24.	CURE - Uganda	CURE - Uganda

Source: UNCST Research Ethics Committee Database

National Drug Authority Research Oversight

The roles and functions of NDA in research oversight are set forth in Section 40 of the National Drug Policy and Authority Act (NDPA) (Chapter 206): "The authority (NDA) can issue a certificate to any person for the purpose of carrying out clinical trials with respect to a drug that may be specified in the certificate"; and "No person can carry out any clinical trial with respect to any drug unless he or she is in possession of the NDA certificate." In this regard, NDA is a key partner when it comes to undertaking clinical trials and ethical clearance and ensuring the safety of humans and animals involved in drug trials.

Oversight by the Uganda National Health Research Organization (UNHRO)

UNHRO is mandated under Section 6(e) of the UNHRO Act "to register, renew, and coordinate different types of health research in Uganda and promote multi-disciplinary and inter-sectoral research collaboration in a bid to establish essential national health research, which is consistent with the National Health Strategic Plan." In executing this function, as a key requirement within the UNHRA Act, the body collaborates with UNCST and NDA to oversee research practices in the health field. This registration process is done centrally at UNCST.

Specialized Technical Committees

UNCST has also put in place specialized technical committees consisting of eminent scientists and in various fields of science such as industry, engineering, chemical and physical sciences, health sciences, information sciences, agriculture, energy, social sciences, and biological sciences. These committees are responsible for advising UNCST on policy and establishing a strategic direction for STI in the country. Under UNCST are task forces and standing committees such as the National Bio-safety Committee, which has the responsibility of minimizing the potential human and environmental harm that may arise from research undertakings in biotechnology. The committee further controls use and exposure to radioactive material and other dangerous substances such as pathogens. The National HIV/AIDS Research Committee reviews and approves the research undertaken on HIV/AIDS participants and other associated research protocols. The Intellectual Property Advisory Committee, in addition to expert panels and consultants, undertakes specific assignments and tasks on behalf of UNCST.

Office of the president

With the hindsight that the execution of research undertakings can result in harmful practices and or security threats such as bio-terrorism and chemical weapons, among others, UNCST, in executing her mandate, is required to liaise with the Office of the President – particularly the Research Secretariat -- to assess all research protocols for any likely threats to security in the country. The Office of the President critically assesses issues of national security together with the researchers, research team, and research protocol. Once the office of the president approves the research protocol, UNCST issues a research permit. The process is usually completed within 14 working days. The research permit grants you permission to collect information and access specified locations as required for your study. The Resident District Commissioners, who are representatives of the president at the district level, are also duly notified either for you to avail yourself of security or to grant you permission.

2.3 REC Chairs' Forum

The ethical environment keeps evolving on a daily basis due to unforeseen challenges. On behalf of UNCST, RECs are the designated custodians of ethics within the scientific institutions. Therefore, UNCST thought it wise to establish a forum for REC chairs countrywide to share experiences in best practices in ethical implementation, share challenges to the ethical standards that can affect the ethical review process, and provide mentorship to upcoming Research Ethics Committees. Since 2010, this forum has been organized annually on a quarterly basis; to date, it has helped strengthen national capacities in research ethics as well as the ethical review process nationwide.

2.4 Research Site Monitoring

Research compliance with the approved research protocols in Uganda is very low despite the existence of institutions such as IRBs/ RECs, UNCST, UNHRO, NARO, NDA, and NDA, among others. Noncompliance with the agreed-upon protocols and violations of Ethics frequently occur due to lack of information by the researcher on the consequences of abuse of the regulations, failure to obtain the anticipated data in the subject area and consequently make changes in the field, failure to complete within the required time frame, and other non-research-related reasons such as spying. This implies the need for constant reviews of the research protocol to assess expiry periods, regular monitoring of the localities where the research protocol was cleared to take place, and conduct of completion checks on the research process. This ultimately minimizes unethical behavior among researchers. In 2007, the UNCST, the National Drug

Authority (NDA), and some RECs agreed to undertake consistently the on-site monitoring of research projects aimed at promoting ethical conduct of research. The process helps to: monitor the progress of research activities at the site, verify information in the study application, and ensure that the research projects adhere to national guidelines for research involving human participants.

2.5 Annual National Research Ethics Conference

ANREC is a platform for engagement and interaction among researchers and other actors involved in the protection of human research subjects in Uganda. The conference is organized annually, bringing together researchers, regulators, policymakers, members of research ethics committees, civil society groups, and research communities to share ideas and best practice principles in order to uphold research ethics. Since 2009 when the first ANREC was held with the theme “Balancing Science and Human Research Protection,” UNCST has embraced the concept of ANREC. The main objective was to provide a platform for sharing experiences, building capacities, sensitizing the public, and identifying options for addressing ethical dilemmas experienced in the conduct of research in Uganda.[7]

ANREC brings together leading scholars in the field of ethics and research to discuss topical and contemporary issues in research ethics due to the increasing sophistication of scientific research, assess ways of community engagement, and determine effective management systems of Institutional Review Committees (IRCs) and standard of care in research. To date, seven ANREC conferences have been organized successfully, and considerable progress in research ethics has been made in terms of policy reforms and establishment of the requisite institutional and regulatory mechanisms at the national level.

[7] UNCST (2009), 1st Annual National Research Ethics Conference Proceedings

3. Research Performance

3.1 Research Registration

UNCST is the apex body for research regulation and management; thus, it is mandated to register and issue research permits to researchers intending to undertake any research investigation in the country. As such, it is a legal requirement for all persons and organizations undertaking research of any form in Uganda to seek permission from the government. Table 2-2 below shows the major types of research undertaken in the country.

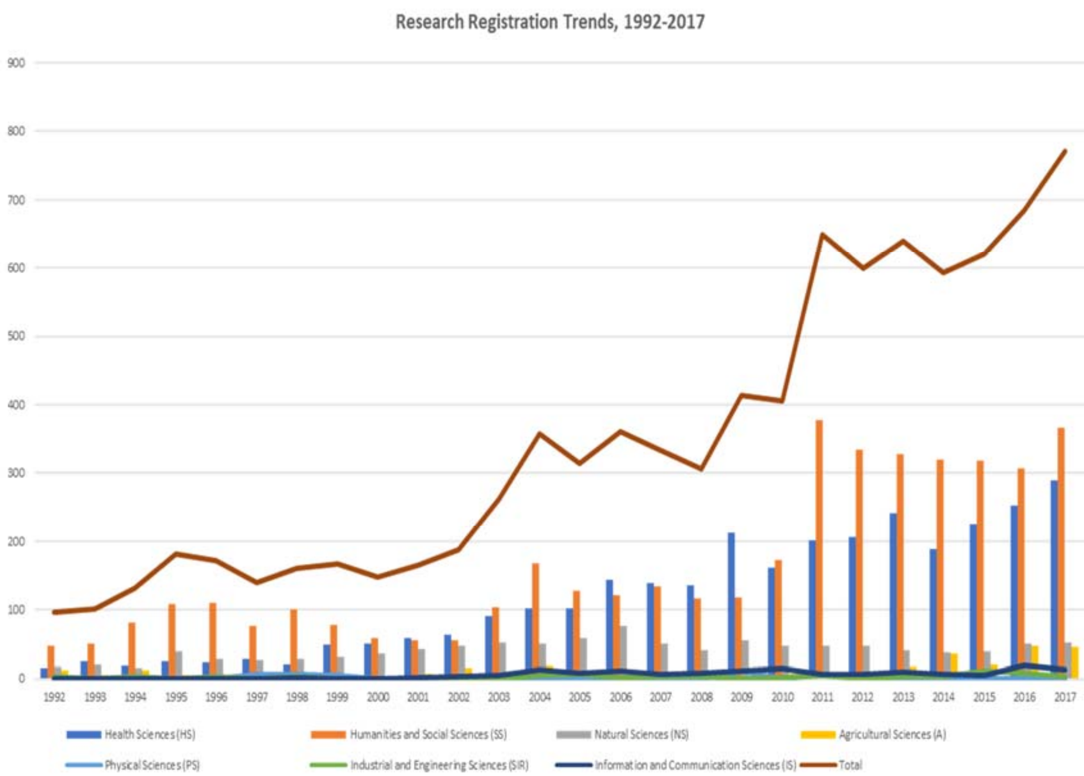
[Table 2-2] Major types of research undertaken in the country

Field	Percentage
Health Sciences	47.3
Humanities & Social Sciences	34.3
Natural Sciences	12
Agricultural Sciences	3.4
Information and Communication	1.7
Industrial & Engineering Sciences	0.8
Physical Sciences	0.5

Source: UNCST Database.

It is noteworthy that majority of the research and development in Uganda is concentrated in the health sciences, followed by social sciences/humanities. The number of research projects registered at UNCST in a calendar year has steadily increased from 98 in 1990 to approximately 800 in 2018. These figures reflect the researchers who have voluntarily registered researches with UNCST but who may not be representative of all research activities going on in the country.

[Figure 2-6] Research Protocols Registered at UNCST, 1992 - 2017



Source: UNCST Research Registration Database.

[Figure 2-6] shows the number of research projects in each research field between 1992 and 2017. It shows a persistent, steady increase in the quantity of research projects registered by UNCST in the fields of Social sciences and Humanities, Health sciences, and Natural sciences. Few projects are registered annually in the fields of Agricultural and Allied sciences, Physical sciences, Information and Communication sciences, and Industrial and Engineering sciences. Research projects in Health and social sciences have ethical implications since they involve humans and communities as such investigators are compelled to seek scientific and ethical approval from UNCST.

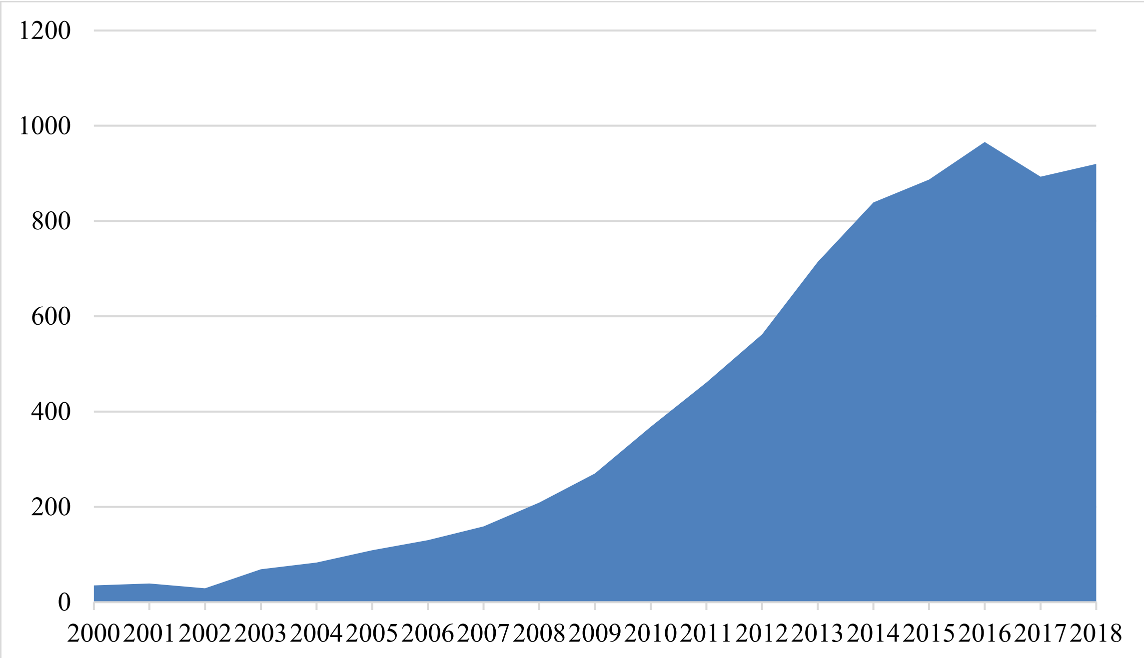
Further, the increased registration of social sciences projects is suggestive of the need to engage societies or communities, which ultimately requires explicit clearance from UNCST as well as the Office of the President. On the other hand, Natural sciences studies are usually carried out in protected areas (game parks, forest reserves, wetlands) and among communities that require facilitation by UNCST in terms of access to these places and the communities. The aforementioned reasons explain UNCST's registration of at least 90% of the research protocols in Health sciences, Social sciences, and Humanities and Natural sciences.

In the case of Agriculture, the large volume of routine research projects/programs conducted by agricultural research institutes could explain the meager number of research projects that have been registered by UNCST over the years. The number of registered projects in Industrial and Engineering fields is very low and is attributable to the difficulties faced by UNCST in registering these studies since most of them are conducted in laboratories and workshops and very few of them involve interaction with people and communities.

3.2 Research Publication Profile for Uganda

Uganda's publications in scientific journals have recorded an upward trend since the year 2000. [Figure 2-7] shows that the number of scientific publications increased from less than 50 publications annually in 2000 to over 920 by 2018.

[Figure 2-7] Scientific and Technical Journal Articles - Uganda



Source: UNCST Database and UNCST Status Report 2014.

In Uganda, majority of R&D undertaken is attributed to donor organizations especially NGOs. They focus not on scientific and experimental research but on program evaluation studies, fact-finding missions through short surveys, and other basic research mainly in health sciences, social sciences, and humanities. On the African continent, Uganda is ranked 10th out of a total of 57 countries with output of 17,406 publications, H index of 169, and 312,337 citations in all. See [Table 2-3] below.

[Table 2-3] Comparison of Publications on the African Continent

	Country	↓ Documents	Citable documents	Citations	Self-Citations	Citations per Document	H index
1	South Africa	272886	247039	3677627	771119	13.48	423
2	Nigeria	90031	84718	645110	134927	7.17	181
3	Tunisia	86600	81933	678610	141681	7.84	174
4	Algeria	65714	63705	444666	94937	6.77	157
5	Morocco	62636	58839	507921	95955	8.11	179
6	Kenya	35120	31548	652942	91803	18.59	233
7	Ethiopia	22934	21479	251468	54208	10.96	136
8	Ghana	20052	18232	236627	31015	11.80	142
9	Tanzania	17484	16072	298625	42246	17.08	161
10	Uganda	17406	15748	312337	45222	17.94	169

Source: SCImago (2019)[8].

[8] SCImago Journal & Country Rank. Retrieved from http://www.scimagojr.com (accessed on September 19, 2019)

4. Conclusion

Research management and evaluation in Uganda have advanced quite significantly since the creation of the Uganda National Council for Science and Technology (UNCST) in 1990. Recent policy reforms have yielded new guidelines for research involving human research participants in Uganda, the latest being the 2014 UNCST guidelines for research involving humans as participants. A total of 24 Research Ethics Committees (RECs) have been instituted to support the research process, uphold ethical standards, and reduce the violation of rights of human research participants through the scientific and ethical review of the submitted proposals.

Despite the commendable progress in advancing research ethics in Uganda, improving efficiency in the protocol review processes, developing research policies, building the capacity of both investigators and members of ethics committees, sensitizing the population and resolving ethical dilemmas that exist in the conduct of research in Uganda remain an uphill task.

Part IV. Emerging Technologies with Focus on ICT in Uganda

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1. Introduction

Uganda has not effectively benefitted from the developments that came with the three past industrial revolutions. For example, most of the agricultural production in the country is carried out using a hand hoe despite the advancement of agricultural mechanization globally. Uganda has not made enough efforts toward research, innovation, technological advancement, and regulatory frameworks to enable her to reap the benefits from the three industrial revolutions.

Although Ugandan President Yoweri Museveni has appointed a team of experts from various sectors to spearhead the country's journey toward the Fourth Industrial Revolution, the task force needs to study carefully the country's chain of technology and advise the Government on how best Uganda can harness the emerging technologies for development.

Therefore, this policy paper is aimed at highlighting key policy issues on emerging technologies with focus on ICT to guide policymakers. Special consideration has been made on the evolution of ICT in Uganda, the progress made thus far, and the strategies to enable Uganda to benefit from the emerging technologies. The paper consists of the background, emerging technologies in relation to the ICT industry, current state of the ICT sector in Uganda, level of implementation of the ICT policy, initiatives toward emerging technologies in Uganda, challenges, and recommendations.

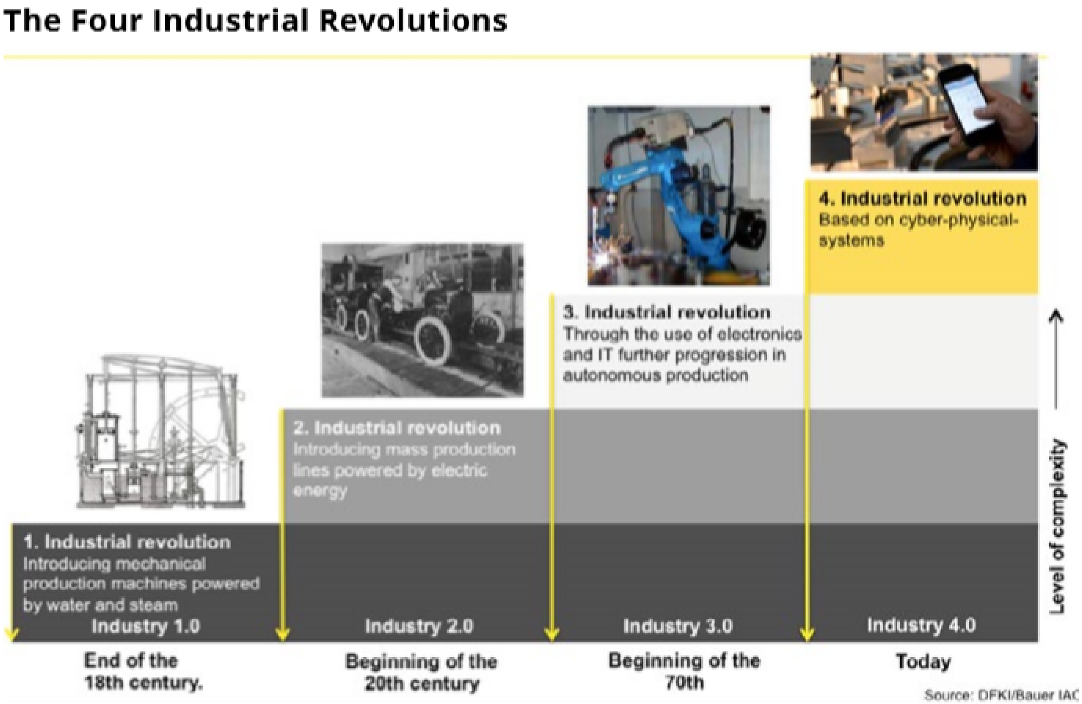
2. Background

All over the world, Information and Communication Technology (ICT) has revolutionized the way the production and distribution of goods and services are organized. This has led to emerging and new business models, making it possible for people to acquire and exchange information in an increasing variety of formats and collaborate with one another across national boundaries. Developed countries have advanced primarily because of the increased awareness of the importance of Information and Communication Technology in achieving their national development objectives. On the other hand, developing countries, including Uganda, have lagged behind largely due to their relatively modest awareness, development, and application of ICT.

ICT refers to technologies that capture, process, and provide access to information through communication technologies such as interconnected computers, telephone services, standardized computing hardware, radio, and television, powered by technologies such as satellite communication, wireless technology, and Internet. ICT was the major breakthrough of the 3rd industrial revolution where information and technology played major roles in production. From 1760 until now, the world has witnessed four major industrial revolutions.

Globally, over the last decade, the dissemination of information and communication technology (ICT) has caused the transformation of the world into an information society. ICT infrastructure such as fixed-line telephones, mobile phones, Internet, and broadband, people, firms, and Government now have much better access to information, knowledge, and wisdom than before in terms of scale, scope, and speed. ICT dissemination has substantially improved the efficiency of resource allocation, enormously reduced production costs, and promoted much greater demand and investment in all economic sectors (Jorgenson and Stiroh, 1999; Vu, 2011; Lee, et al, 2012; Grimes, et al, 2012; Pradhan, et al, 2015).

[Figure 2-8] Four Industrial Revolutions



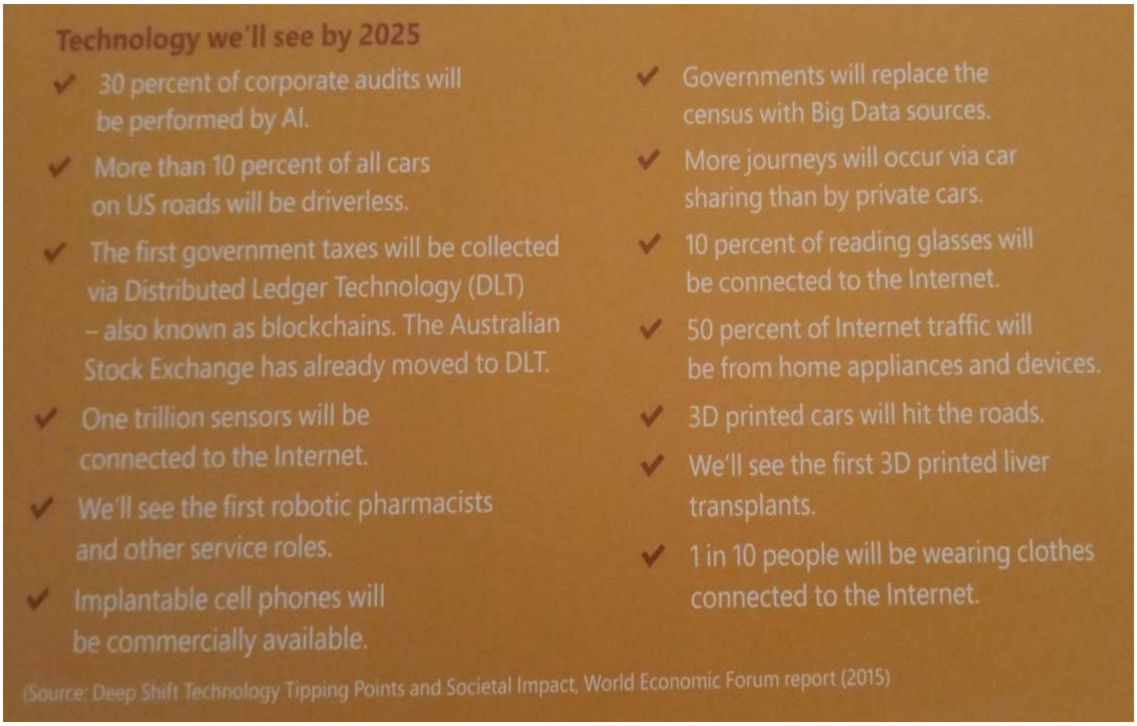
Source: DFKI/Bauer IAO[9].

As shown in [Figure 2-8] above, the first industrial revolution (1760) was triggered by the need to address the rampant poverty that arose from low productivity levels. This saw the birth of the steam engine, which led to the mechanization of agriculture to improve productivity.

The second industrial revolution happened at the beginning of the 20th century and led to increased productivity thanks to the use of electric power. The third revolution (21st century) used data, electronics, and information technology to improve productivity through the automation of production processes. The fourth revolution, dubbed 4IR, is building on the third one, combining technological and digital aspects, having started in the middle of the 21st century. This is a dramatic transition that is expected to create several changes in every sphere of life.

[9] Ojok, Donnas, et al.(2018), *Managing the 4th Industrial Revolution in East Africa: Insights from the First Kampala Digitalization Forum*, p. 33.

[Figure 2-9] What is to be Expected under 4IR?



[Figure 2-9] above shows the confluence of emerging technology breakthroughs, covering wide-ranging fields such as artificial intelligence (AI), robotics, Internet of Things (IoT), autonomous vehicles, 3D printing, nanotechnology, biotechnology, materials science, energy storage and quantum computing, etc.

3. Emerging Technologies in Relation to the ICT Industry

In the 21st century, we have witnessed rapid technology changes as to which trend is expected to increase as the fourth Industrial Revolution unfolds. Changes in technology enable fast progress, at the same time creating demand for the evolution of technology-related careers. This implies that the technology worker of the 21st century will, out of necessity, constantly keep learning, projecting the future to know the kind of skills that will be needed. [Table 2-4] below highlights the various emerging technologies that are closely linked to ICT and will be easily applied in real life.

[Table 2-4] Emerging Technologies and their Applications

Technology	Potential Use
Big Data Telecommunication networks generate huge amounts of data on customer location, demographics, mobile money, and spending. Anonymized and aggregated data can be analyzed to create valuable insights to accelerate efforts by governments, development agencies, and NGOs to address various socioeconomic, environmental, and governance challenges as well as achieve the SDGs. In Uganda, UN Pulse Lab is testing the use of mobile big data to track the outbreak and spread of diseases, such as cholera, in collaboration with the government and mobile operators.	Insights on population location, mobility, and demographic makeup can support efforts to; • Prevent or control the spread of diseases • Prepare and respond to natural disasters • Plan urban settlements and infrastructure investments, e.g., schools, hospitals, and transport • Improve tourist destinations and services Other than people, the same can apply to agricultural plants and animals, minerals, geology, soils, water points, and many other fields to provide real-time data to inform planning.

Technology	Potential Use
Blockchain – This is a secure platform -- based on distributed ledger architecture -- that lets people and organizations share information with each other with a high degree of trust and transparency. By automatically distributing the blocks of information across the whole network, the blockchain ensures that every user sees the most up-to-date information, the database has no single point of failure, and no single institution can control how the information is recorded, audited, or managed. Note, however, that blockchain technology is usually mistaken to mean crypto-currencies such as Bitcoin.	• Blockchain could be used by the Government and its partners in the following areas: • Health – address security, incompatibility, and portability issues of health records, resulting in greater standardization • Identity – enhance KYC processes to speed up business registration and drive G2P, G2B, and B2C transactions; growth • Agriculture – secure land transactions and verify ownership • Oil and gas – enhance fidelity of production and distribution information • Bioeconomy – appreciate issues that relate to bioengineering, biosecurity, and biosafety • With a high level of security in blockchain, you do not need a trusted third party to oversee or validate transactions. For that reason, blockchain is used for crypto-currency, and it can play a significant role in protecting information such as personal medical data.
Artificial Intelligence (AI): Artificial Intelligence (AI) is in its early stages of development. AI refers to computers systems built to mimic human intelligence and perform tasks such as recognition of images, speech, or patterns and decision making. Increasing trends in artificial intelligence point to significant economic disruptions in the coming years.	• Artificial systems that rationally solve complex problems posing a threat to many kinds of employment as well as offer new avenues for economic growth. AI can do these tasks faster and more accurately than humans. • A report by McKinsey & Company found that half of all existing work activities would be automated by currently existing technologies, thereby enabling companies to save billions of dollars and to create new types of jobs. • AI is also used in the management of time-bound systems such as aviation management, scheduling of trains, and prediction of equipment maintenance where there is no room for error.

Technology	Potential Use
	• Other modern applications of AI include assessment of business risks, navigation apps, streaming services, smartphone personal assistants, ride-sharing apps, home personal assistants, and smart home devices.
Machine Learning: Machine Learning is a component of AI that enables computers to be programmed to learn to do something they are not originally programmed to do. This is done by discovering patterns and insights from data to draw conclusions.	• Machine Learning applications are used for data analytics, data mining, and pattern recognition, which creates a base for network intrusion detection, web search results, and real-time adverts. The computer is able to make certain conclusions based on the algorithms it has been fed.
Robotic Process Automation (RPA): A technology that automates repetitive tasks such as processing transactions, dealing with data, and even replying to emails.	• Every line of profession has repetitive tasks that can be automated.
Edge Computing: Edge computing technology has emerged to address weaknesses identified with cloud computing especially having to move data to the data center. This technology has grown to a level of attracting major industry players like Google and Amazon. Cloud computing may not easily be classified as an emerging technology, but it is still growing and attracting more businesses.	• Edge computing is very useful in processing time-sensitive data especially in remote locations with limited or no connectivity to a centralized data center. This is possible because edge computing technology provides mini data centers to enable seamless transactions.
Virtual Reality and Augmented Reality: These two technologies work hand in hand. Virtual reality introduces a user to an environment, whereas	• This technology was initially useful in gaming applications but has now been enhanced to include teaching / learning experiences and training and simulation in fields such as aircraft

Technology	Potential Use
Augmented Reality (AR) enhances their environment and the experience.	and army operations. • Virtual Reality and Augmented Reality have enormous potential in various fields including marketing and rehabilitation after an injury, and they could be used to train doctors to do surgery and offer museum goers a better experience.
Cyber Security: Cyber security may not be considered emerging technology because it has been around for some time but is evolving just like other technologies. That is because cyber threats are constantly evolving, and new ones are emerging at any time. Cyber security will continue emerging as long as we have hackers because they keep trying to access illegally data and they are not going to give up anytime soon but will continue to find ways to get through even the toughest security measures.	• Cyber security technology helps prevent cyber attacks, data breaches, and unauthorized access and manipulation of data of an organization. The world has become more dependent on data since the evolution of digital systems and ICT. Cyber security is as important as the data itself.
Internet of Things (IoT): Several things like fridges, cameras, music systems, etc. are being built with Wi-Fi connectivity to enable connection to the Internet; hence the Internet of Things (IoT). Internet of Things is an emerging technology that will enable devices, home appliances, cars and others to connect to and exchange data over the Internet.	• With the help of IoT, we are able to monitor activities at home, lock our doors remotely, preheat our ovens on our way home from work, track our fitness, etc. • In business, IoT will enable better safety, efficiency, and decision making for businesses as reliable data will be collected over said connections. • IoT will enable efficient predictive maintenance, speed up medical care, and improve customer service, among others.
Mobile and Web applications: In Uganda, there is still a lot of untapped potential. The Global System for Mobile	• With the largely open setup of the technology ecosystem in Uganda, software applications are expected to find increasing utility in sectors such

Technology	Potential Use
applications used by all the major telecom companies in Uganda (MTN, Airtel, Africell) were all developed by/with contractors/individuals from Asia. Ugandans usually develop the front-facing applications for telecom companies.	as e-commerce, payments, services sector, (services delivery, health, etc.) • E-commerce: The changing shopping behaviors of Ugandans create vast opportunities for developing mobile and web applications for shopping. E-commerce also creates many other jobs across the entire value chain -- logistics, payment processing, warehousing, among others. • Health Sector: The uptake and success of mobile and web software applications are dependent on good design, cutting-edge efficiency, and well-thought-out business models that meet the needs of clients/users. • Agriculture to increase labor productivity: There are numerous web applications on the Ugandan market to, for example, predict weather patterns that guide farmers, guide farmers on genuine farm inputs, and provide market information for agricultural produce. • Employment opportunities, increased mobile applications and products brought to market, increased revenues from mobile and web technologies, and improvement of socioeconomic livelihood

Source: Duggal (2019), Top 8 Technology Trends for 2020.[10]

[10] https://www.simplilearn.com/top-technology-trends-and-jobs-article (accessed on 2 December, 2019).

4. Status of the ICT Sector in Uganda

The Uganda Government, through its National Development Plan II, identified ICT as one of the high-potential areas for stimulating growth through business, socioeconomic, development and employment creation due to its cross-cutting nature. According to the Financial Year 2016/17 Budget Speech, the ICT sector contributed 2.5% of the GDP in 2015, while the number of Internet users stood at 18.5 million in 2018. The ICT sector share stood at 6.0% of the National GDP, which signifies the level of commitment from the Government to promote the ICT sector.

The National ICT Policy (2014) aims to transform Uganda into a knowledge society by 2025 with ICT at the center of all spheres of life. It builds on the key elements of the National ICT Policy Framework (2003) and several other policies, such as the E-government Framework policy 2010 (draft) and the Telecom Policy 2011 (draft), which are aimed at increasing access to and usage of digital services. The National ICT policy is a key enabler for the Digital Uganda campaign launched in July 2017 to foster innovation and create a positive socioeconomic impact by empowering people through ICT-based services. [Table 2-5] outlines the four key action areas of the National ICT policy.

[Table 2-5] National ICT Policy Action Areas

Key Action Area	Priority Actions
Expansion of ICT infrastructure and its integration in the Country	a) Extension of the national backbone infrastructure to cover the entire country including addressing last mile challenges b) Integration of communication, broadcasting, and information infrastructure and systems c) Promotion of reliable, affordable ICT infrastructure in rural and remote areas
Deepening utilization of ICT services by government, private sector, non-profit ICT organizations, and wider citizenry	a) Implementation of the national e-government strategy and master plan b) Awareness creation and mindset change c) Increasing penetration of ICT equipment, services, and applications
Enhancement of research and innovation in ICT products, applications, and services	a) Development and implementation of an ICT research and innovation strategy b) Promotion of industrial production and assembly of ICT products c) Promotion of software and application development d) Setting up of ICT parks to support research and development as well as innovation
Improvement of ICT governance and environment in Uganda	a) Consolidation of reforms in the institutional, policy, legal, and regulatory environment for the ICT sector b) Setting of requisite standards and regulations

Source: Ministry of ICT & National Guidance.

5. Level of Implementation of the ICT Policy

5.1 ICT Infrastructure Development

The government of Uganda has made substantial investment in developing the ICT backbone infrastructure across the country. The government of Uganda, through the National Information Technology Authority of Uganda (NITA-U), implemented the National Data Transmission Backbone Infrastructure and e-Government Infrastructure Project (NBI/EGI). The major aim for the NBI/EGI was to connect all major towns within the country onto an Optical Fiber Cable-based Network and to connect Ministries and Government Departments to the e-Government Network. This project has already laid over 2,400km of fiber across Uganda, with several towns connected including Kampala, Mbale, Entebbe, Gulu, and Mutukula. Such investment by the Government has led to the reduction in Internet bandwidth pricing from $190 per Mbps to a new pricing of $70 per Mbps for government agencies. The digital connection of East Africa with Europe by the laying of optical fiber in Mombasa, Kenya is expected to raise the quality of telecom infrastructure in the region and bring down telecom costs. According to the budget speech FY 2019/20, the total optical fiber network covers 49% of all districts and 24% of sub-counties as well as all the border points. The number of Internet users has increased from 13 million in 2015 to 18.8 million in 2017, translating into a penetration rate of 45.4%. The National Backbone Infrastructure for ICT will eventually be extended to cover all districts. In addition, Internet costs will be reduced through the implementation of a new national broadband policy.

5.2 E-Governance

The Government has had plans to implement several initiatives to enable citizens to attain the full benefit of ICT through the implementation of e-government. These include:

- Government ePayment Gateway: To facilitate electronic payments for Government services to improve service delivery efficiency and provide solutions to the needs of the citizens. The ePayment gateway will also promote online business by providing an interface for the private sector.

- Government System Integration: This Integration of all Government systems will ensure simplicity and efficiency of access to Government services in a secure environment. Using a single User ID (National Identification Number), citizens will have easy access to Government services. Additionally, this is expected to promote transparency for Government as data sharing across agencies will be seamless and digitized.

- E-Services: Refers to an online platform to provide quick, efficient access to Government services such as tax assessment and payment, registration for licenses, application for government documents such as passports, National identification, etc. The Kampala City Council Authority manages some of their service delivery through a platform called eCitie, and the government also launched the ebiz platform, a one-stop center for government services.

- Other initiatives include the e-tax platform operated by the Uganda Revenue Authority to shift as many tax transactions and operations online for faster service delivery and transparency. Regarding the management of the country's finances, there is a system called IFMS through which all government agencies' budget allocations are made and tracked.

- These services are still handled in an isolated manner due to the lack of centralized data. Each entity such as the Uganda Revenue Authority, Kampala City Authority, and registration bodies all have independent data that does not link to the other. Multiple data sources create space for manipulation and forgery.

5.3 Enhancement of Research and Innovation in ICT Products

Low policy implementation: Technology training, research, and development in Uganda are all still low in capacity and quality that universities do not play any significant role in the transfer, dissemination, adoption, and development of emerging technologies into

Uganda. Because of the low research and technology development coupled with a low level of private sector participation in R&D activities, Uganda ranked as low as 152nd in the **ICT Development Index (IDI) in 2017.**

Nonetheless, there are some efforts to boost the growth of ICT including:

- **Support to ICT Private Enterprises:** The government continues to provide support to the ICT sector through the provision of grants to help private small and medium ICT enterprises become more competitive. Such support is in form of technical assistance, quality certifications, and capacity building. With such support, there is noticeable growth in the use of locally produced solutions especially mobile apps for personal savings, payment for utilities, money transfer, and other information systems.
- **ICT Innovation Awards:** In a bid to spur ICT innovation, the Government put in place an ICT innovation fund wherein annual innovation competitions are held and winning innovators are given grants to help them develop their innovation into commercial products and build enterprises.

5.4 Role of the Academe in the Promotion of ICT

In Uganda, scholars are introduced to elementary computer studies as early as primary school, where basic concepts are taught to the children. Most of the schools especially around the city have computer laboratories to help introduce the concept to the children. In secondary school, ICT is one of the elective subjects that students are able to choose for consideration at higher levels of education. Those who take up computer studies in university are introduced to more advanced studies including systems engineering, computer software development, and innovations around the ICT industry. Nonetheless, the challenge is that most of the training undertaken at all these levels is mostly theoretical, aiming only at passing examinations. The interviews with a number of software development firms in the country confirmed that most of the university graduates cannot produce something tangible in terms of work output. They are usually required to undergo at least one year of intensive training and mentorship so as not to create conflict with business. In many cases, these are students who have scored highly in the examinations.

Another initiative particular to Makerere University is holding various school competitions to boost the innovativeness of the students. Although these events attract active participation, no efforts have been made to transform these innovations into viable products.

5.5 Private Sector Engagement

There is limited engagement in research and development in the ICT field in Uganda. Note, however, that there are a number of companies dealing with technology-related solutions, business process outsourcing (BPO), ICT training solutions, ICT infrastructure solutions, software and hardware solutions, enterprise solutions support, and business management solutions.

Some of the products include the following, among others:

- Hotel reservation systems
- Centralized procurement systems
- Real estate management and maintenance systems
- Bluetooth social network tools
- Asset management systems and Payroll systems

6. Initiatives toward Emerging Technologies in Uganda

Uganda has not registered much progress in the three industrial revolutions except for a few isolated cases in the field of ICT. There are a few isolated cases of Artificial Intelligence programs in Makerere University. The same can be said of other emerging technologies such as sensor systems, Internet of Things, mobile technologies, data science, etc. Note, however, that most of these programs introduce students to theoretical knowledge with limited practical engagement. The one closest exception is with the field of robotics; the few robotics development programs, rudimentary as they are, are operated within a few universities such as Islamic University of East Africa, Makerere University, and Kampala International University-Ishaka. As such, universities still cannot be considered the main drivers of robotics in Uganda -- the adoption of robotic systems in manufacturing, home gadgets and appliances, etc.

As early adopters, governments play a big role in the growth and advancement of emerging technologies in any country. This is exemplified in countries such as the USA (developed and pushed Internet technology for adoption by the private sector), Estonia (big data and data science in e-government), Dubai (with its 10-year plan), Sweden, Norway, etc. Uganda is undertaking a number of technological efforts that could spearhead the advancement and development of emerging technologies. Government agencies and institutions are implementing projects such as: eCitizen portal, e-procurement for government agencies, Government e-Payment Gateway, e-Voucher project for small farmers, national ID registration project, Government SMS gateway.

These projects can form the foundation for the development and fast-tracking of technologies of the 4th Industrial Revolution. For example, data from the eCitizen portal can be processed with artificial intelligence techniques to determine the best or most suitable service delivery mechanisms.

Technologies such as distributed ledgers can be incorporated into the e-procurement platform to increase transparency and public trust. Telecom user data could be integrated with the national biodata system to streamline service delivery, minimize duplication of roles, and reduce costs.

6.1 Development of Web and Mobile Technologies

The web and mobile technologies space in Uganda is the most diverse sector in the ICT sector, involving every kind of participants from individuals, startups, and small and large businesses. There are many individual mobile app developers, small business built around mobile applications, and big companies with applications they use for service delivery. Application development for large companies is done largely by non-Ugandan contractors. For example, the Global System for Mobile Communications (GSM) applications used by all the major telecom companies in Uganda (MTN, Airtel, Africel) are developed by/with contractors/individuals from Asia. Ugandans usually develop the front-facing applications for telecom companies, many times with limited functionality and of questionable quality.

There are some startup companies that have sprung up in Uganda and which offer their services through mobile and web applications. Most are involved in payment processing, with a few in utility services, bookings, and provision of information. Such examples are mobile apps like Xente mobile app for the payment of school fees and utility bills, Axess for mobile micro-credit online, Airsave for promoting savings using a mobile money app, etc.

6.2 Efforts toward Artificial Intelligence

There are a few players in Artificial Intelligence & Data Science in Uganda. One of them is the Artificial Intelligence Research (AIR) group at Makerere University. The group is largely a research outfit; they have published some papers on work done in crop disease diagnostics, traffic flows, famine prediction, and causal discovery in disease data, among other things. The work seemingly focuses on locally relevant challenges. Their techniques, however, are not necessarily ground-breaking – they are a rehash of conventional techniques used at universities in other countries, akin to university research projects aiming at academic publications as opposed to commercial products. The productivity of research groups is measured by the number, quality, and utility of tools they put out. By this measure, AIR is not scoring highly beyond a few academic outputs. Certainly, very few of its outputs have been adopted, let alone widely for general use. [Figure 2-10] shows one example of a viable artificial intelligence innovation by AIR at Makerere University.

[Figure 2-10] Innovation to Diagnose Malaria based on Artificial Intelligence[11]

With regard to sensor systems, there is some local innovation happening in Uganda. There is a group that has developed a gadget for "detecting" breast lumps, another has developed a non-intrusive "malaria diagnosis" tool, another has reinvented the mobile

[11] https://www.africanexponent.com/post/9535-new-app-to-fight-malaria.

phone-based incubation for newborn babies, and another has developed a fetal scanning tool that works with a smartphone. There are many other tools, gadgets, and devices that utilize sensors to perform a function better, faster, or at lower cost. This is a field that demonstrates a lot of local research potential and development activity, although locally developed commercial products have yet to take root. If such innovators are given the necessary support, they can put out solutions to some of the most pressing problems in Uganda.

6.3 Efforts toward Big Data

There are also a few user/practice groups for Data Science and Artificial Intelligence that operate within Kampala. These regularly bring together prospective, novice, and people with some experience in data science and Artificial Intelligence to work on practice challenges and generate knowledge. This is an encouraging prospect that, on its own, fosters the expansion of a critical mass of practitioners from which highly skilled professionals can emerge.

The big player is UN Pulse Lab Kampala, one the three “Pulse Labs” (New York, Jakarta, Kampala) created under the UN Global Pulse program to “harness big data for development and humanitarian action.” In Uganda, it is run under the United Nations Resident Coordinator in Uganda, and they “bring together data scientists, data engineers, partnership specialists, academicians, and technical experts to generate high-impact data analysis tools to address development challenges.” Their projects/works have development/ humanitarian/governance overtones, and some are conducted in conjunction with government agencies. Their flagship work is a word analysis tool for radio programming content. This is a novel application, but the material societal benefit of this application and much of the other work is hard to conceptualize.

6.4 Efforts toward the Development of Robotics

In a bid to encourage young students to pursue innovation, Makerere University has been holding science innovation competitions over the years. The latest robotics competition was held in 2017, targeting children in secondary schools. Organized by iLabs@Mak, the robotics competition was held at the College of Engineering, Design, Art, and Technology (CEDAT). The competition attracted students from nine different schools across the country under the theme “Using Science and Technology to Increase the Efficiency of Small-Scale Manufacturing Industries.”

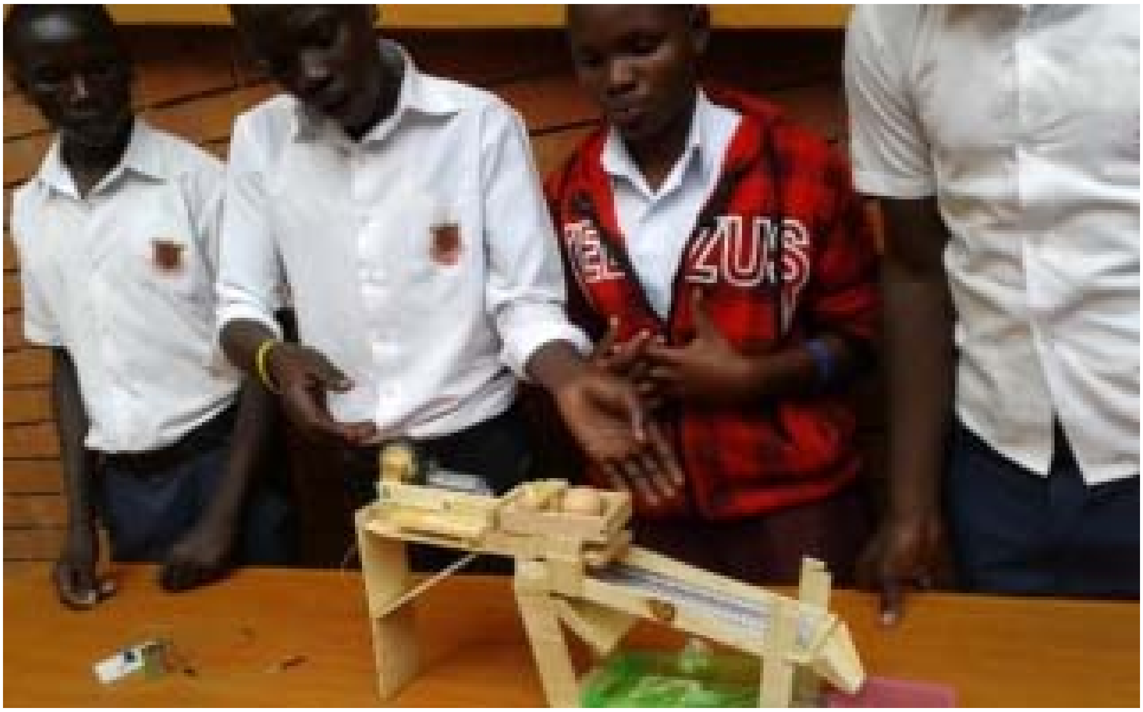
Students of Gulu High School got a trophy for developing the best prototype of an Egg Crack Partition Machine. Reagan Okumu, a student of Gulu High School, is among the students who worked day and night to develop the model. Reagan comes from a poor family in Omoro district, located more than 30 kilometers to the East of Gulu town. Every holiday, he sets up a small business stall and makes chapat and “rolex” (a snack made of fried eggs, cabbages, green peppers and tomatoes rolled in a chapati), to raise money for his tuition and scholastic materials.

[Figure 2-11] The Rolex[12]

Reagan has been using a knife to crack eggs, a major ingredient in his chapatti and "rolex" business. But using a knife has caused him to cut himself many times, besides performing under capacity. Being part of the group makes him feel lucky as he has gained skills during the training and competition. Reagan's wish is to have the prototype of their egg crack and partition machine developed into a functional device to help Reagan increase his production and minimize the risk of injuries.

[12] Source: https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTjOa5R9Dv1vdusLwXobOlhBREqcpT62wu-IlE1hzf_LNRSRTpV&s.

[Figure 2-12] Students of Gulu High School Demonstrate the Egg Crack Partition Machine Prototype



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Another scientific innovation challenge by Makerere University iLabs saw students from various schools come up with interesting innovation in the fields of robotics, artificial intelligence, and data mining. The exhibited innovations included the following: St. Mary's College Kisubi (Rapid Response Fire Brigade Car and Hydra 256, a water harvesting and treatment tank), Gayaza High School (ATM monitoring machine and Water conservation machine), Mt. St. Mary's Namagunga (Conveyor belt and smart voting booth), Namilyango College (Security system and Traffic lights system), Kings College Buddo (Smart seed planter and paper gluing machine), and Makerere College School (Smart rubbish truck and Smart lavatory).

[Figure 2-13] Students from St. Mary's College Kisubi Demonstrate the Rapid Response Fire Brigade Car[13]

6.5 Data Analytics Development in Uganda

There are some businesses in Uganda that are building data analytics capacity. An example is solar power solutions provider Fenix International, which operates in several countries and has an active data science team that carries out data engineering, analytics, and visualization for the data coming from the solar Global System for Mobile Communications (GSM) machines. Their in-house solutions are geared toward improving the quality of their product offering. While majority of this team are foreigners, this is a glimpse of the ideal role of artificial intelligence and data science in society – a marketplace product offering solutions. While it may sound insensitive, it is true that people's willingness to pay for a product is the biggest testament of its utility.

[13] https://cedat.mak.ac.ug/wp-content/uploads/2013/09/dsc_0680-1200.jpg.

Other micro-services providers involved in micro-insurance and micro-credit services have data science forming a large part of their operations. Telecom companies -- the entities that generate the largest amounts of data -- have also begun employing data science practitioners. This underlines the growing tendency among Ugandan companies that generate large amounts of data in their operations to leverage such data for increased productivity. It is unclear at this point what the complexity of data analytics these companies are carrying out, i.e., are they using cutting-edge production-grade AI techniques/tools, or is it basic database-based aggregation techniques coupled with visualization?

Nonetheless, this is a sector that is on the up -- it will take the continuing expansion of the mass of data scientists, development of world-class expertise, and development of commercially viable products in the AI space for this technology to become a reasonable contributor to the economic development in Uganda in terms of jobs created, revenues, and social value.

6.6 Internet of Things (IoT)

There are many Internet of Things (IoT) products/devices on the market in Uganda. Quick examples include surveillance cameras connected to smartphones, security systems linked to mobile phones, solar power machines controlled from mobile phones, etc. All of these commercial products are imported, although limited local expertise to develop systems for interconnecting existing gadgets and systems has been demonstrated. Apart from solar systems, applications of IoT systems have yet to become widespread in Uganda partly due to limited knowledge, expense, and connectivity issues.

6.7 Efforts toward Blockchain Technologies

From a conceptual point of view, blockchain technologies can have applications in a broad range of fields as shown in Table 1 above. In Uganda, however, the most widely known application is crypto-currencies, so much so that, to most Ugandans, blockchain is synonymous with crypto-currencies. The highly speculative nature of crypto-currencies and their unregulated state has exposed the public to scams and Ponzi schemes. That has eroded public trust in digital currencies and damaged the prospects of the merits of digital currencies being considered by regulatory authorities.

The proliferation of a myriad of other digital assets has not helped the situation as they have demonstrated themselves to be little more than software backed up by hope.

In 2018, an organization that calls itself Blockchain Association of Uganda organized the "Africa Blockchain Conference" in Kampala, which brought together enthusiasts, government, and business leaders "to discuss the role of blockchain in Africa's Transformation." The usual prospective applications were parroted at this conference; as the futility of blockchain has been demonstrated all over the world, however, the gathering did not the lead to any substantial product or policy developments. Interestingly, the same conference has been organized for July 2019, with the Fourth Industrial Revolution, not blockchain, as theme. Clearly, the organizers themselves, the "Blockchain Association of Uganda," have pivoted away from blockchain. It is fair to conclude that much of the hullabaloo around blockchain is little more than pizazz driven by the desire of the proponents to justify its existence – this is true of Uganda as it is of many other countries. There is no indication of any technology/product development work going on in the blockchain space in Uganda. As mentioned above, all the focus seems to be crypto-currencies; companies operating online crypto-currency exchanges are founded in Uganda.

6.8 Growth of Innovation Hubs

In a bid to advance technology, private sector businesses have invested in innovation hubs working on various fields and technologies. Majority of them focus on IT solutions, mobile and web applications, and other engineering technologies. A few of them have made forays into supporting the development of emerging technologies. Going by outputs, not many products in emerging technologies seem to have come from such innovation centers. Most of the innovation hubs act as entrepreneurship centers for small startups; not from a business sense, but their role is mainly the provision of a conducive working environment in the technology sector, emergence of new technologies, operational models, funding sources, and technical capacity as well as what they have done (with examples), including their challenges and third-party analysis. In the emerging technology space, startup companies in Uganda are often created by innovators turned entrepreneurs. Note, however, that a number of them -- while they may have interesting concepts -- never manage to bring quality products to the market or fail with their business models. Consequently, apart from mobile and web application, startup companies in Uganda have not been responsible for any other emerging technology in Uganda.

7. Challenges

Despite the aforesaid interventions by both government and private players, the ICT sector in Uganda has not grown enough to bring about the desired revolution to transform the Ugandan society. The situation gets worse with the coming of the fourth industrial revolution, which is expected to build on the existing structures and technologies of the third industrial revolution. The low development of ICT can be attributed to, among others:

- Lack of critical infrastructures: The Government has made some progress in laying the optic fiber cable (under the NBI project) across some towns in the country. On the other hand, the NBI/EGI has not yielded the much needed impact as it targets Government Ministries, Departments, and Agencies. The private sector has not yet had a chance to take advantage of the initiative to boost innovations in the ICT industry. Second, the last mile connectivity to fiber optic is very costly, and it has left many Government entities off the grid. Accessing the Government data center for data storage is also an expensive venture.
- Lack of investment in research and development in the ICT sector to develop innovations to provide solutions for today and in the future. According to the National budget speech for the financial year 2018/19, no specific intervention was planned in line with research. Research helps generate new knowledge that informs of future investments.
- Bad policy decisions: According to the 2017/18 Uganda National Information Technology Survey, social media platforms are some of the popular avenues for citizens to engage with each other and to pursue businesses and education opportunities. In 2018, the government introduced the Over-the-Top service tax popularly known as OTT. At least 76% of the survey respondents cited the price of Internet subscription as a key limitation to their Internet use. This was followed by concerns over slow Internet speeds and lack of connectivity in some areas.

- Low ICT skills development: Graduates with IT training are produced by training institutions, including universities, without the level of advanced skills necessary in the market especially the IT engineering skills required for computer applications, electronics, and software engineering. This has undermined Uganda's ambition to increase productivity in the technology field and has consequently led to majority of advanced technical jobs being taken by foreigners. A key example of this phenomenon is observed in Uganda's telecom sector where majority of the top experts creating the telecom technologies used in Uganda are foreigners. Uganda's IT graduates do not have the skills to create marketable products and technologies. The poor quality of IT applications produced in the country has made Uganda's IT products uncompetitive in the market and consequently caused the low growth of Ugandan IT-based companies.
- According to the May 2019 Inter-University Council for East Africa (IUCEA) report, at least half of the graduates produced by East African universities are "half-baked" for the job market. Uganda is in the lead with the worst record of at least 63% of graduates found to lack job market skills. This challenge results from the low quality of training and the non-rigorous training methods that students are subjected to. Thus, while thousands of young people are graduating each year, their qualifications are unable to secure jobs for many of them. Moreover, employers are increasingly shunning new graduates in favor of the highly skilled; thus complicating the problem of youth unemployment in the region.
- Insufficient ICT Business Development Facilities: Plans to establish a National ICT incubation hub have been on the table for quite a while with very little progress made. Most of the incubation centers around the country are privately owned, levying charges to access them, which makes it difficult for SMEs. Even then, incubation centers have major weaknesses such as: insufficient business support services; inadequate physical and operational infrastructure; and costly Internet and inadequate capabilities to exploit opportunities in the emerging sectors such as ICT, biotechnology, and new materials.

8. Recommendations and Next Steps

We are going to enter this very different world. It will affect our identity and all the issues associated with it -- our sense of privacy, our notions of ownership, our consumption patterns, the time we devote to work and leisure, and how we develop our careers, cultivate our skills, meet people, and nurture relationships. It is already changing our health and leading to a "quantified" self, and it may lead to human augmentation sooner than we think. The list is endless because it is bound only by our imagination.

8.1 Policy Action Points

An important objective for policymaking is seeking to encourage the efficient use of data in the economy to promote good practice through transparent, clear, fair, and consistent data protection policies.

There is a need for evaluation and refocusing of the second National Development Plan's five priority areas to match the emerging realities in the face of the fourth industrial revolution. Harnessing the power of mobile technology will be critical to influencing progress across all development goals. This is because mobile is the first, among all Information and Communication Technologies (ICTs), to reach across geographies, income levels, and cultures, enabling access to basic services where traditional brick and mortar firms have often failed -- financial services, support to farmers, access to health information, education, and clean energy and more. Mobile technology also enables the most widespread means of accessing the Internet, the foundation for Uganda's digital future.

With more people using mobile services today in Uganda than at any other time in the past, the technology has a direct impact on social and economic activities and, by extension, supports progress toward the achievement of national and global development goals. As for digital entrepreneurship and emerging technologies, around 4 out of 5 tech startups utilize one or more mobile platforms in their solution, with mobile connectivity enabling the development of blockchain and other emerging technologies in Uganda.

The government should empower the relevant agencies (such as Ministry of ICT and Ministry of Science, Technology, and Innovation) to effect the required changes to respond to the data revolution including the promotion of sustainable guidelines for efficient data sharing between mobile operators and other stakeholders in the ecosystem, including across government agencies.

There is a need to update the Uganda Data Privacy bill to bring it in line with international best practices and expedite its passage into law, in order to maximize the potential of using mobile big data to address environmental, social, and governance challenges. The bill should address issues surrounding data collected by non-government bodies and businesses to enable government access and usage of said data.

The establishment of research and innovation infrastructure to facilitate research and development of ICT products calls for the development and implementation of an ICT research and innovation strategy, promotion of industrial production, and assembly of ICT products and promotion of software and applications development. Investment in skills and infrastructure are also required to support a viable blockchain ecosystem.

To ensure that Uganda's youth have skills and that they are employable, the education system will have to adapt to prepare individuals for the changing labor market. The supply-side and demand-side (Governments and Development agencies) parties should invest in the necessary capacity, skills, and infrastructure to support the new revolution solutions. Sufficient human capital development for scientists and research manpower is also needed. Training abroad enables knowledge and tech transfer. In Uganda, blockchain company Crypto Savannah has committed to training a talent pool of more than 5,000 developers over the next five years.

Entrepreneurs and innovators in the ICT area need two primary ingredients: ready product-development service and business support. For software applications coming out of Uganda to meet this standard, the quality of software created by developers in

Uganda needs to be improved. A product development service should include a collection of trained developers that offer readily available free or low-cost software product testing and improvement service. For business support, developers and innovators need to be provided with material support such as computational infrastructure for testing and hosting their software applications.

In May 2018, the Ministry of ICT announced plans to appoint a taskforce to explore how the Ugandan government can utilize blockchain technology. The taskforce should seek to implement sustainable business models to support the long-term use of mobile big data solutions as opposed to one-off pilots. Technology Transfer and homegrown solutions must be promoted to support the use of data.

There are many developed countries that are willing to collaborate with Uganda to support her in the form of transfer of knowledge, technology transfer, and capacity building in various areas such as research, ICT, agritech, healthcare, fintech, and other emerging technologies. Collaborations in research and development should be pursued by establishing bilateral agreements with multinational companies for a win-win partnership wherein the companies share funding for research and technology development, technology transfer, and piloting.

Regulatory Aspects: Uganda generates most of the data especially through telecommunication. Note, however, that the collected data is not owned by the government. For example, telecommunication companies collect a lot of data about the citizens, but that data remains a private property. This means that the government cannot make use of the data for analysis and planning purposes.

8.2 Detailed Action Plans

As a way of upgrading Uganda's emerging technology policy, the following specific action plans and relevant stakeholders have been identified:

[Table 2-6] Specific Action Plans and Relevant Stakeholders for Each Policy Item

Policy Issue	Actions	Stakeholders
Refocusing of the second National Development Plan’s five priority areas	i. Develop a working paper on which partners can support Uganda in this field. The paper should state exactly what Uganda wants to do as well as the kind of support sought. ii. Define the domain of focus for Uganda, e.g., agritech, healthcare, space, fintech, engineering, smart city, etc. iii. Sensitize private research institutions, researchers, and private sector on the emerging technologies’ paradigm shift iv. Organize a conference of scientists to identify new areas of research focus for the country v. Rally the leaders to champion policies and initiatives toward emerging technologies	• National Planning Authority • Ministry of Science, Technology, and Innovation • Presidential Task Force on Emerging Technologies • Ministry of ICT and National Guidance • Ministry of Finance, Planning, and Economic Development
Regulatory Environment	i. Update the Uganda Data Privacy bill to bring it in line with international best practices and expedite the passage into law, in order to maximize the potential of using mobile big data and other technologies ii. Develop and implement a National Strategy on emerging technologies iii. Review the Electronic Transactions Act and other related laws to address the emerging issues	• Ministry of Finance, Planning, and Economic Development • Ministry of ICT and National Guidance • Parliament of Uganda • Presidential Task Force on Emerging Technologies
Research and Innovation	i. Support and promote the establishment of research and	• Ministry of Science, Technology, and Innovation

Policy Issue	Actions	Stakeholders
Infrastructure	development centers to support research in emerging technologies ii. Establish National Information Technology Skills Development iii. Establish incubation centers for technology transfer and to support the commercial viability of startup technology firms	• Ministry of ICT and National Guidance • Presidential Task Force on Emerging Technologies •
Human Capital Development	i. Develop supportive curriculum, human resources (teachers/ instructors), and teaching methods at all levels of education ii. Improve practical skills development and increased private sector involvement in ST&I vocational training and in higher education courses for science and engineering iii. Develop and implement special training programs including training abroad to enable knowledge and technology transfer	• Ministry of Science, Technology, and Innovation • Ministry of Education and Sports • National Curriculum Development Center • Presidential Task Force on Emerging Technologies • •
Collaboration	i. Rally regional and international partners to support the projects under this domain	• Ministry of Science, Technology, and Innovation • Presidential Task Force on Emerging Technologies

Part V. Establishing Science and Technology Parks in Uganda

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1. Introduction

1.1 STI Conditions in Uganda

The National Science, Technology, and Innovation (STI) Policy (2009) seeks to create a coherent, comprehensive policy environment that supports -- on one hand -- the integration of traditional aspects of Science and Technology and, on the other hand, innovation and entrepreneurship in the national development processes that improve the country's competitiveness and better the standards of living for Ugandans.

The policy further prioritizes and provides for increased funding for the STI sector from 3% to 5% of the National Gross Domestic Product (GDP) per annum. In addition to the foregoing, the development and upgrading of research and development (R&D) infrastructure including technology incubators and Science and Technology Parks (STPs) around the country to support the commercialization of research results is at the core of this policy. This is a particularly important undertaking of the policy to enable the development of Uganda's manufacturing sector by developing and nurturing technology entrepreneurship for industrialization, improving competitiveness, and creating decent employment.

The National Development Plan (NDP II: 2016/17-2019/20) and the Uganda Vision 2040 set the Technology Achievement Index (TAI) for the STI sector as its key result measure. The TAI is expected to improve from 0.15 in the marginalized group of countries to 0.21, ranking Uganda among the dynamic adopters. By achieving a TAI of 0.21, Uganda would have achieved Sustainable Development Goal (SDG) 1 of eradicating extreme poverty and, consequently, middle income country status. In addition to the TAI, enhanced research and development intensity measured by increased Gross Expenditure on Research and Development (GERD) from 0.23 to 1% as recommended by the African Union (STISA, 2024) is another key result area. To create an enabling policy environment for the STI sector to achieve these targets as set forth in Uganda's Vision 2040, National Development Plan (NDP II: 2016/17-2019/20), and STI policy (2009), the Ministry of Science, Technology, and Innovation (MoSTI) was created in 2016.

Below are selected scientific research and development indicators for Uganda, which is showing its comparative position both regionally and globally in the technological landscapes.

[Table 2-7] Summary Showing Research and Innovation Indicators

Category	Indicator	Value
Research and Development Expenditure	Overall R&D intensity (GERD as a % of GDP), 2016	0.23
	Business R&D expenditure as a % of GDP, 2016	0.01
	Government R&D expenditure as a % of GDP, 2016	0.11
	Higher education R&D expenditure as a % of GDP, 2016	0.10
	Private non-profit R&D expenditure as a % of GDP, 2016	0.01
	Percentage of government R&D financed from abroad, 2016	43.9
Human Resources	Total R&D personnel (head count), 2016	2881
	Total R&D personnel (full-time equivalent)	1612.7
	Mean years of schooling, (age 2>=15 years), 2017	6.1
	GNI per capita (2011, PPP$), 2017	1658
	Human development index (HD value), 2017	0.516
	Life expectancy at birth, 2018	62.5
	Expected years of schooling	11.6
	Gross tertiary science enrollment (%)	0.022
	Total R&D personnel per 1,000 total employment (FTE), 2016	0.05
	Women researchers in proportion to the total researchers (HC) (%), 2016	29.8

Category	Indicator	Value
Innovation Indicators	Global Innovation Index, 2016	25.3
	Technology Achievement Index, 2016	0.15
	Uganda's competitiveness rank, 2018	117
	Uganda's global competitiveness index, 2018	46.8
	No. of patents granted for Uganda, 2018	117
	No. of innovative enterprises	3002
	Royalties and license fees received (million USD), 2016	1.53
	Payments for the use of intellectual property (million USD), 2016	19.74

Source: UNCST, Uganda's Research and Innovation Outlook, 2019

A TAI of 0.15 (2014) places Uganda among the marginalized group of countries (TAI<0.2) remaining a net importer of medium- and high-technology goods and services. The research intensity of Uganda as measured by GERD (0.23 per cent of the Gross Domestic Product (GDP)) of 2018 is still below the African Union recommendation of 1 per cent of GDP. The low tertiary enrollment in science at 0.022 per cent is representative of the generally very low ratio to the total enrollment in tertiary institutions of learning (just 3%), ranking the country 106th out of 125 countries. Note, however, that Uganda's quality of scientific institutions ranks high in the world at no. 34, second only to South Africa in the Sub-Saharan country group.

[Table 2-8] Comparative Analysis of the Global Innovation Index (GII) of Selected African Countries

Country/Year	2012	2014	2016	2018
Uganda	25.6	31.14	27.14	25.32
Kenya	28.9	31.85	30.36	31.07
Rwanda	27.9	29.31	29.96	26.45
South Africa	37.4	38.25	35.85	35.13

Source: WIPO, Global Innovation Index Reports (2012-2018)

In Africa, South Africa has the highest GII, although it has been declining from 37.4 in 2012 to 35.13 in 2018, whereas that of Uganda fluctuated between 25.6 to 25.32 in 2018. Uganda's GII improved only between 2012 and 2014 from 25.6 to 31.14, respectively, declining back to 25.32 in 2018.

1.2 Manufacturing Sector in Uganda

Uganda's national development policies and strategies are aligned around transforming the country from a low to lower middle income country by 2040 as specified in the Uganda Vision 2040, National Development Plan (NDP II: 2016/17-2019/20), National Industrial Policy (2008), and National Strategy for Private Sector Development (NSPSD 2017/18-2021/22). These policies and strategies give priority to adding value to local raw materials and agro-based processing with a performance target of increasing the contribution of the industrial sector to the GDP from 25 per cent (2010) to about 31 per cent, with manufactured exports increasing from 24.1 per cent to 50 per cent. The other key performance targets include increasing the following: labor productivity in the industrial sector from USD 3,550 to USD 24,820; labor force in the industrial sector from 7.6 to 26 per cent; technology up-take and dissemination as measured by the TAI from 0.24 to 0.5; GERD as a share of GDP from 0.1 to 2.5 per cent; and Innovation as measured by the number of patents registered per year from an average of 3 to 6,000 by 2040.

Traditionally, transforming an economy from a mainly agrarian to an industrialized one is associated with manufacturing increasing its contribution to the GDP. Currently, the contribution of manufacturing to Uganda's GDP is still small at an estimated 8.2 per cent, with its biggest part based on agricultural and mineral resources (Africa Development Bank Group, 2014). In addition, Uganda's competitive advantage in the manufacturing subsector has not been harnessed fully because of some policy and structural challenges such as limited access to finances, transport infrastructures, energy, R&D infrastructure (both physical and human), and ICT networks, among others (UNECA, 2017). Uganda's Research and Innovation Outlook 2019 showed that the business sector expenditure on

R&D is still low at 0.01; University-Industry R&D collaboration was estimated at 44.5%, and technology adoption by firms (measured on a scale of 1=low; 7=High) was 4.1, suggesting that majority of Ugandan firms acquire technology through technology transfer from abroad but not through local R&D endeavors.

Cognizant of the fact that agriculture is a major contributor to Uganda’s economy and the interdependencies and linkages between industrial development and agriculture, Uganda’s policies and strategies provide for an agriculture-led industrialization by seeking to create a favorable environment to reduce economic and technological bottlenecks affecting the agricultural and manufacturing sectors. To create such favorable environment that supports the growth of Uganda's manufacturing sector in moving toward high value-added products and service (e.g., after-sales services offered by manufacturing companies) with high technology content, the government, through the National Development Plan (NDP II: 2016/17-2019/20), seeks to develop regional STPs to facilitate and spur research, development, and innovation for industrialization, improved competitiveness, and decent employment creation for its youth population.

2. Policy on Science and Technology Parks

2.1 Definition of Science and Technology Parks

The International Association of Science Parks (IASP) defines a Science Park (interchangeably used with Science and Technology Park) as any of the establishments managed by specialized professionals with the aim of increasing the prosperity of its community by cultivating a culture of innovation and the competitiveness of businesses and knowledge-based institutions within its boundaries.

To achieve the intended goals, STPs encourage and control how knowledge and technology flow among Universities, R&D institutions, business enterprises, and markets. STPs also facilitate the formation and growth of knowledge-based companies by creating an enabling environment in terms of infrastructure and economic policies including high-quality space and facilities, incubation, and spin-off processes.

2.2 Why Science and Technology Parks in Uganda?

The second National Development Plan (NDPII) for Uganda proposes the establishment of Science and Technology Parks as one of the ways to: support and enhance the assimilation of science and technology into the processes of national development; increase the transfer and adoption of technologies; and improve the coordination and guidance of research and development in the country. Other reasons include:

- Developing and upgrading the necessary STI research and development infrastructure that could support and facilitate the operations of science and/or technology-based enterprises
- Creating an attractive economic environment for attracting foreign direct investment (FDI) to encourage technology transfer, adaptation, and utilization by attracting large foreign technology companies

- Fostering the formation of startups and spin-offs through business incubation, coaching, and mentoring via the commercialization of research results and innovations from universities and research institutes
- Facilitating and supporting the growth and competitiveness of existing and new modern Small and Medium Enterprises (SMEs) by utilizing the technological expertise, innovations, and knowledge available at Universities and Research Institutes
- Developing competency among Ugandans especially University faculty and students through consultancies, joint research projects, internships, part-time jobs, and Universities using the employees of STPs tenants as adjunct faculty
- Generating a sizeable number of decent jobs that are knowledge-based and able to absorb the country's highly skilled young professionals and which also attract those working abroad to come back and work in Uganda in order to contribute to national development
- Transforming the mainly agrarian Ugandan economy into an industrial and knowledge-based one by encouraging and creating a favorable environment for invention, innovation, and knowledge-based enterprise development.

2.3 Global experience in establishing Science and Technology Parks

All over the world, Science, Technology, Engineering & Innovation have played a vital part in the development of countries such as Korea, China, India, and Singapore, which are considered the world's biggest economies today. China, for example, has the second biggest and fastest growing economy in the world with a 10 per cent average rate over the last 30 years. China also ranks number one in the world in terms of manufacturing (currently the world's manufacturing capital) and export of goods and services and plays a prominent role in international trade. The per capita income of Singaporeans is one of the largest in the world even though the country has no natural resources or raw materials of its own. Singapore has focused on scientific and technological development as the main driver for economic growth. Its main source of revenue comes from the exports of high-end electronics, chemicals, and services. The revenue generated is used to purchase natural resources and raw materials that are processed into high value-added manufactured products and services (e.g., after-sales services for manufacturing companies) for export. Singapore was ranked 2nd overall in the Scientific American Biotech ranking in 2014 due to its Biopolis International Research and Development Center for Biomedical Sciences. Singapore trade to GDP level is now among the highest in the world, averaging 400% during the period 2008-2011 (Krishna and Sha, 2015).

These countries have developed their economies through targeted scientific and technological developments to refine inputs into exports and have now become the largest trading nations in the world. In addition to targeted scientific and technological developments, favorable and linked (connected) policies such as STI, Industrialization, Education, and other trade and economic development policies were put in place to support the use of science and technology in spurring economic growth.

Against this backdrop, the Ugandan government recognized the need to integrate science, technology, and Innovation (STI) into the processes of national development if the country is to achieve the Uganda Vision 2040 goal of attaining lower middle income status within 30 years with per capita income of USD 9,500. In accordance with Paragraph 219 of the Uganda Vision 2040, the government promises to set up science and technology parks, engineering centers, and technology and business incubation centers that meet international standards and which would reduce the cost of product development and innovations. According to the 2nd National Development Plan (NDP II: 2016/17-2019/20), “No progress has been registered regarding the construction and development of regional science parks and technology incubation centers (SPTIC) envisaged under Uganda's National Development Plan (2010/11-2014/15).”

Therefore, before the end of the 2nd National Development Plan (NDP II: 2016/17-2019/20), the government is taking its initial steps toward the process of establishing the envisaged regional STPs that will act as a platform for building in a phased manner the STI infrastructure for the Silicon Valley-like science and technology parks in the four main regions of Uganda (Central, Northern, West Nile, Eastern and Western). As the government embarks on this mega undertaking in terms of the capital investments required, responsible institutions should consider the following factors as good practices for the successful establishment, development, and management of STPs in Uganda:

a) Availability of qualified and skilled professionals who are able to carry out high-quality research and development within the focus areas (disciplines) of the STPs
b) Capacity of the STPs to provide expertise in the area of business development and management to nurture the tenant companies, especially small and medium-sized companies that may lack such resources
c) Good policy and legal framework are important in creating an enabling economic environment for startups and attracting large foreign technology companies to STPs
d) STPs should have a clear criterion for the admission and graduation of the tenant companies in compliance with the STPs' identity
e) Branding and marketing by STPs of their services to target customers to generate sales
f) The management of STPs is key, i.e., managers should ideally have a wealth of experience in finance and operation management of similar or closely related entities
g) STPs should have strong and stable support in terms of funding, government political will, and institutional support, e.g., attached to universities or research institutions

3. Organization and Legal Framework of Regional Science and Technology Parks

National Development Plans I and II propose the development of STPs as a way to integrate science and technology into the national development process to support national development. Therefore, to be able to establish the proposed regional STPs, preparing a detailed STP master plan should be the first practical step in this journey for Uganda.

The proposed five regional STPs should be hosted at the regional government Universities, which have large chunks of undeveloped land in their names. While the host University can provide land and allow its facilities such as laboratories, workshops, and libraries to be used by the STP tenant companies, the government should provide funding for infrastructure development and to cover the continuing operational costs of the STPs.

The proposed regional STPs could be incorporated as non-profit institutions co-owned by the host Universities and the government represented by the Uganda National Council for Science and Technology as the designated competent authority (promotion agency) and which is consequently responsible for STP management.

The designated STP competent authority could be promoted to report directly to the office of the Prime Minister or Vice President, creating the necessary governance and direct coordination with the political leadership to enable direct support in terms of resource allocation and create the necessary policy environment to facilitate the smooth establishment and management of these STPs.

The physical infrastructure in the STPs should be built around the concept of “Work-Life Integration” to create a conducive environment where today’s knowledge workers can live, work, and play, e.g., be built near regional municipalities that offer good services such as good schools, transport and communication network, shopping malls, restaurants, sports facilities, cinema, etc.

In addition, the research and teaching faculty and students of the host and neighboring non-host Universities could be encouraged to participate in the STPs' activities through consultancies, internships, and joint research projects. Universities could also be encouraged to tap into the expertise of the employees of the tenant companies in the STPs for adjacent teaching positions to improve the experience of learning for their students while strengthening the collaborations with these tenant companies. The research institutes, laboratories, and centers that have gone beyond the proof of concept stage to a working prototype stage of developing products and services with commercial potential could also become the first tenants of the STPs.

STPs should initially focus first on biosciences (Health, Agriculture, Flavor, Fragrance and Cosmetics, Agro-processing), Agro-chemicals, ICT, Renewable Energy, and material science where Uganda has comparative and competitive advantages in terms of raw materials and human capital. This is very much in line with the East African Community Industrialization policy adopted in November 2011 (EAC, 2012) and which aims to facilitate the structural transformation of the manufacturing sector in the region through high-end value addition and product diversification based on comparative and competitive advantages.

There should be a clear, deliberate plan to enable the available qualified and skilled professionals (especially at MScs & PhDs) to carry out high-quality research and development (R&D) within the focus areas (disciplines) of the STPs to support the National Industrialization Plan. Such human capital development plan could include attracting foreign experts and encouraging Ugandan technical industrial experts and research scientists in the diaspora of specific scientific disciplines into the country.

There is a need to establish industry-focused Government Research Institutes (GRIs) /or program to support the growth of local industries/ manufacturing sector through applied research and, where possible, for reverse engineering to be encouraged. Such GRIs could, in the beginning, focus on applied research supporting a particular industry sector, e.g., the Fragrance, Flavor, and Cosmetics Research Institute, to support the Ugandan companies producing beauty and cosmetics products to grow and remain competitive in the local and regional markets at the same time by encouraging the transfer of knowledge from the GRIs to the industry. In addition, reverse engineering could be encouraged in these GRIs whenever possible (looking at expired patent documents and those not protected in Uganda) to fast-track the country's technological catch-up. The GRIs' research and development activities should also emphasize product design more than production processes as the largest share of value is created through the design and development of new products. To ensure that GRIs fulfill their mandates, there should be a deliberate plan for GRIs to tap into and utilize the talented graduates from universities in their R&D activities and programs.

Specialized technology business incubation (such as Makerere University's Food Technology and Business Incubation Center and Renewable Energy Business Incubator) and acceleration programs should be established in each regional STP to help new and struggling early-stage businesses develop into financially stable, sustainable business enterprises through the provision of technical and business support services. In addition, admission to these programs should be bundled to propose a project producing a new product or service based on an R&D result with commercial potential.

The design /or planning of the regional STPs should take into consideration creating science and technology-focused industrial parks appropriate for such enterprises (e.g., tenant companies graduating from the STP's incubation programs) because technology products designed and developed at universities, research centers, and research institutes must be produced as industrial products to be able to contribute to national development. In addition, STPs could gain from the approach of encouraging the creation of open innovation networks to foster collaborations across the research and industry ecosystem.

Good policy and legal framework (relevant laws and regulations) should be emphasized (or put in place where they do not exist) as they are important in creating an enabling economic environment for startups, micro, small, and medium enterprises (MSMEs) and attracting large foreign technology companies into STPs. For example: (1) STPs' tenant companies could be exempted from income taxes, import, and export duties for at least the first three to five (3-5) years of their operations; (2) A strong Intellectual Property Rights (IPR) protection regime could be attractive to large foreign technology companies and startups whose products (or services) are based on IPR and which could easily attract investments; (3) Linkages/Linked policies such as STI, Industrialization, Education, and other trade and economic policies could help create the necessary economic environment to support MSMEs' growth and encourage innovation and companies to invest back in R&D; (4) The regulatory environment should be sufficiently company-friendly to ensure that products developed through R&D could freely be placed on the market.

The regional STPs should adopted the Triple Helix model of innovation (with a few modifications as needed) of university-industry-government interactions. The triple helix model of innovation can help spawn novel institutional and social formats vital for the creation, adaptation, and application of knowledge and skills necessary to support Uganda's industrial growth.

4. Financing of the Regional Science and Technology Parks in Uganda

STPs' financing can be broken down into three aspects:

- Initial financing for infrastructure development: The development of infrastructure including civil works and equipment could be directly funded by the central government through the development projects' funding facility, and regional Universities could provide the land on which these STPs can be developed. The regional Universities could also be expected to avail themselves of their existing infrastructure such as laboratories and workshops to support these STPs. In addition, common user equipment infrastructure facilities for researchers from both industry and research institutes to carry out indispensable verification tests during the R&D process should be considered. Furthermore, the Uganda National Council for Science and Technology should build a research equipment database for equipment available in the country.

  STP continuous operations: There are costs involved in continuing operations including maintenance of buildings and other related infrastructure, laboratory and workshop equipment, management salaries, utilities, provision of security, and STPs' promotions. The continuing operations of the STPs should also be funded by the central government for at least 15 years, with a review done every 5 years to establish whether the STPs are on track toward self-sufficiency in terms of financing their operations. After 15 years of continuous funding by the central government, the regional STPs should be expected to have become self-sustaining to cover their operating costs mostly through rent of space, common facilities, and sales of STP services.

  Startup capital financing for the tenant companies of STPs: The National Innovation Fund should be the first-line financing mechanism for startups specified in the proposed regional STPs. This is because access to credit is one of the major constraints to startups in Uganda, partly because of the prohibitive high lending rates (EAC, 2017) and also because most financial institutions do not want to provide funding for technology-based startups as new ventures are seen as high risk and venture capitalists (a relatively new concept in Uganda) are hard to come by. Therefore, as a way of supporting the commercialization of research outputs and the formation of new knowledge-based SMEs, the government should continue operating the Innovation Fund.

- The second proposal is bank financing especially through Uganda Development Bank Limited (UDBL), which can be an alternative financing mechanism for the startup tenant businesses in the regional STPs. Uganda Development Bank Limited (UDBL) is a Development Finance Institution (DFI) focused on accelerating socioeconomic development through sustainable financial interventions in line with the country's development priorities. To activate funding from UDBL, STPs should develop a special arrangement with the bank to assist tenant companies in preparing loan applications and put in place arrangements to service the bank debts as the STPs could provide loan guarantee for their tenant companies.

- The third proposed financing mechanism is through a venture fund. Although this concept is relatively new in Uganda, it could be one of the ways some STP tenant companies can be financed. Attracting a venture capitalist should be a direct function of the management of STPs.

5. Vision for Science and Technology Parks for Uganda

Regional STPs should be established as national technological hubs just like Daejeon in South Korea, Biopolis in Singapore, and Silicon Valley in the USA for technological development, product commercialization, and attraction of large foreign technology companies. These could provide a favorable economic environment to attract large foreign technology companies and spur the growth of new and existing science and technology-based SMEs, which in turn create high-value employment. These STPs should also provide the STI infrastructure (physical and human) necessary to support Uganda's industrialization plan where manufactured goods can contribute at least 50 per cent to the country's export and domestic trade by 2040.

6. Technology and Business Incubators in Uganda

In this section, the Food Technology and Business Incubation Center and Renewable Energy Business Incubator Limited are discussed as examples of specialized incubators in Uganda, where the proposed regional STPs could be built.

6.1 Food Technology and Business Incubation Center – foundation and perspectives

The Food Technology and Business Incubation Center (FTBIC) is owned by the School of Food Technology, Nutrition, and Bio-Engineering of Makerere University and was established in 2009. FTBIC was the first technology and business incubator based in the University in the East and Central Africa region. The economic rationale for developing this incubator is to transform research innovations, knowledge, and skills into viable enterprises. The FTBIC programs particularly target the youth, women, and young graduates with innovative science and technology ideas that have commercial potential. Its vision is to be a center where innovative world-class agribusiness enterprises and products are born, whereas its mission is to nurture and sustain food and allied businesses especially among women and young graduates by providing innovative research, practical solutions, linkages, entrepreneurship development, and outreach leading to wealth creation and nutrition enhancement. FTBIC offers the following services: (1) Processing infrastructure and Office space; (2) Research and product development support; (3) Technology transfer services; (4) Enterprise development and outreach; and (5) Hands-on skills development and training.

FTBIC is well-equipped with many different processing units (meat processing, dairy processing, fruit and vegetable processing, extrusion processing, confectionery and baking, thermal processing and fermentation) that most food processing operations can be carried out.

Although the incubator is funded by the central government under the Presidential Support for the Science and Technology program, as a way of sustainability in the future, the center engages in the following for profit: (i) contract processing and manufacturing especially through its mobile fruit and vegetable processing unit that supports farming processing; and (ii) short, specialized hands-on skills training on different technologies through the Skills Training Program for SMEs (STRAP 4 SMEs). In addition, small affordable rental fees have been introduced for all tenant companies and are charged on a case-by-case basis (FTBIC, 2019).

6.2 Renewable Energy Business Incubator (REBi) Limited – foundation and perspectives

As a private, non-profit company limited by guarantee, Renewable Energy Business Incubator (REBi) Limited was incorporated in September 2016. REBi was established in 2011 as part of the Nordic Climate Facility (NCF) project titled "Sustainable Renewable Energy Businesses in Uganda." The project was implemented by Makerere University in close collaboration with The Royal Norwegian Society for Development. The vision of REBi is to become the leading one-stop center of excellency for Renewable Energy business needs in Uganda, and its mission is to build sustainable renewable energy businesses that transform lives through the engagement of resourceful stakeholders. REBi focuses on four main aspects of renewable energy: (1) Business incubation (nurturing) of renewable energy startups; (2) Provision of business development support services for Micro, Small, and Medium Enterprises; (3) Skills development; and (4) Piloting of technologies with the aim of technology transfer.

REBi supports tenant companies admitted to its incubation program through the provision of a number of services:

- Individual assistance to develop business plans and applications for funding
- Network with financing institutions/capital providers, authorities, industry, and universities

- Training to upgrade managerial, operational, technical, financial, and legal capabilities
- Office space for a limited period, depending on the business idea
- Investment support based on application
- Training to upgrade managerial, operational, technical, financial, and legal capabilities
- Joint research projects with universities, focusing on tenant company challenges such as market and barrier analysis

7. Conclusion

The role of manufacturing in the socioeconomic transformation and development of nations is well-documented. For Uganda to transform from an agrarian economy to an industrial one, the contribution of manufacturing to its GDP must increase to a substantial level to include a large share of high-end technology products and services (such as after-sales services provided by manufacturing firms) traded.

Therefore, in addition to pursuing non-neutral sectoral policies in support of manufacturing, STI infrastructure development (both physical such as STPs and human resources) should also be fast-tracked to ensure that any gains made in the manufacturing sector can be sustained in the future through local innovation to guarantee good future economic performance. It is also worth emphasizing that the process of establishing STPs to support the industrialization process in Uganda could benefit from the lessons learned from comparative successful case studies of STP models in emerging economies.

CHAPTER 3

Sharing Korea's Experiences in Industry-University Cooperation

Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (Ⅱ)
– Policy Capacity-building Program

Chapter 3. Sharing Korea's Experiences in Industry-University Cooperation

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1. Experiences in Industry-University Cooperation in Korea

1.1 Introduction: Three Academic Revolutions in the Korean History of Science Culture

Industry-University Cooperation (IUC) is a source of competitive advantage of nations, regions, and communities. This chapter mentions the three academic roles, the performance indicators of UIC, and the cultural gap between industry and university. IUC's roles in the STP -- for instance, Taiwan, Korea, and USA – are explained through the case of Korean UIC policy and history. We suggest the implication for Uganda and expect the real performance to contribute to economic development and quality of life. The methodology for this part consists of literature review and in-depth case studies in Daedeok Innopolis and Research Triangle Park (RTP), USA.

Historically, universities and institutions of higher learning have gone through three academic revolutions: teaching, research, and Industry-University cooperation (innovation and entrepreneurship) [Table 3-1].

[Table 3-1] Role of University

1. Teaching	2. Research	3. I-U Collaboration (innovation)
Teaching in specialized higher educational institutions continued until the beginning of the 19th century	Advancement of knowledge via research with particular emphasis on science and technology disciplines	Getting involved in knowledge transfer and economic development and job creation

Source: Kim and Choi (2005)

Universities and Higher Educational Institutions (HEIs) in the last two decades have come to occupy an important part in national innovation systems (NIS), which is a complex of “all important economic, social, political, organizational, institutional, and other factors that influence the development, dissemination, and use of innovations.” From a broader perspective, universities, together with public Research and Development (R&D) labs and science agencies, public policies (on industry, research, innovation and higher education, etc.), and business enterprises are now considered important actors in the NIS of Asia-Pacific economies. The rise of Asia in the global knowledge-based economy from the mid-1990s is closely associated with the rise of knowledge institutions of higher learning and scientific research performance.

Every Asia-Pacific country embraced and introduced policies related to innovation in a variety of forms. With consultancy and collaborative links with industry assuming traditional forms of engagement, new policy and institutional measures for technology transfer and innovation to engage with society and business enterprises are gaining prominence. Policies for incubation, startups, and spin-offs, technology transfer offices (TTOs), and science and technology parks (STP) have gained tremendous prominence in leading Asia-Pacific universities. Different national innovation systems in the Asia-Pacific region have given rise to varying roles of universities. While universities in Southeast Asian countries and India continue to play a traditional role of teaching and generating human capital, in countries such as Korea, Singapore, China, Taiwan, and Japan, universities are being transformed into entrepreneurial universities. Science and innovation policies in these countries have orchestrated the goal direction of universities as frontiers of innovation. Universities in Australia and New Zealand have so far been quite successful in marketing higher education to neighboring Asian countries. They have, in recent years, begun to embark on innovation and commercialization of research. The paper focuses on Southeast Asia and draws some comparison with more dynamic university ecosystems in East Asia. Doing so brings into focus the emerging innovation landscapes across the region (Krishna, 2019).

According to a recent UNESCO report, higher education enrollment in the world at large saw tremendous growth from 32.6 million in 1990 to 182.2 million in 2011. A little less than half of this (46%) expansion is attributed to East and South Asia. As Table 2 shows, higher education, particularly gross enrollment ratios for bachelor's-level students, witnessed tremendous growth between 1990 and 2011, particularly in China, India, Malaysia, and Thailand. In some of these countries, enrollment increased by three to six times between 1990 and 2011. In most Asian countries, the role of private higher education has been an important factor in the expansion of higher education. In some countries such as China, Malaysia, Vietnam, and Cambodia, private institutions were not permitted. After the 1990s, however, all these countries opened higher education to private players. In the last fifteen years, private higher education has been the most dynamic growing sector in Asia than in other regions of the world. In 2011, over 40% of all students in higher education were enrolled in private institutions. In Korea, Japan, Singapore, the Philippines, and Indonesia, the figure ranges from 62 to 81 per cent. Expansion of higher education is also associated with the growing public expenditure on higher education. As Table 3-2 also shows, most Asian countries increased their public higher education expenditures in proportion to the total publication education expenditure from 3 to 13 per cent.

[Table 3-2] University Education: Gross Enrollment Ratio (GER) and Expenditure

Country	GER for Bachelor Level Students		GER in Higher Education	Public Expenditure on Higher Education as % of Total Public Education Expenditure		Public Expenditure on Higher Education as% of GERD
	1990	2011	2015	2000	2011	2015
China	2	11	42	na	Na	8
India	5	15	25	20	33	4
Japan	24	43	62	15	20	12
South Korea	28	69	87	15	18	9
Malaysia	3	19	na	32	35	29
Philippines	20	25	na	14	12 (-)	na
Thailand	10	38	na	20	13 (-)	20
Indonesia	Na	19	23	Na	20	34
Vietnam	Na	18 (2008)	na	Na	15	5
Singapore	Na	Na	na	20	37	28

Source: Krishna (2019), UNESCO (2014), and World Bank (2017-2018)

2. Industry-University Collaboration

University-Industry collaboration (UIC) refers to the interaction between any part of the higher educational system and industry aiming mainly at encouraging knowledge and technology exchange. UIC has had a long history as a means of building organizations' knowledge stock. There has been a substantial increase in these collaborations in several nations including the United States, Korea, Japan, Singapore, and European Union Countries. Such performance has been attributed to a combination of pressures on both industry and universities.

For industry, pressures have included rapid technological change, shorter product life cycles, and intense global competition that have radically transformed the current competitive environment for most companies. With regard to universities, pressures have included growth in the new knowledge and challenge of rising costs and funding issues, which have imposed enormous resource burdens on universities that they seek relationships with firms to enable them to remain at the leading edge in all subject areas. In addition, there is increasing societal pressure on universities for them to be seen as engines for economic growth and less as addressing the broader social issues (i.e., education and generating knowledge) they have had in the past. These pressures on both parties have led to an increasing stimulus for developing UICs that aim to enhance innovation and economic competitiveness at institutional levels (e.g., countries and sectors) through knowledge exchange between academic and commercial domains. Moreover, UIC has been widely perceived as a promising tool for enhancing organizational capacity in open innovation — where an organization employs external networks in developing innovation and knowledge — as a complementary option to traditional internal R&D (Ankrah and AL-Tabbaa, 2015).

The performance of industry–university cooperation is measured by various indicators such as university research capacity, resource capacity, and education capacity. In detail, it was primarily measured by the number of patents, participants, and technology transfers and subsequent revenue. In addition, the number of student startups and university's educational support, such as capstone design programs, are considered important factors (Nam, et al, 2019).

[Table 3-3] Performance Indicator of Industry–University Cooperation

Researcher	Independent Variable	Dependent Variable
Byun Chang Ryul (2004)	Investment by government, private, and university Management (inventor's share in royalties, size of organization dedicated to technology transfer, history of dedicated technology transfer organization) Technology infrastructure (university's science and technology capacity, regional economy, size of university)	Academic achievements (SCI(E) * papers, students' internship, number of graduate students) Technical achievements (technical guidance, technology in possession) Business performance (number of start-ups, number of technology transfers and royalty income)
Kim Cheol Hoe-Lee Sang Don (2007)	Research competency (SCI(E) * papers, international and domestic patent registe) Management competency (number of personnel assigned to technology transfer, number of experts in technology transfer organization)	Technology transfer income Number of technology transfer Number of spin-off companies
Han Seung Hwan-Kwon Ki Seok (2009)	Organizational characteristics (university establishment, establishment type, size) Research capacity (domestic and overseas papers published) R and D funding system (funding scale, funding ratio) Research environment (location)	Achievement of cooperation between industry and academia (number of patent applications in domestic and foreign countries, number of technology transfer and profit)
Kim Gyeong Jin (2010)	Internal competence (the size of the organization dedicated to technology transfer, the number of patent applications, funding) External environment (location, Connect Korea participation, ndustry recognition)	Technology transfer income
Joh Hyun Jeong (2012)	Resource capacity (the number of graduate school students, the total amount of research expenses) Organizational competence (number of personnel assigned to technical transfer, number of professionals)	Patent performance (domestic and foreign patent applications for 3 years) Technology transfer performance (number of technology transfer, revenue) Entrepreneurial achievements (number of professor founding a company, sales of professor start-ups)
Lim Eui Ju-Kim Chang Wan-Jo Geun Tae (2013)	Number of personnel ssigned to technical transfer Number of personnel dedicated to incubation	Achievement of industry–university cooperation (number of technology transfer, income, number of start-up and its revenue)
Nah Sang Min-n Chang Wan-Lee Hee sang (2014)	Tenured professor's experience in private sector The number of professionals Number of faculty support students start-up Number of universities with technology holding companies Number of courses for founding	Technology transfer and commercialization achievement (technology transfer fee per tenured faculty, number of student founders of students' start-ups)

Source: Nam, Gyeong Min, et al (2019), How Resources of Universities Influence Industry Cooperation, J. Open Innov. Technol. Mark. Complex.

[Table 3-4] suggests that there is a gap between university and industry focus; hence the need to understand their different culture and focus for mutual collaboration.

[Table 3-4] Identified Cultural Gap

University Focus	Industry Focus
- Cooperative Knowledge Creation and Generation - Academic Freedom - Common Information Exchange - Students, Post-Docs Who are Not Quite "Employees" - Legal Constraints on Preference for Sponsors - Science & Tech Discovery	- Knowledge Utilization and Competitive Advantage - Proprietary Information to Exploit & Control - Information as Asset - Research-for-hire - Disclosure Normal - Market Hazards - Sales & Profit

Source: Mariann Jelinek.

[Table 3-5] shows that the history of Korean University-Industry Collaboration (UIC) consists of four stages: Industry-Academe Cooperation centered on vocational high schools and professional schools (1960s-1970s); Industry-Academe Cooperation centered on Government-funded Research Institutes (1970s-1980s); Commencement of University-centered Industry-Academe Cooperation (1990s); and Emphasis on the university's role and Industry-Academe Cooperation for Balanced National & Regional Development (2000s).

[Table 3-5] History of Korean I-U Collaboration (UIC)

Time	Policy	Contents
1960s-1970s	Industry-Academe Cooperation centered on vocational high schools and professional schools	Providing customized training for companies through industrial high schools and commercial high schools Establishment of (vocational) college for mature technology manpower; establishing and fostering Government-funded Research Institutes and University Research System in the process of industrialization Training and supplying professional manpower and promoting Industry-Academe-Research Institute cooperation as an intermediate digestion and absorption channel for foreign

Time	Policy	Contents
		technology introduction rather than joint R&D.
1970s-1980s	Industry-Academe Cooperation centered on Government-funded Research Institutes	Establishing a Government-funded Research Institute to lead industrial technology development; creating an atmosphere of private company's own research institute; promoting R&D projects centered on Government-funded Research Institutes. Daedeok Research Complex was established to lead industrial technology development.
1990s	Commencement of university-centered Industry-Academe Cooperation	Government R&D investment in universities was increasing, and various projects were promoted Therefore, the university emerged as the subject of Industry-Academe cooperation. In addition to the venture boom, the role of the university as a technology entrepreneur was strengthened. Enhancement of universities' ability to cultivate research personnel as private R&D investment expands.
2000s	Emphasis on the university's role and Industry-Academe Cooperation for Balanced National & regional Development	Promoting new Industry-Academe cooperation centered on technology as a technology user; introducing the Industry-Academe cooperation agency system (2004); promoting the NURI project for regional university innovation (2004); supporting the first Technology License Office (TLO) support project (2006); introducing a technology-holding company system for Industry-Academe-Research Institute cooperation (2008); supporting Leaders in the Industry-University Cooperation (LINC) support project (2012) with $4 mil. per university; emphasizing Industry-Academe-Research Institute cooperation and regional linkages to create an innovative economic ecosystem (2013).

Source: Choi, Jong-in (2019).

[Table 3-6] presents the factors that facilitate or impede University-Industry Collaborations: Capacity and Resource; Legal issues (IPR); Management & Organization

Issue; Technology issues; Political issues; and social issues.

[Table 3-6] Factors that Facilitate or Impede UICs

Category	Contents
Capacity and Resource	Adequate resources (funding, manpower, and facilities)— Incentive structures for university researchers— Recruitment and training of technology transfer staff— Capacity constraints of SME
Legal issues	Inflexible university policies including intellectual property rights (IPR), patents, and licenses and contractual mechanisms — Treatment of confidential and proprietary information Moral responsibility versus legal restrictions (research on human resources)
Management & Organization Issue	Leadership/Top management commitment and support, Collaboration champion, Teamwork and flexibility to adapt, Communication, Mutual trust and commitment (and personal relationships), Corporate stability, Project management, Organizational culture (cultural differences between the world of academe and of industry), Organizational structure (university administrative structure and firm structure), Firm size (size of organization), Absorptive capacity, Skill and role of both university and industry boundary spanners, Human capital mobility/personnel exchange
Technology issues	Nature of the technology/knowledge to be transferred (tacit or explicit; generic or specialized; academic rigor or industrial relevance)
Political issues	Policy/Legislation/Regulation to guide/support/encourage UIC (support such as tax credits, information networks, and direct advisory assistance to industry
Social issues	Enhancement in reputation/prestige
Others	Low level of awareness of university research capabilities, Use of intermediary (third party), Risk of research, Cross-sector differences/similarities, Geographic proximity

Source: Ankrah, Samuel, and Omar AL-Tabbaa (2015).

3. Science and Innovation Parks and UICs

As part of the third academic revolution, science and innovation parks and university-industry collaboration within close proximity to universities present us with an important innovation landscape. In Europe, the 1985 publication of The Cambridge Phenomenon: The Growth of High-Technology Industry in a University Town by Segal Quince Wicksteed emphasized the importance of science and innovation parks in Europe. Much of this story deals with the success of science parks around universities in igniting innovation.

Founded by Trinity College in 1970, Cambridge Science Park is the oldest science park in the United Kingdom. It is a concentration of science- and technology-related businesses, having strong links with the nearby University of Cambridge. Later, this led to the construction of the well-known St. John's Innovation Center (SJIC) in a 21-acre area in 1987. Subsequently, the Cambridge Technopole Group was established to promote innovation and convert Cambridge's research results into commercialization. The SJIC houses about 85 companies employing over 400 people. The Cambridge Science Park employs 6,500 and houses about 104 companies. In 2018, the park alone was reported to have contributed 2.4 billion pounds to the UK's economy.

Similarly, there are two science parks connected with Oxford University: Oxford Science Park -- which is owned by Magdalen College -- and Begbroke Science Park, which is owned by the University. The University's science parks were estimated to have contributed £135 million and 2382 jobs in Oxford City, £155.4 million and 2,762 jobs in Oxfordshire, and £166.7 million and 3,043 jobs in the UK. Much of the success of these science parks is also attributed to the proximity of Oxbridge Universities. Given the success of science parks in Europe, Asia's leading economies began to give high priority to building science parks in proximity to universities.

We have the conspicuous examples of Tsukuba Science City, National University of Singapore and Biopolis, Daedeok Innopolis (formerly known as Daedeok Science Town), innovation hubs near Tsinghua and Peking Universities, Beijing, and science park of IIT Madras and Hsinchu Science Park of Taiwan (Krishna, 2019).

Three cases -- Hsinchu Science Park (Taiwan), Daedeok Innopolis (Korea), and RTP (USA) -- are discussed below.

3.1 Hsinchu Science Park: Taiwan

Let us look briefly at the case of Hsinchu Science Park, which emerged in the 1980s in close proximity to two universities, i.e., National Tsing Hua University (NTHU) and National Chiao Tung University (NCTU). The National Space Organization, Industry and Technology Research Institute (ITRI), and other public research institutions are located within the close radius of Hsinchu Science Park, which is now spread over 653 hectares of land area. The significance of the innovation complex involving Hsinchu Science Park (HSP) and two universities has made Taiwan the world's leading country in the field of semiconductor technology from design, fabrication, and packaging, to sales since the 1990s.

HSP houses some 570 firms dealing mainly with semiconductor-related technologies but has expanded to biosciences, ICT, and other new technologies to date. Between 1991 and 2013, for instance, the total number of firms related to semiconductor technology (semiconductors, PC & Peripheral, communications, optoelectronics, precision machinery, etc.) increased from around 120 ~ 480 firms. During the same period, sales of semiconductor-based products increased from NT $5000 (million) to NT $800,000 (million). The semiconductor sector has played a significant economic role for Taiwan and as the main knowledge hub since the mid-1990s. At the end of 2016, HSP employed 147,624 persons and recorded revenues of over NT $1 trillion in exports.

The Biomedical Park was established at HSP in 2012, attracting 40 firms and investment of NT $15 billion. By 2019, HSP has emerged as one of the world's leading knowledge hubs in high technology and new technologies of the 4th Industrial Revolution, such as Internet of Things and smart technologies. In the science park, applied research and development research are organized in collaboration with ITRI through strong focus on semiconductor devices, and the relevant basic research and skilled manpower are supported by the academic sector of two universities (NTHU and NCTU). The university, industry, and government linkages in the Hsinchu innovation region have been rapidly growing since the 2000s following the establishment of technology licensing offices and incubation centers in NTHU, NCTU, and ITRI. Hsinchu Science Park in the 21st Century has emerged as one of the model science parks in Asia. HSP had demonstrated several exemplary pathways to other countries in the region for developing knowledge hubs, leading to technological leadership in some niche technology sectors (Krishna, 2019).

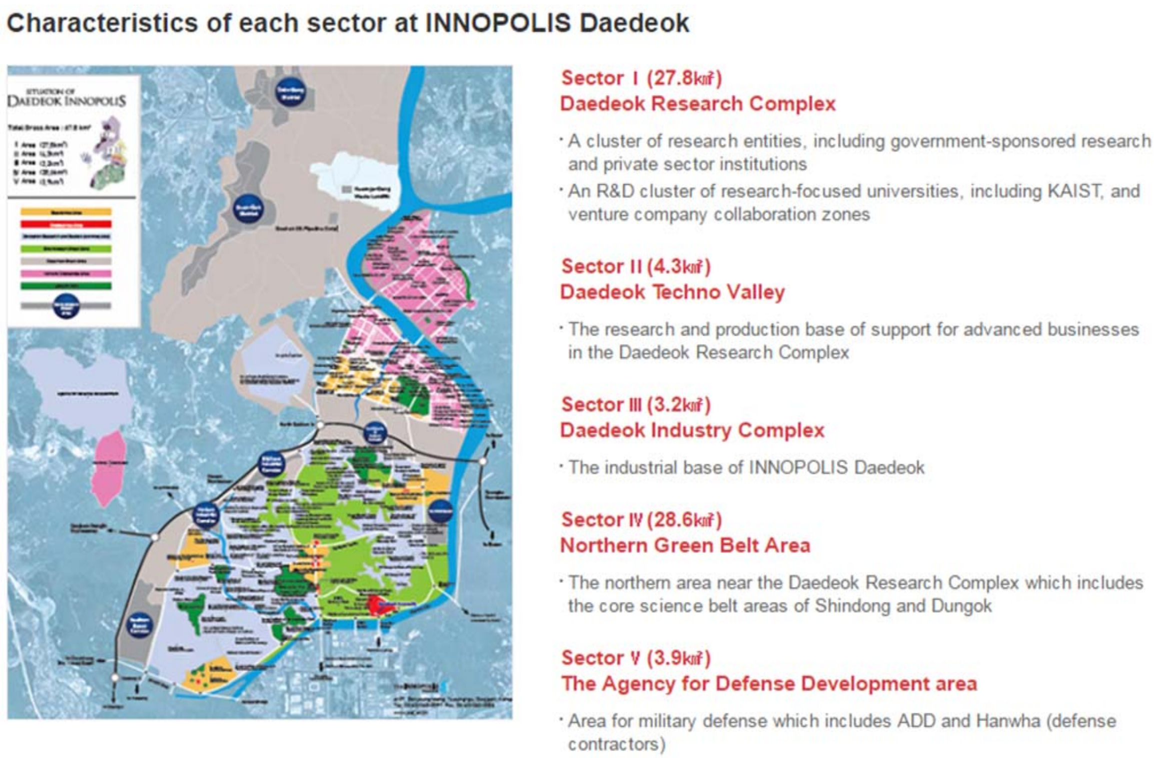
3.2 Daedeok Innopolis: Korea

Formerly known as Daedeok Science Town (DST), Daedeok Innopolis is the research and development district in Yuseong-gu District in Daejeon, which is 150 km away from Seoul. Daedeok Innopolis grew out of the research cluster established by President Park, Chunghee in 1973 with the opening of KAIST. Over 20 major research institutes and 40 corporate research centers make up this science and technology cluster. In the last few years, a number of IT venture companies have sprung up in this region, which boasts of a high concentration of doctors in applied sciences. There are 232 research and educational institutions that can be found in Daejeon, with many in the Daedeok region, among them ETRI (Electronics and Telecommunications Research Institute) and KARI (Korea Aerospace Research Institute). The Daedeok area has provided the core for the International Science and Business Belt. The Daedeok Innopolis logo was created by industrial design company INNO Design in Palo Alto, USA.

Daedeok Innopolis is located in 32 legal administrative districts – or dongs -- of Yuseong-gu and Daedeok-gu, Daejeon Metropolitan City, and the area spans about 70.4 ㎢ based on the legal administrative districts as regulated in the enforcement ordinance. Currently, there are 17 business incubation centers in research institutions and universities. Besides, the industry-academe-research exchange center and techno business centers are located as a park operation facility, and the Daedeok INNOPOLIS Foundation is also located at the center of Daedeok. The park is divided into a dedicated general residential area, a semi-residential area, a commercial zone, a green zone, an education/research zone, a business facility zone, and an industrial zone, and restrictions are set on construction, business activities, and development activities (Seo, 2013).

The following are the entities making up Daedeok Innopolis: 29 government-sponsored institutions (18 of which are GRIs); 14 government and public agencies; 7 public institutions; 5 education institutions; 17 venture incubation centers; and about 1,100 firms (about 580 venture firms). The number of employees at Daedeok Innopolis was about 45,526 in 2010. Daedeok Innopolis is an area of top research capacity in Korea focusing on government research institutes. There are 29 government-sponsored institutions in the area, including 18 GRIs under both KRCI (Korea Research Council for Industrial Science & Technology) and KRCF (Korea Research Council of Fundamental Science & Technology). The R&D expenses of government-sponsored institutions at Daedeok Innopolis accounted for 31.9 % of the total government R&D budget for 2017 and 10.4 % of gross expenditure in R&D (GERD). The R&D expenses of Daedeok Innopolis were about 7 billion US dollars in 2017.

[Figure 3-1] Map of Daedeok Innopolis, Korea



Source: Daedeok Innopolis (2019)

Supporting programs: Innopolis Foundation

Since 2005, the Innopolis Foundation has grown Daedeok Innopolis to become home to over 1,100 SMEs through its major programs as described below after the Daedeok research park was transformed into an innovative cluster with Zone 2, a newly developed industrial area. With these programs, the Innopolis Foundation directs substantial effort to transform research results at Daedeok Innopolis into companies and products generating revenue. The supporting programs consist of four main sub-programs such as venture incubation project, three-up program, special clustering project, and total design project. These supporting programs are used for activities that promote the research institute spin-off mechanism (Seo, 2013).

a) Venture Incubation Project

It has solved difficulties and cultivated the business incubation and star business through the systematic business support of small and middle venture enterprises in Daedeok Innopolis. Such technology business creation and management services provide customized consulting service and solutions as to the difficulties that the enterprises might encounter with regard to the commercialization of research performance.

b) Three-Up Program

As a lifetime business support program for entrepreneurs, the Innopolis Foundation operates the three-up program consisting of 3 sub-programs such as startup, high-up, and lift-up. Startup is to educate prep entrepreneurs starting hi-tech-based businesses to grow the enterprise and make a basic business plan. High-up supports early production through continuous business plan monitoring. Lift-up supports the implementation of the business expansion strategy with the active commercialization of excellent business items identified through special technology joint development projects.

c) Special Clustering Project

The special clustering project has been facilitated in connection with enterprises with big demands based on the technologies that public research institutes have, facilitating the early commercialization of performance of research and entry into the market. It has been strategically positioning Daedeok Innopolis as the center of new business creation through hub-spoke-type business promotion in connection with other strategic business areas.

d) Total Design Project

This project seeks to maximize the success of advanced technology commercialization and to facilitate the success of enterprises through the operation of a comprehensive support system from the planning of new business model, design development, and marketing support. Business model design was developed, merging the excellent technology of research institutes or enterprises in the Special District. Marketing support was given in connection with prominent enterprises at home and abroad as well as production of prototype including technical design for the developed model. The project has facilitated the commercialization and transfer of public technology by supporting the technology transfer and marketing for the technology jointly developed by private commercialization enterprises and public TLO (Technology Licensing Office) as an excellent technology identification project.

As a patent packaging-based technology marketing project, it has contributed to the activation of technology transfer and reduction of technology transaction cost to enable purchase of the patents required for demand technology in bulk. As a strategic business commercialization project, it has supported the solution required for business advancement or new business model utilizing public technology.

3.3 Research Institute Spin-off Companies

Spin-offs from either research institutes or universities are not a new idea for technology transfer. The survival rate of spin-offs — 3 years after starting their businesses — stands at 75% (Chong 2012). It is much higher than 46% as the normal startup survival rate after 3 years. Spin-offs are a means of technology transfer from a parent organization that represents a mechanism for creating jobs and new wealth (Steffensen, et al, 1999). Steffensen (1999) suggested that there are two types of spin-offs: (1) planned, when the new venture results from an organized effort by the parent organization; and (2) spontaneously occurring, when the company is established by an entrepreneur who identifies a market opportunity and who establishes the spin-off with little encouragement (Seo, 2013).

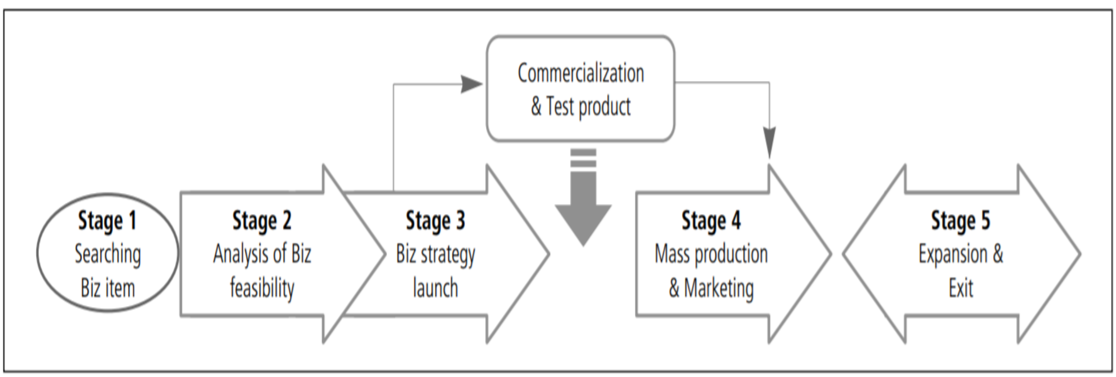
Performances for setting up spin-offs vary significantly across organizations, however. Cross-organization variation in startup activity can be explained by four reasons: understanding the source of knowledge spillovers; proximity to economic development and agglomeration economies; support from TLO; inventor and CEO of organization; and norms and policies of organization (Gregorio and Shane, 2003). Based on the Special Act on Fostering Daedeok Innopolis, Daedeok Innopolis is authorized to have a special policy to create spin-offs from government research institutes in Daedeok Innopolis. Such special policy allows spin-offs from GRIs to have tax benefits after meeting several conditions like ownership and company location. The “Research Institute Spin-off” was prescribed legally and defined as in the table below.

“Research institute spin-off” refers to a company established in Daedeok Innopolis with investment in kind of more than 20% by appropriate entities set forth in the law — on its own or jointly — for the purpose of starting a business with the technology developed and held by government research institutes (Article 13 of the Special Law of Daedeok Innopolis, Articles 9, 3 of its Enforcement Act). The movement of creating the policy of research institute spin-offs originated from the criticism that the achievement from R&D of government research institutes does not lead to production and business in an actual industry field, whereas R&D investment by administration has been increasing consistently since 1990. The underlying view indicates that the low rate of successful technology commercialization based on the public R&D results is attributable to the inefficiency of business mechanism in the R&D institutes, e.g., unified commercialization approach simply focusing on a license. In improving such environment, the need for appropriate support is being emphasized with regard to the strategy of optimizing the success rate of business by activating a wide range of technology commercialization methods, e.g., by technology capital investment, research joint company, professor or researcher's startup, etc. considering the characteristics of the developed technologies and qualification of business entity collectively.

In addition, it is expected to promote the creation of innovation network as both private companies as the user and government research institutes as a technology provider work together by combining excellent government R&D results, private capital, and management know-how. The entities entitled to establish the research institute spin-offs in accordance with the Law of Daedeok Innopolis are "government research institute, educational & industrial cooperation tech-holding company, and new-tech startup supporting company." Research institute spin-offs get two main benefits, the biggest and most important of which is the tax benefits as shown below (Seo, 2013).

National tax benefits include income tax and corporate tax exemptions (100% exemption for 3 years and 50% reduction for the next 2 years). Local tax benefits include acquisition and property tax exemptions (100% exemption for 7 years and 50% reduction for the next 3 years). The other is the full process business support program, which helps spin-offs prepare a business action plan and commercialize the technology from a research institute with supporting matching fund. This program supposedly provides lifetime support to spin-offs for the entire business process from stage 1 to stage 5 as shown below.

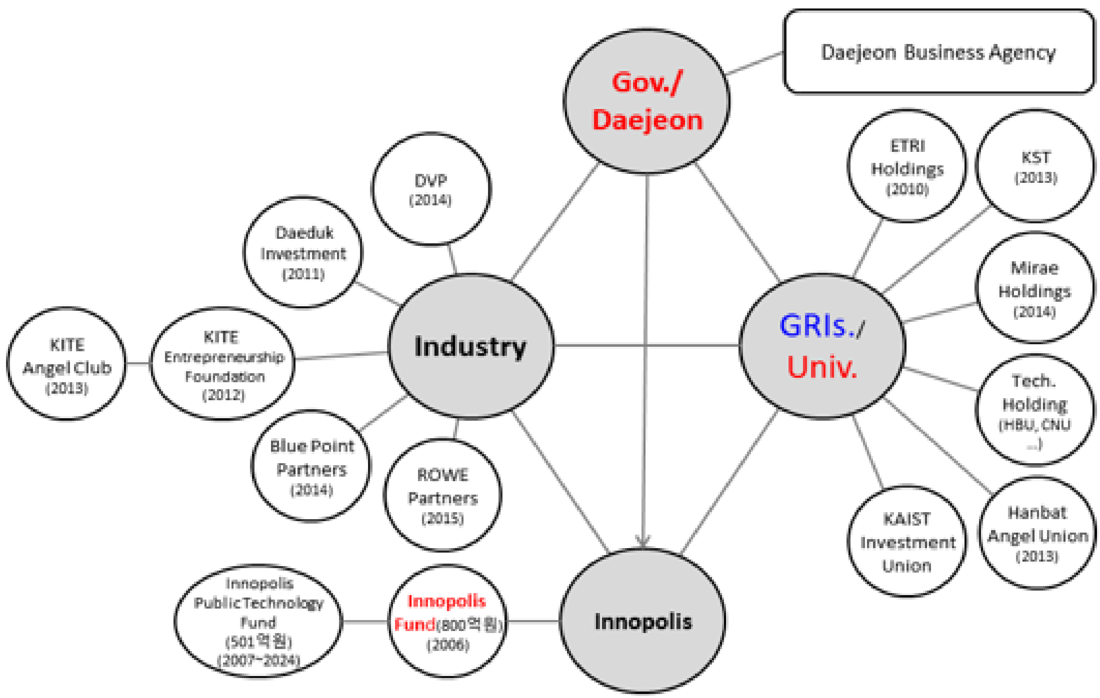
[Figure 3-2] Research Institute Spin-off System in Daedeok Innopolis, Korea



Source: Seo (2013).

The funding ecosystem in Daedeok Innopolis has evolved with the efforts of the government, industry, and university (Figure 3-3). The main performance of Daedeok Innopolis for 40 years lies in the diverse technology and economic development.

[Figure 3-3] Funding Ecosystem in Daedeok Innopolis, Korea



Source: Choi, Baik, and Jung (2017).

[Figure 3-4] Main Performance of Daedeok Innopolis, 1973-2013

Source: Daedeok Innopolis (2019).

4. RTP: Regional entrepreneurship ecosystem, USA

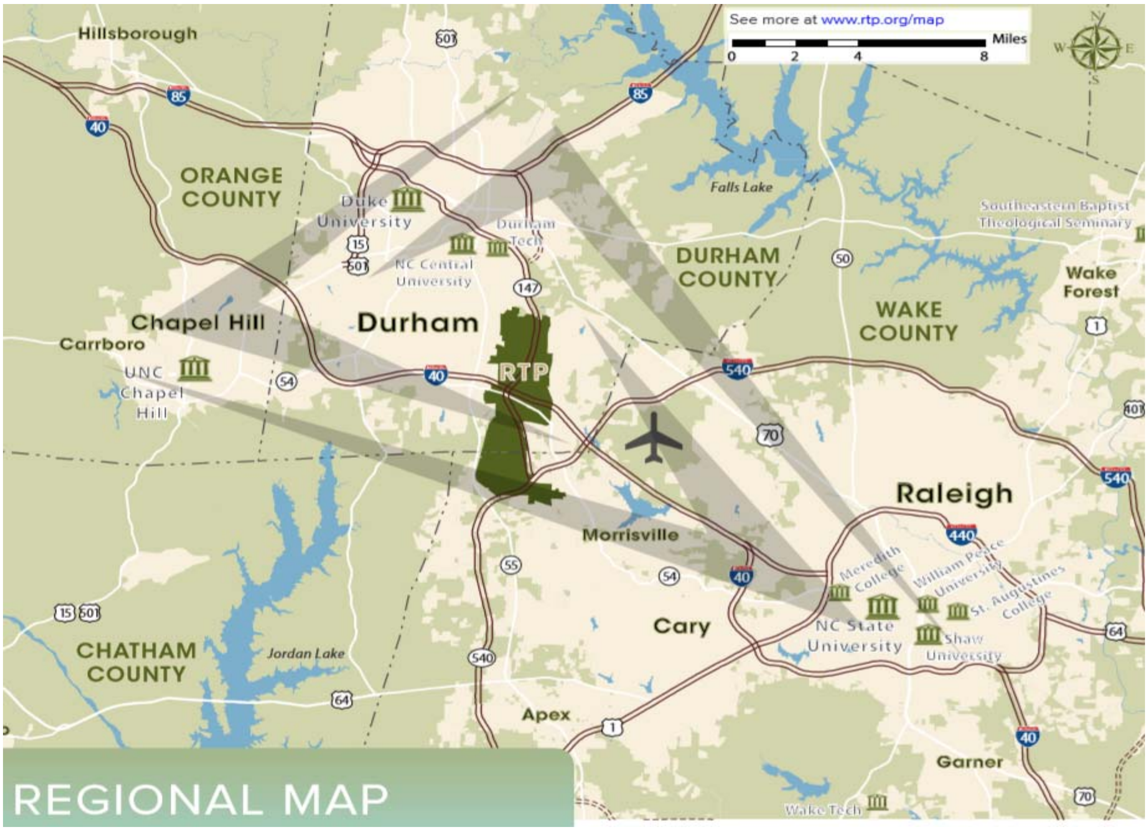
The Research Triangle Park (RTP) of North Carolina, USA is one of the few technology startup locations and is located in the top 3 universities: Duke, UNC-Chapel Hill, and NC State University. The presence of corporations in the RTP has often occurred in the form of development divisions since 1959, and they are often tasked with creating new value. Therefore, they have a strong need for entrepreneurial thinking. Uniquely positioned as a land grant university with a new campus that co-locates companies and academic units, NC State is one of the top ten universities supported by corporate grants. This combination of development-based innovation in both RTP companies and universities makes it a unique location for studying corporate entrepreneurship. Before making corporate entrepreneurship education recommendations, we will include examples of regional entrepreneurship ecosystems and university examples. These give concrete examples wherein companies can participate and provide guidance (Choi and Markham, 2019).

4.1 Research Triangle Park (RTP)

Governments seeking to establish economic engines and companies wishing to participate in corporate entrepreneurship should study the RTP. Unlike some other larger technology areas (Silicon Valley, Boston), the RTP is the result of careful planning from the very inception. The RTP did not just happen, and neither is it serendipity left over from military funding during the Cold War. Because of the deliberate, well-documented actions of the founders and subsequent managers, the RTP is uniquely replicable in other areas (McCorkle, 2012).

The RTP is one of the world's largest hi-tech research parks. The founders saw the potential for collaboration among universities, industry, and government in 1959. Today, RTP has more than 200 companies employing over 50,000 employees with expertise in microelectronics, telecommunications, biotechnology, nanotechnology, pharmaceutical, chemical, and environmental sciences. The park is home to the headquarters of the following organizations: IBM; GSK; Biogen Idec; Syngenta; United Therapeutics; Cisco; Bayer CropScience; Eisai; BASF; US EPA; NIH's National Institute of Environmental Health Sciences (NIEHS; the only NIH institute/center outside of the DC metro area); and RTI International, the first research institute to help launch the park (Kroll, 2014).

[Figure 3-5] Research Triangle Park Map, USA



Source: www.rtp.org.

The RTP can be considered a Moneyball area; its success ignores traditional wisdom. The RTP region has attracted more corporate investments than traditional venture capital. Corporate investment seeks relevant strategic technologies and promising opportunities. Entrepreneurial outcomes are more likely to be acquired by a large company than initial public offerings (IPO). In the RTP region, corporate R&D, mergers, and acquisitions have led to greater employment growth than venture capital investment in the region. The acquiring company has recognized that the assets acquired were more valuable in its place than in dismantling operations. Acquisitions often make additional investments in the region (Feldman, 2014). In 2014, the CED Reported 59 Mergers & Acquisitions (M&As) and 6 IPOs. Large corporate investments and entrepreneurial startups in the RTP share similar end points. This natural link between startup and acquisition is one aspect that helps integrate corporate and traditional entrepreneurship.

[Figure 3-6] Main Resident Companies in Research Triangle Park

Source: www.rtp.org

4.2 Research Triangle Foundation

The Research Triangle Park Foundation manages RTP and operates three buildings: the Frontier, the Lab, and the Archie K. Davis Conference Center. The Frontier is an innovative and collaborative space that was first opened in January 2015. The Lab is a full-service lab and office space with several research and development companies. The Archie K. Davis Conference (AKD) Center is a 6800-square foot event space and is also home to the Research Triangle Park Foundation headquarters. There are numerous programs for large and small companies sponsored by the RTF (Research Triangle Foundation); for example, "RTP 180°" invites speakers from three founding universities and local companies as well as the community at large. The TEDxRaleigh series draws between 250 and 300 people on the Foundation's third Thursday evening.

Frontiers

The RTF also runs a centrally located branch of RTP to plug in and collaborate and plan the next big move. The Frontier is the main building of the Park Center and is an important step for redevelopment efforts. It includes free collaborative work and drop in meeting space, mixed-use space including a third-floor tenant office, and event space. As part of the free offerings, Wi-Fi, counter-culture coffee, community garden, and programs are all free of charge and open to everyone. The success of the Frontier depends on the community to create a vibrant, rich space with innovative ideas. Weekly features and programming, such as Happy Hour, RTP/Fidelity Food Truck Rodeos, 1 Million Cups, and RTP 180 lighting talks are held in this place.

4.3 Council for Entrepreneurial Development (CED)

A mainstay of entrepreneurship in the RTP is CED. Established in 1984, CED is the largest corporate support organization in the United States with more than 5000 members representing more than 1100 companies. Located in AU (American Underground) since 2010, CED's Board of Directors (64 people) are committed to the Triangle's entrepreneurial future. CED is based on the entrepreneurially inspired individual and business community and is a network of national networks that helps Triangle entrepreneurs build and grow successful businesses. CED provides education (or training), mentoring, and capital formation resources to technology-based, high-growth entrepreneurs. Activities include the CED Life Science Conference (March), CED Tech Venture Conference (September), CED Venture Mentoring Service, entrepreneur-only workshops, and investor introductions.

4.4 Co-working Space

Old buildings in downtown Raleigh and Durham City of RTP have been changed into a co-working space and a space for entrepreneurship education, such as American Underground and HQ Raleigh.

American Underground. In 2010, the basement of the American Tobacco Campus — hence the name American Underground — bloomed into two spaces in Durham and one in Raleigh, one of which has 240 founders with 70 employees in a solo entrepreneurial business.

HQ Raleigh. It opened in 2012 in the downtown warehouse known as Raleigh. In 2015, it added an additional 5000 square feet of space across the street and acquired an adjacent site, where it plans to build a new 43,000-square foot building or more than triple its current space. Currently, 128 startups are housed in the HQ Raleigh space. American Underground and HQ Raleigh are clearly the bold newcomers that have made a large impact on the entrepreneurial scene, fostering dense clusters of startups that are unprecedented in the region.

The RTP, together with the many programs and organizations to support existing and entrepreneurial companies, and many technical specialties form the ecosystem that companies of all sizes can leverage for growth.

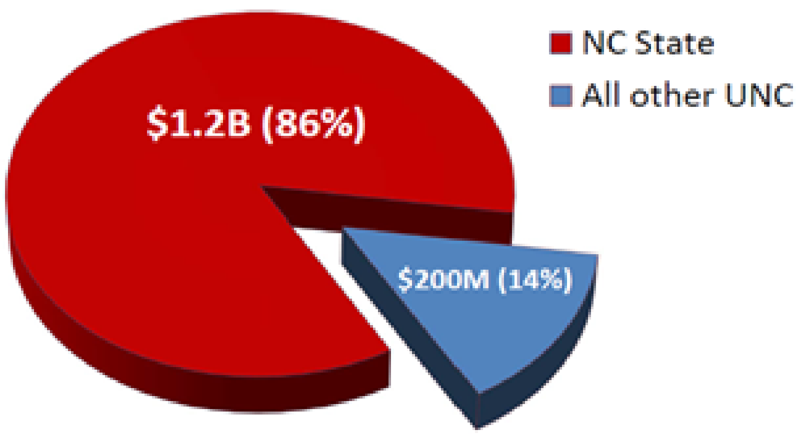
4.5 North Carolina State University

The RTP is a major corporate, university, and government research and technology center. North Carolina State University (NC State) has a number of characteristics that make it an attractive model for companies wishing to partner with universities for corporate entrepreneurship and entrepreneurial thinking. NC State is the largest STEM (Science, Technology, Engineering, and Mathematics) institution in North Carolina and one of the largest in Southeastern US. The STEM orientation is derived from being a land grant institution originally chartered in 1881 to focus on agriculture and mechanics, both technical and designed to interface with the industry.

According to the Association of University Technology Managers, NC State is #1 in licensing to companies — besting MIT, Berkeley, Stanford, and all other notable universities — and #6 in attracting corporate funding. This is not just due to an effective technology transfer office but also reflects the long-standing relationships developed for decades between NC State and the business community. Faculty, staff, students, and administrators are all deeply immersed in relationships with large companies.

From an economic perspective, NC State has a disproportionately large impact on the state through both entrepreneurial and corporate activities. [Figure 3-7] shows the relative contribution of NC State compared with all the other state universities in North Carolina (UNC).

[Figure 3-7] UNC System's Statewide Business Startup Impact 2012–2013



Source: Economic Modeling Specialists International (2015); Choi and Markham (2019)

Business startup impact

NC State creates an exceptional environment for innovation and entrepreneurship as proven by the number of startup companies associated with NC State in the region. In FY 2012–2013, a university-based startup company generated $1.1 billion in additional regional income for the Economic Development Partnership Region economy, the equivalent of creating 4,984 jobs.

Established in 2008, the entrepreneurship center at North Carolina State University currently offers 42 entrepreneurship-related undergraduate courses. Over the last five years, its graduates have started 104 companies and have collectively raised over $365 million in funding. At least 77 per cent of its undergraduate entrepreneurship faculty has started, bought, or run a successful business.

[Table 3-7] Rankings of Entrepreneurial Universities: Undergraduate Level

Rank	University	No. of entrepreneur-related courses	Students enrolled in entrepreneurship classes	Per cent of faculty with entrepreneurial experience	Startups launched by grads in the last 5 years	Funding raised by grads in the last 5 years ($M)	Center name	Year established
1	University of Michigan	68	4,554	63%	807	20	Ann Arbor	1999
2	University of Houston (TX)	34	2,268	100%	476	57	Cyvia and Melvyn Wolff Center for Entrepreneurship	1991
3	Babson College (MA)	38	2,342	100%	334	40	Arthur M. Blank Center for Entrepreneurship	1978
4	Brigham Young University (UT)	49	5,823	96%	361	733	Rollins Center for Entrepreneurship & Technology	1989
5	Northeastern University (MA)	37	1,537	76%	657	100	Northeastern University Center for Entrepreneurship Education	2012
6	Baylor University (TX)	31	1,800	95%	1,095	678	The Entrepreneurship Center at Baylor University	1977
7	Washington University in St. Louis (MO)	28	584	62%	84	329	Skandalaris Center for Interdisciplinary Innovation & Entrepreneurship	2003
8	University of Maryland	80	9,754	41%	226	49	Academy for Innovation and Entrepreneurship	1986

Rank	University	No. of entrepreneur-related courses	Students enrolled in entrepreneurship classes	Per cent of faculty with entrepreneurial experience	Startups launched by grads in the last 5 years	Funding raised by grads in the last 5 years ($M)	Center name	Year established
9	University of Utah	36	882	88%	266	33	Lassonde Entrepreneur Institute	2002
10	Tecnológico de Monterrey (Mexico)	80	20,000	84%	1,044	29	Institute for Entrepreneurship Eugenio Garza Lagüera	2001
11	North Carolina State University	42	3,433	77%	104	365	NC State Entrepreneurship	2008

Source: https://www.entrepreneur.com/slideshow/322898#11.

Impact of Spin-off Companies

Spin-off companies include businesses created by faculty, students, or graduates at NC State as well as startup business development through NC State's programs. In FY 2012–2013, the combined impact of spin-offs associated with NC shares was $596.4 million in additional regional income, which is equivalent to 10,166 new jobs. This impact is important because it constitutes a significant part of the regional business environment.

Centennial Campus

This state-of-the-art campus provides proximity like no other research campus in North America. Corporate tenants and university departments coexist in the same campus and even in the same buildings. This university, industry, and government partnership provides a powerful environment for entrepreneurial education to include the needs of corporations. The first land of 385 acres (1.6 km^2) was handed over to North Carolina State University in 1984 by North Carolina Governor James B. Hunt, Jr. Another 455 acres (1.8 km^2) was submitted to NC State in 1985 by Governor James G. Martin. Claude McKinney, dean of the School of Design, developed the Master Plan, and the first building was completed and occupied in 1989. In 1991, ABB first moved in. That same year, the College of Textiles moved to the Centennial Campus. In 2000, the Centennial

Biomedical Campus was established. The College of Engineering has started moving to the Centennial Campus since 2002. The Centennial Campus added a fourth dimension to the trinity of teaching, research, extension, and partnerships with the private sector in pursuit of public good.

Approximately 70 companies, governments, and non-profit organizations are located at the Centennial Campus with academic departments. To lease space for property, prospective "partners" must be programmatically linked to NC states, such as through conducting joint research with faculty or use of students for internships or part-time jobs. About one-third of our partners are venture companies or early-stage companies and are located at Centennial's Technology Incubator. Another 20% are research and development departments of large enterprises, and the rest are small businesses, state and federal agencies, and nonprofit organizations. Current partners include ABB, GlaxoSmithKline, the US Department of Agriculture, and the Regional Office of the National Weather Service.

STEM-Focused

NC State has the fourth largest engineering undergraduate program in the US, one of the oldest computer science departments in the US, and the fourth largest in terms of graduate degrees awarded. It is the only NSF (National Science Foundation)-funded smart grid research center in the US, and one of the world's largest bio-manufacturing training and educational facilities. Moreover, it has the third highest-ranked veterinary college in the US, the largest textiles college program in the US, the only next-generation wireless test bed system in the US, and the only research park in North America with a world-class research library (complete with automated book retrieval system and about 100 technology-equipped, group study spaces).

Intellectual Property Management

Of the 164 individual institutions responding to the 2014 Licensing Activity Survey conducted by the Association of University Technology Managers (AUTM), NC State ranked 6th in terms of number of Licenses and Options Executed. In fact, OTT (Office of Technology Transfer) of NC State consistently ranks among the top 10 technology transfer offices at universities without medical schools in terms of invention disclosures (6th), US patents issued (7th), startups formed (5th), and licenses and options executed (1st). The university has engaged multiple corporations in unique intellectual property management agreements, and it can accommodate a wide variety of corporate needs.

Sponsored Research

NC Sate provides a one-stop shop environment for industry and government partners and ranks 6th in industry-supported research among peers. The university provides a positive environment for corporations to engage with faculty and university programs at all levels. The combination of sharing space and sponsored research provides benefits to faculty, students, and corporations engaged in joint projects.

Chancellor's Faculty Excellence Program

This program is committed to the interdisciplinary endeavor of NC State University and to addressing the most important issues in the world. Following a strong strategic plan and an aggressive vision, the cluster recruiting program has added 77 new faculty members in 20 disciplines to add to the breadth and depth of NC State's already strong efforts. For example, faculty clusters are Bioinformatics, Carbon Electronics, Data-driven Science, Emerging Plant Disease and Global Food Security, Environmental Health Science, Forensic Science, Geospatial Analytics, and Synthetic and Systems Biology.

Center for Innovation Management Studies (CIMS)

This center is involved in the creation, synthesis, and dissemination of industry-related information on innovation management and development of current and future generations of innovation management researchers and industry workers. In addition, the center connects industry needs and university resources, and it has been sustainable since 1984.

4.6 Entrepreneurial Education Programs

The university uses both coordinated and decentralized programs of entrepreneurship as follows:

Entrepreneurship Initiative (EI)

The NC State Entrepreneurship Initiative was established to promote student innovation and entrepreneurship. At NC State, EI strives to create unique learning opportunities for students to grow into leaders and job-generators of the next generation. As a program, EI takes an interdisciplinary approach to learning. Faculty and staff come from a variety of backgrounds and industries and actively recruit students from across the campus. The desire is to build a community within and outside the university where everyone is thriving with creative and immersive entrepreneurship. Several levels of involvement are offered, depending on individual interests and entrepreneurial goals (Miller, et al, 2011).

Technology Entrepreneurship and Commercialization (TEC)

The Technology Entrepreneurship and Commercialization program (TEC) was initially developed at the College of Management from 1994 to 1999. Development was supported by the National Science Foundation and Kenan Institute for Science and Engineering as well as a number of corporate sponsors. The TEC program engaged MBA and technical graduate students to create viable business cases for new technologies (Barr et al., 2009 and Markham and Mugge, 2015).

E-Clinic

The Entrepreneurship Clinic is held in collaboration with the Poole College of Management at NC State. This clinic is designed to help business college students build a hands-on experience by implementing projects submitted by startup companies within the HQ. NC State E-Clinic is a place where students, faculty, entrepreneurs, and service providers are teaching, learning, and building the next generation of business at Raleigh, inspired by a model of university teaching hospitals. "For years, faculty, students, and entrepreneurs have been working together to envision a physical space where they can learn from each other and, ultimately, to develop commerce and create special value for the North Carolina economy," said Lewis Sheats, NC State E-Clinic Executive Director: "HQ Raleigh's collaborative workspace is an ideal place for providing these resources to Triangle's entrepreneurial community."

Wolfpack Investor Network

With an estimated 30,000 accredited investors in the alumni base, the Wolfpack Investor Network (WIN) provides a new mechanism for engaging these individuals with NC State's entrepreneurial ecosystem. Investors, many of whom are corporations, support technologies that come to the market after technical and business proofs of concept are demonstrated. These technologies can then become available to corporations. Students and faculty are engaged in preparing commercialization plans that meet targeted industrial needs.

RTP companies and universities have grown by having symbiotic relationships with each other. Relationships and funding mechanisms provide mutual support. The RTP is a specialized example of how universities, government, and industry partners work together to create value through technical innovation. A large part of that is conducted through corporate entrepreneurship.

5. Experiences of Technology Commercialization to Catch up with a Developed Country

Dr. Jong Heung Park, Vice President,
Electronics and Telecommunications Research Institute

5.1 Introduction

There are a couple of ways by which a country or a company can acquire technology: direct research from basic to advanced technology, or directly from a developed country or a technology-holding company. At the national level, technology-creating companies are usually industrial developed countries, and beneficiary companies are underdeveloped countries, i.e., developing countries or LDCs.

While there may be various ways to catch up with developed countries, such as the relationship between countries, content of technology, ability to absorb technology, and structure of the industry, it is efficient for technology-driven countries to receive technology transfers through international organizations such as advanced countries under the support of the United Nations and to conduct research and development using the manpower and resources they possess at their bases.

Over the past 60 years, Korea's development has become a model for these latecomer countries. The introduction of technology from developed countries in Korea, establishment of technology embodied naturally as multinational corporations entered Korea, R&D performed as a fast follower, activities of government-funded research institutes, and role of regional clusters will be helpful.

As part of the Uganda K-Innovation Project (2018-2019), this report aims to discuss the process and method of technological commercialization that should be known for Uganda's economic development and share experiences in our country's industry-academe cooperation system, methods of technology transfer between countries, R&D

methodologies, startups, and technology commercialization policies to foster new industries and businesses. In addition, local experts in related fields will investigate and summarize the current state of Uganda's technology and the role of universities and startups.

In this report, as part of the Uganda K-Innovation Project (2018-2019), we discuss the process and methods of technological commercialization that we need to know for economic development in Uganda including Korea's industry-university cooperation system, methods of technology transfer between countries, and R&D. We will share our experience in methodology, entrepreneurship, and technology commercialization policies to foster new industries and companies.

The transfer of technology between countries is achieved through representative channels of international trade, technology licensing, foreign direct investment (FDI) and transnational corporation (TNC) activities, and people's movements (UNCTAD, 2014). After cross-border technology transfer, beneficiary countries will improve their technology through R&D. As a means of improving the technology of companies or nations, R&D is used to develop new products or improve competitiveness and invest resources (human and other resources) to develop new technologies in the process of copying or reverse engineering, even in the case of technology transfer. Therefore, it needs to go through several months to years of research and development. Commercialization-related technology development (R&BD) can be regarded as a combination of existing research and development (R&D) and business and is a case of developing a new type of product or technology in connection with the business strategy of a country or a company.

Based on the discussions regarding international technology transfer, capacity building including R&D, and technology commercialization in this report, the suggestions for Uganda's technology commercialization will also be discussed as a result.

6. Catch-up Strategy

6.1 Catch-up Overview

Uganda has some difficulties in commercializing new technologies except for some agriculture and livestock industries, research for most of which is basic, few applications. In addition, Uganda lacks its own production capacity, so it is difficult to transfer technology effectively to Uganda since there are few companies to absorb technology from advanced countries. Economic development is generally associated with increased flows of knowledge and technology as well as cross-border capital and product flows. Typically, technology flows from developed countries or companies with regional-based innovation systems to developing countries, LDC, regions, or companies as part of the development process. Narrowing the technology gap with developed countries and joining the ranks of the developed countries make up the catch-up process.

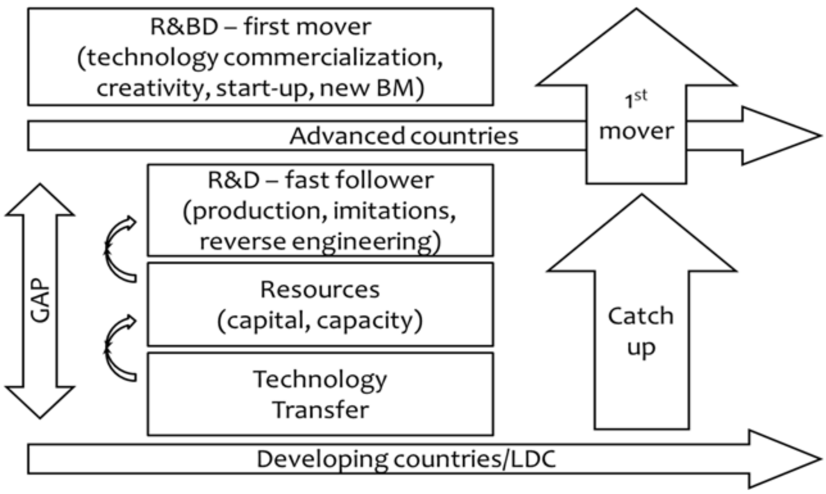
It is necessary to understand how underdeveloped countries use technology commercialization as a means of catching up with advanced countries as part of Uganda's capacity building and, to do so, to explain to policymakers such as Uganda's science and technology sector "what it means to catch up" including Korea's cases of technology transfer, technology business, and research and development. In this report, the process of catching up with these advanced countries will be introduced with several steps: “technology transfer,” R&D management, and technology commercialization.

6.2 Catch-up Process

A country or a company secures technology by creating it directly from scratch and by transferring it from an advanced country or from a technology-holding company. At the national level, the technology-generating entity is usually an industrial advanced country, and the acquiring entity is a developing country or one of the poorest.

There may be various methods depending on the relationship between countries, content of technology, ability to absorb technology, and structure of the industry, but Korea's example can be explained as shown in [Figure 3-8]. There is a way of creating technology directly from the beginning using the resources (labor, capital, etc.) possessed by the state, but this method is time-consuming and costly to catch up with developed countries. Therefore, R&D is usually carried out using the resources possessed by the advanced entity after receiving the technology transfer. Initially, R&D can be developed by entering the market with cheap labor through simple imitation or by differentiating the product by developing a function reverse design or an additional function. It has similar experiences in the past auto industry and semiconductor industries in Korea.

[Figure 3-8] Catch-up Process



Source: Park (2019)

By accumulating technology through the performance of R&D as a fast follower based on the transferred (received) technology and incorporating this method into many industries, the national economy will ultimately improve gradually. Being a prosperous country for years or decades can be almost equivalent to a developed country. From then on, technology must be developed in a new way, in the form of R&BD that develops business models instead of mere R&D. This development approach, called the first mover type, requires a more complex process of adding creativity to the newly developed technology and commercializing the technology, wherein startups play a lot of roles (R&D and R&BD will be discussed in detail in the next section). That is why the government supports a large number of startups or startup incubation programs worldwide. Any country/firms that can play their role as first movers will join the ranks of advanced countries/firms from a technical point of view.

7. Technology Transfer

7.1 R&D Eco-cycle and STI in Korea

Korea has 25 GRIs (government-funded research institutes) dealing with science and technology, each developing a science technology with its own station, whereas those developed in GRI are being transferred to various industrial sites and companies (NST Homepage https://www.nst.re.kr/nst_en/).

In Korea, GRIs in the field of science and technology have made tremendous contributions to the rapid economic growth and level of Science & Technology since the mid-1960s.

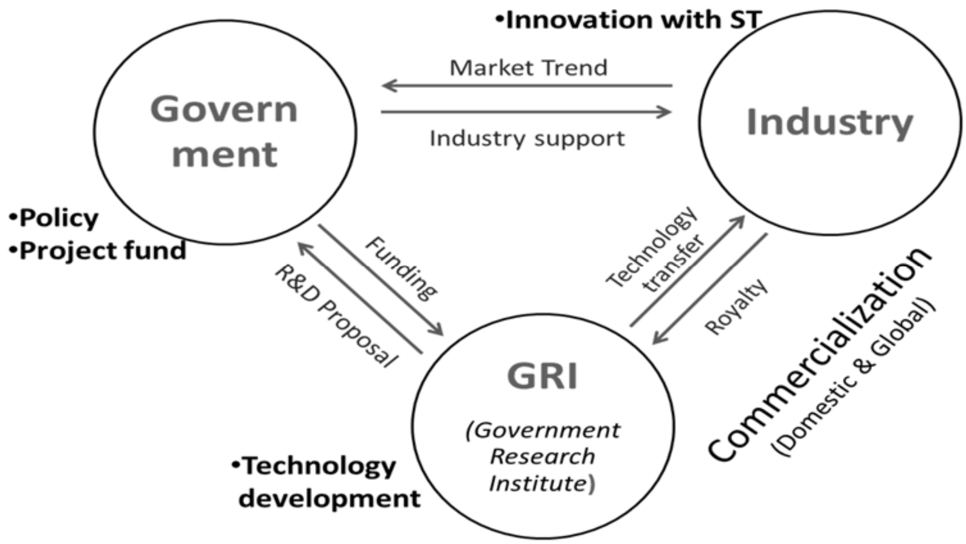
For continued growth and structural change, the government should push for institutional changes to promote innovation while offering incentives through appropriate intervention. In addition, the development of human capital across the country is important in enabling technology development and absorbing the transferred technology, becoming the main character in carrying out R&D activities.

State-funded agencies such as R&D institutions, universities, and Tech Transfer Offices (TLO) can make great contributions to utilizing and accumulating the transferred technology and practical experience.

Technology transfer in Korea is usually a case of universities or technology holding institutions -- GRIs -- acting as technology suppliers with ordinary companies (such as big group companies, small and medium-sized enterprises, venture companies, startups, etc.) as the beneficiaries and of technology transfer between companies not happening very well due to competition.

To keep pace with industries, Korea has unique eco-cycle and STI policy rarely found all over the world as shown in [Figure 3-9]. Governments, industry, and research institutes have their respective roles as the industry leaders. In particular, the government has the authority to select the technologies needed by the industry and funds the GRIs to develop them. GRIs develop new technologies and transfer them to the industry, which then develops competitive products and enters the market. The industry makes revenue through this, and some of the profits are provided to GRIs in the form of royalty and some serve as incentives for developers, who in turn stimulate new research and development activities; others involve developing more advanced technology development.

[Figure 3-9] R&D Eco-cycle and STI in Korea



Source: Park (2019).

7.2 International Technology Transfer

According to UNCTAD (2014), technology transfers between countries are made through representative channels: foreign direct investment (FDI) and multinational corporate (TNC) activities, international trade, licensing of technology, and people's migration. Knowledge and resources or imitation (e.g., reverse engineering) may work among technology transfer channels, and there may be different mixed channels; in either case, however, it may change according to the technology level and conditions.

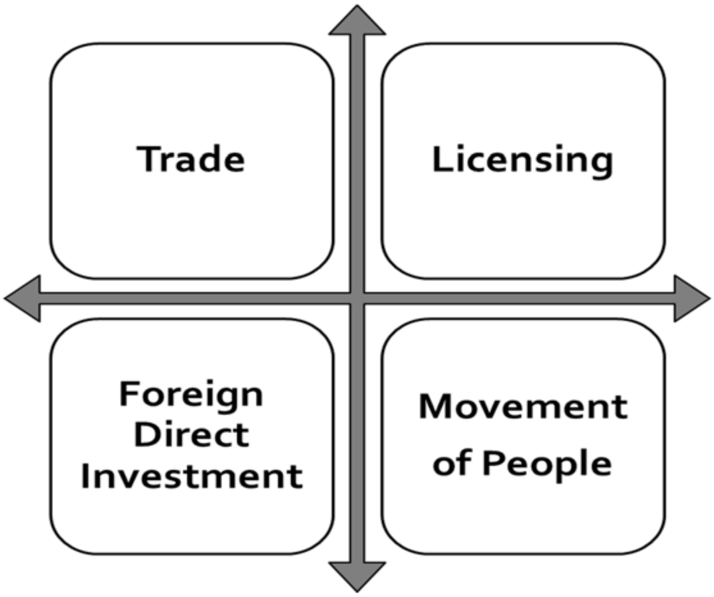
Because of such complexity of technology transfer between countries, successful technology transfer requires significant investment or staffing in order to utilize the technology, and it is also why the resources (capital, manpower capacity, etc.) in the technology transferred are required. Complete understanding of technology and new R&D can help companies and industries become more competitive in the market including products with strong performance.

Four ways of international technology transfer by UNCTAD (2014) are introduced with some cases successfully deployed by Korea.

Trade

International trade acts as a channel for the distribution/transaction of resources — which affects the development of national industries — and for the transfer of technology. When goods or products are traded, the technology with hi-tech elements implemented in them is also transferred. In particular, Korea has been successful in the process of technology transfer by specifying the terms and conditions for technology transfer in the process of introducing hi-tech products in the past and by receiving education and training, provision of technical documents, and know-how of product development from advanced countries. ETRI can see many examples of this in the course of past TDX development and communication satellite development.

[Figure 3-10] 4 Ways of International Technology Transfer



Source: UNCTAD (2014).

Licensing

Licensing is an important technology transfer channel between countries or between firms. Note, however, that this is likely to fail when the counterparty has significant differences in access to information and knowledge or in its ability to accept technology. The primary concern of the licensee is royalty income, and it expects incentives to cover the potential risks, such as product leaks to unlicensed markets, reverse design of proprietary technologies and so on. Technology transfer through licensing is widely used in developed countries and in some emerging ones as a means of technological advancement; in order for this method to work properly, it must have the innovation systems of the recipient countries and the appropriate absorption capabilities of individual companies.

In case of licensing, however, technology transfer is not a good tool if the absorption capability of the recipient country is not sufficient. Since licensing does not convey detailed technical information or know-how, using it to make products takes more and other efforts even if it is licensed.

FDI (Foreign Direct Investment)

FDI, along with the local production of capital, employment, and international value chains, is a medium of technical transfer and a channel for conveying many capabilities, such as management skill, knowledge of marketing, or standards and regulations in global markets. Technology transfer due to FDI takes place through knowledge acquisition via product utilization, technology acquisition by labor provision, and connectivity between companies through value chain. Companies in a developing country or LDC will acquire knowledge or skills from developed companies and improve the capacity by imitation or reverse engineering. Likewise, the transfer of engineers from multinational companies is a form of technology transfer that occurs through the process of being employed by local companies or establishing their own companies. This refers to the process of improving products and services, including adopting improved technologies and improved processes, as local companies meet the necessary technical specifications and quality standards when engaging in multinational companies' value chains.

Technology licensing and corporate acquisition are critical tools for companies in developing countries to get closer to the technology landscape. There have been many acquisition cases for foreign companies and licensing agreements with TNCs in developed countries wherein the acquisition company has no production site (UNCTAD, 2004).

Practical technology transfer is the most efficient means; in Korea, foreign capital and technology have been successfully introduced into FDI for the past 60 years.

As a global group of companies initially dealing with integrated circuits, Samsung has accumulated technical skills through licenses of multinational companies and functions obtained from GRI and internal development (Kim, 2003).

Technology transfer through the movement of people

Human capital is a factor determining the ability to absorb the benefit from transferred technologies as well as a key element in the work of the technology transfer channels of trade, licensing, and FDI as mentioned in the previous sector. The movement of people, scientists, engineers, or highly educated professionals increases science and technology development capacity. Absorptive capacity for technical support through an effective national innovation system is important, and it is certain that experts from developing countries who work abroad will further develop their knowledge and skills, and networks will help improve national technical skills. Therefore, developed countries should regard these movements as soft technologies related to the export of knowledge and processes of production, management, entrepreneurship, marketing, etc. – the mechanisms for transferring technology to developing countries (UNCTAD, 2014).

Sometimes, the permanent overseas migration of educated professionals may result in losses in human capital inventory and capacity. Still, it will also help increase human capacity such as consulting in the end.

Absorptive Capacity for Technology Transfer

Regarding technology development, technology transfer from a developed country is particularly important in terms of contributing to the technology capabilities of the country and firms. Technical competence refers to the ability of industry (big group company, SMEs, startups) or GRIs to identify and use technology and to create new technologies. According to STEPI (2018), absorption capacity means how the internalization of the newly transferred technology can be achieved by an organization; in the effective learning of technology by a country (or individual), the absorption capacity consists of two elements: the basis of holding knowledge and the strength of effort.

Strong potential knowledge increases the ability to understand and use new knowledge, and strength of effort is the amount of energy that scientists and engineers put into solving the problem; these efforts enhance the interaction among the engineers, thereby facilitating technical learning at the developed country's level or higher. Thus, the absorptivity (intensity of effort) can be seen as a more important factor than the heritage of company. Such absorptive capacity and success story of the Korean industry are documented well in the STEPI report.

7.3 Intellectual property rights

An essential consideration in the technology transfer process is IPR (intellectual property rights), the key element of trading and technology transactions in the global world. Intellectual property right includes copyrights, patents, trademarks, and trade secrets. IPR has economic characteristics, terms and conditions of legal protection, and impact on technology transfer depending on the beneficiary countries. In terms of technology transfer, IPR is used as a means of technology protection for developed countries; it can also be an obstacle to technology transfer in recipient countries if the royalty burden is high. Although technology-providing (usually developed) countries protect technologies using IPRs and seek profits in the process of technology transfer between countries, developing countries and LDCs may use weak IPRs claims and expired IPRs to develop products or come up with creative imitation and establish their technological and industrial capacities. This was how Korea had performed R&D including the strengthening of the IPR regime from the 1980s.

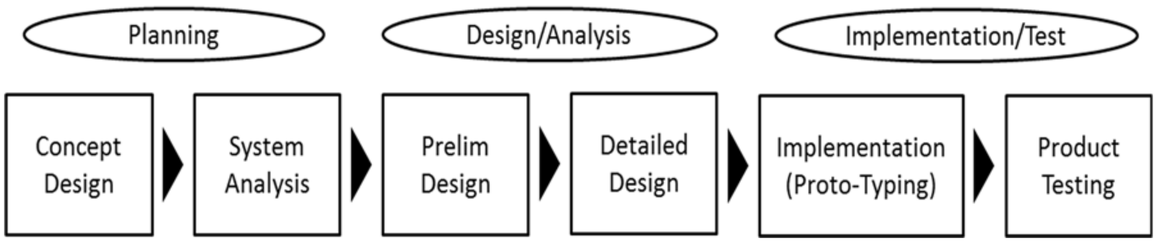
8. Research and Development to Catch-up

8.1 R&D Life Cycle and Management

R&D is used to develop new products or improve competitiveness as a means of improving the technological capabilities of the companies or the country. Even when technology transfers are received between countries, resources (people and resources) must be invested to develop new technologies in the process of imitating or retro-designing them for months or years. R&BD (Related Technology Development) can be viewed as a combination of existing R&D and business and is the case when a new type of product or technology is developed in conjunction with a national or a corporate business strategy.

Normally, R&D is a technology push type, wherein R&BD goes through a rather complex process of exploring potential markets, verifying ideas, switching technologies on a small scale, producing on a small scale, verifying the market's response, and fully commercializing the product by adding additional features or enhancing the competitiveness of the product by adding additional features. This is because the market is a means of reducing the risk of resource-intensive R&D in an untested business world and of combining technology development with commercialization through main feedback, even if it takes a little longer. In the process of catching up with advanced countries, R&D is the path of technology development of a fast follower, such as simple production, imitation, and reverse design by investing resources based on the technology transfer received from developed countries. It is advantageous to carry out the R&BD strategy as first mover in order to secure the same semi-height or better technological capability as developed countries and occupy the market.

[Figure 3-11] Typical R&D Life Cycle



Source: Park (2019).

A typical R&D cycle can be divided into 3 ~ 6 stages depending on the technology being developed; still, the more difficult the technology is, the more steps required, and the lower the risk in the interim process. The six-step R&D sequence is a relatively modular project management system that allows the development process to be broken down and operated flexibly and is typically composed of the following (in some cases, an assessment may be made with seven steps, if necessary):

- Concept Design
- System Analysis
- Preliminary Design
- Detailed Design
- Implementation (prototyping)
- Product Testing
- Optional) Post-project evaluation

- Concept Design: At this stage, the primary objectives of the project are set. We set goals by using the words "explore," "problem," and "optimal" to describe the R&D project objectives. There are some differences in project management between non-R&D projects and R&D project management. It may be good to set flexible goals in non-research and development projects that require setting goals early in the project and follow them throughout the entire business cycle. This is to address uncertainty in research and development activities, inspire creativity among researchers, and ensure the continuation of the project.

- System Analysis: Project specifications regarding the necessary resources, availability of technical knowledge, and team technology mapping should be developed prior to detailed planning. It is necessary to map the available know-how and additional knowledge required for the startup project prior to planning because it has significant impact on project success.
- Preliminary Design: This phase is finished with a time schedule when the project period is broken down step by step with indicators for specific tasks and success evaluations. This step ensures balance between the unrealistic goals that researchers want to achieve and the practical implications of being able to achieve them with the available resources over a given period of time.
- Detailed Design: In this phase, the overall system is mathematically validated by predicting the respective shape and performance of the project/product through specific modeling, drawings, and specifications of the product as well as procurement of materials. In this phase, the major activities that mainly contribute to the essential outcome of the project, i.e., problems related to intellectual property, should also be considered at this particular stage.
- Implementation (prototyping): Production planning and tool design consist of how to mass-produce the product and the tools to be used in the manufacturing process. The tasks to be completed in this step include material selection, production process selection, work order determination, jigs, fixtures, metal cutting, and tool selection, such as metal or plastic forming tools.
- Product Testing: This task involves verifying the different performance that the product must have and repeating additional prototype tests as a process for ensuring that the mass-production version meets the qualification test standards.

8.2 TRL (Technology Readiness Level) for R&D

Developed at NASA during the 1970s (ESA Handbook, TRL 2018), technology readiness levels (TRLs) are a method for estimating the maturity of technologies during the acquisition phase of a program. The use of TRLs enables consistent, uniform discussions of technical maturity across the R&D process of technology. The concept of TRL has been generalized and is widely used in the typical R&D process.

The generalized concept is as follows:

<Fundamental Research Stage>

- TRL 1, Basic Theory/Experiment: Basic Theory Establishment Phase
- TRL 2, Conceptualization of Ideas, Patents, etc. for Practical Use: Establishing Concepts for Technology Development and Applying Patents for Ideas

<Experimental Stage>

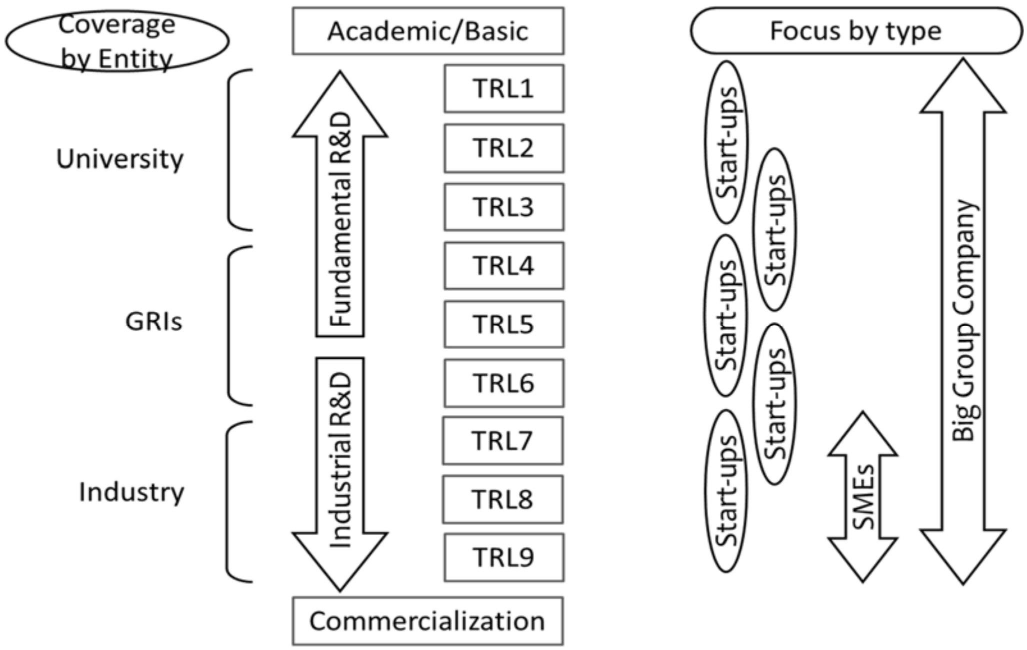
- TRL 3, Verification of Basic Performance of Laboratory Scale: The stage wherein basic performance can be verified through experimental or computerized simulation in the laboratory environment, the stage wherein the basic design drawings of components/systems to be developed are obtained.
- TRL 4, Evaluation of Core Performance of Materials/Parts/Systems on a Laboratory Scale: Test Sampling is completed to select optimal results from the various ones derived from Step 3, and computer simulation is completed, if possible.

<Implementation and Prototyping>

- TRL 5, Finalized Material/Part/System Startup Production and Performance Evaluation: The stage wherein the laboratory starts product production and performance evaluation of fixed material/part/system, number of samples of actual startup designed considering the production target but not the economy, the stage wherein the target performance is achieved to the extent that only those displaying the core performance of the technology can be sold.

- TRL 6, Production and performance evaluation of the starting products of the pilot scale: Completion of production and evaluation of the starting products of the pilot scale (some one-tenth of the mass scale) and presentation of production, production capacity, defect rate, etc., and large-scale investment for the production of the pilot, with the production company conducting its own field test to achieve the target performance of the company and the results of public certification of the performance.

[Figure 3-12] TRL (Technology Readiness Level) for R&D



Source: Park (2019)

<Commercialization and Practical Application>

- TRL 7, Reliability Assessment and Demand Enterprise Assessment: In case of the stage wherein performance verification is performed in a real-world environment, onsite assessment of the beginning of the pilot by the supplier (performance and reliability assessment) and, if possible, submitted by the certification authority.

- TRL 8, Product Certification and Standard

- TRL 9, Stage of mass production and commercialization, 6-sigma, etc. is important.

One of the GRIs in Korea, ETRI (Electronics and Telecommunications Research Institute), is in the middle of university and industry, transferring over 200 technologies to the industry for commercialization every year. University focuses on core and basic research studies, whereas industry develops advanced applied technologies. Sometimes, ETRI works with universities for the industry and sometimes works with industries for universities. As a result of this strong collaboration with the academe and industry, ETRI has gained a reputation for its patent prowess in Korea.

Each year, every R&D project in ETRI sets TRL as a goal and undertakes research from the beginning phase of technology development. Depending on the nature of the technology, the TRL goal is set differently. Basic researches naturally begin at the lower TRLs; on the other hand, industrial or commercial R&D begin at higher TRLs. Typically, there are two stages of TRL up to three years or four stages of TRL up to five years after the development is completed.

Technologies with any TRL can be transferred according to the demand from the industry, but successful technology transfer requires clear decisions based on a comparison of the technology's productivity and maturity.

As such, it is very important to set goals for the TRL in the early stages of research and development; likewise, when introducing advanced technologies from developing countries, it is also important to determine which level of technology will be brought in and further developed and commercialized for the regional industry. In Uganda, due to weak infrastructures and resources, higher TRLs will be better and effective to transfer and commercialize hi-tech products for rapid economic growth even if they are not cutting-edge technologies.

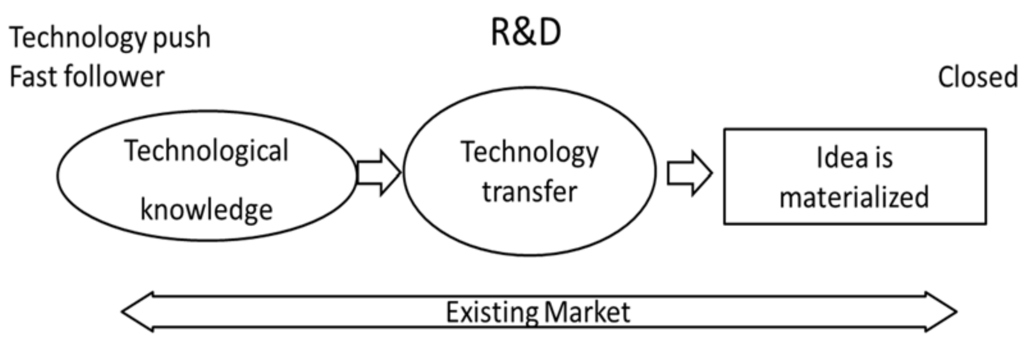
For several decades, Korea has become much less dependent on GRIs due to the improved technological capabilities of companies, especially a large group of companies as shown in Figure 3-12. Big companies such as Samsung are capable of reaching Technology Readiness Levels 1 (university level) to 9 (industry/real world). They have lots of high technology, people, and resources. In contrast, SMEs, even if they want to, will not be able to cover that area. They cannot afford to do so because of limited resources and manpower, so they just focus on manufacturing the product and making a profit on a very short-term basis. SMEs are interested in much mature TRL7 through TRL8 technology from GRI to start production and enter the market right away. A variety of business models are available for startups, where TRLs are often patent businesses for source technologies with very hi-tech skills. On the other hand, when a TRL is intended for matured technologies, it often directs the developed technology to produce and sell the product in the market. Lots of business opportunities and ideas for the intermediate level of TRL are also created by startups.

8.3 R&D vs. R&BD

To catch up with advanced countries, we can carry out R&D as usual by taking advantage of the resources we have after receiving technology transfer and develop competitive products or products with additional functions and enter the market. When these R&Ds settle in several industries in a country, the national economy will be solid with products that can enter the existing markets.

Nonetheless, these usual technological developments have limitations in developing new business models or products beyond advanced technologies. Because there are no products to imitate, and even if we develop creative products, we cannot guarantee the market.

[Figure 3-13] Typical R&D for the Fast Follower



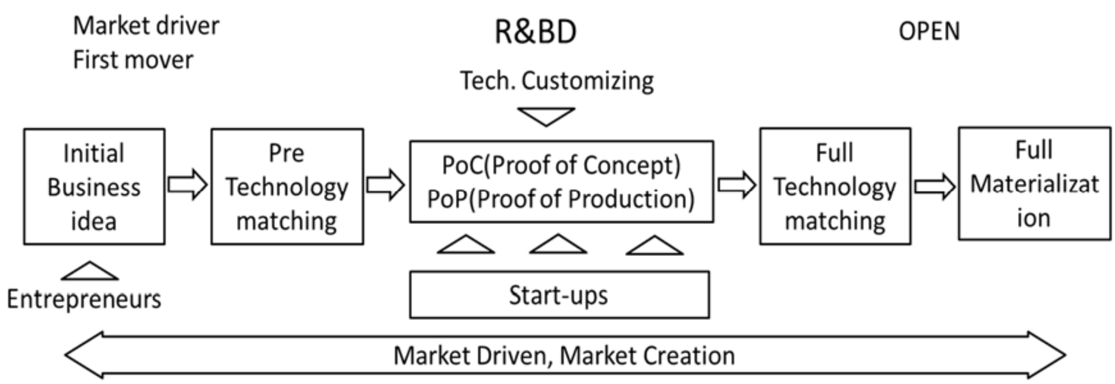
Source: Park (2019)

Conventional R&D is a simple process as shown in Figure 3-13, wherein new products can be developed through additional R&D for technologies that have already been transferred or transferred to create products that meet the needs of the market. Therefore, it is sometimes called the technology push type because the usual R&D is called fast fallower type and is manufactured using technologies that meet the market demand.

On the other hand, creative R&D, which pioneered non-existent markets and created new demand for technology, is specifically called R&BD (Research and Business Development); as shown in [Figure 3-14], there are a number of new elements that work here.

First of all, even if entrepreneurs have a good business idea, the market is not certain. It is necessary to examine the feasibility of success in advance and to reduce the risk of investment.

[Figure 3-14] R&BD to be a First Mover



Source: Park (2019)

When it is determined to be feasible, entrepreneurs as founders of startups may go through the PoC (Proof of Concept) process and PoP (Proof of Product) process; if it is deemed successful, entrepreneurs will carry out full-scale technology transfer or technology development, and then produce the product. The role of many startups in such PoC and PoP process is necessary because existing companies have less flexibility, whereas startups have the flexibility to change direction more quickly and easily. The reason governments of both developed and developing countries around the world support and continue to nurture startups is that startups are beneficial to new markets. This is why R&BD-type technology development is called 1st mover type and Open type.

According to Christine (2018), it is best to be first to market with innovative products and services as first movers get a head start to carve out a dominant market share, citing the many advantages of being the first mover. Naturally, however, fast followers are more successful in the existing market due to the advantages of imitating or reverse-engineering the pioneer product. They can even improve the product or service by adding more functions or making it lighter and cheaper, finally sharing the market with the first mover.

Even if this is true and easy to follow, it is inevitable to act as a first mover to be a developed country as we can easily see in the cases of Amazon (first major online bookstore), eBay (first online auction), Kleenex (first facial tissue), etc.

9. Technology Commercialization

9.1 Technology commercialization process

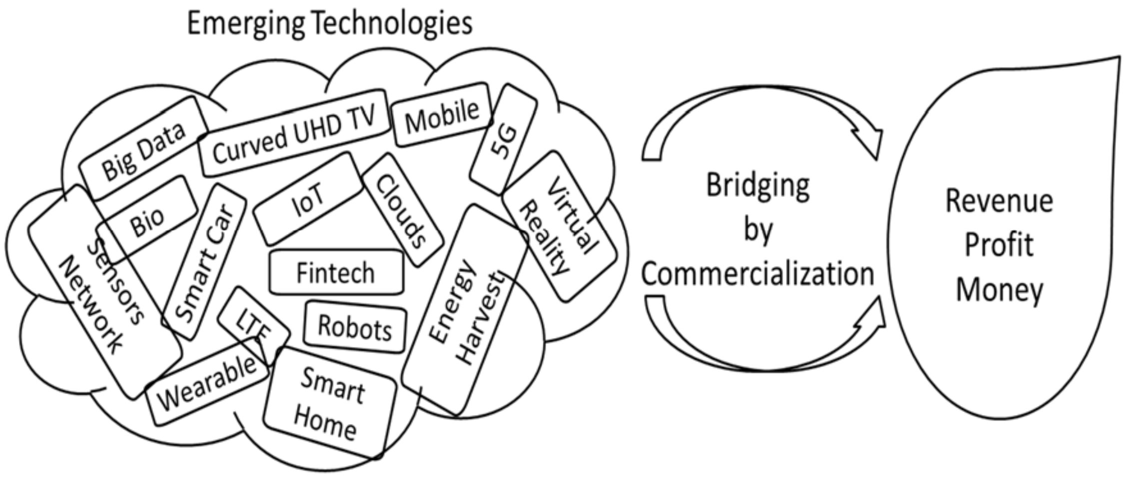
To commercialize the technology developed by universities or GRIs, one must cross the Death Valley – the chasm between innovation and real-world products and services. SMEs and startups that are creative and innovative may hinder the existing market by introducing a new business model with cutting-edge technology. Unfortunately, it is well- known that most new technologies die in the "Valley of Death" and never get commercialized. To cross this valley between laboratory prototypes and commercial products, the bridges between technology and business opportunity are necessary so as to overcome the Death Valley as shown in Figure 3-15.

Recent developments in ICT technology have led to new technologies coming out almost every day and old ones disappearing. In Korea are thousands of technologies developed at GRIs and universities each year, not all of which are successful in commercializing; only about 20 per cent of them are known to be used properly by companies. The reasons may vary, such as the immaturity of technology, market mispredictions, and emergence of competitive technologies, and the situation is similar in most advanced countries.

The final process of catching up with advanced countries is technology commercialization to become the first mover. Creativity in R&D and R&BD, especially R&BD, based on the aforementioned technology transfer is an important process to catch up with developed countries and become the 1st mover.

There can be many ways to commercialize the technology, and the aforementioned technology transfer is the most common and convenient method. Although technology licensing methods are similar, they are more likely to fail if there is no technical literacy. Technology developed through technology transfer or technology licensing methods will have difficulty crossing the Death Valley.

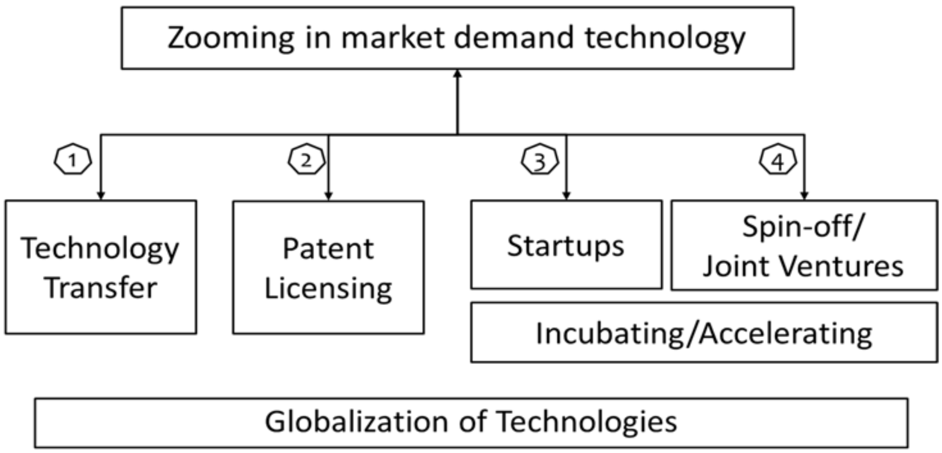
[Figure 3-15] Bridging for Technologies and Business



Source: Park (2019)

Four technology commercialization methods are shown in Figure 3-16. Technology transfer and licensing are already mentioned in the previous section and are the most common and easy way for many countries. It is used as a technology transfer channel between companies or GRIs and companies. A most active way is for the technology holder to start his/her own business. It is a suitable method for developing new products because the details of technology are understood compared with competing technologies, and it directly refers to the market conditions in the technology development process. It also has many advantages because decision making is fast since it is a startup company. A method similar to a technology partnership is the spin-off of the technology holding department or the establishment of a joint venture with an existing company.

[Figure 3-16] Technologies' Commercialization Processes



Source: Park (2019)

9.2 Technology Startups

Technology-based startups mean entrepreneurship based on advanced technology, expertise, and new business model. Unlike general startups, factors such as innovative technology creation, high R&D concentration, and technological superiority are characterized. Technology-based startups are attracting attention for the same reason that they are likely to create new industries and markets based on new products and services. Countries around the world are emphasizing technology-based entrepreneurship as an alternative to overcoming the age of weak employment growth after the global financial crisis. Korea also presented technology-based entrepreneurship as a way of changing the paradigm of job creation, based on the recognition that the growth model centered on major industries and large corporations no longer works. In addition, the rapid development of technology, such as ICT, increases the chances of realizing this idea.

Process for creating a tech startup

According to a literature survey, technology-based startups can hardly be said to have a typical set process, but they have different characteristics depending on the types of technologies in different fields and countries.

Conner Forrest (2015) suggested the following ten steps, which, unlike general startups, specified the importance of technology and the process of securing IPR:

- Do your market research
- Secure intellectual property
- Decide on branding
- Incorporate
- Choose a co-founder
- Write a business plan
- Pick a workplace
- Find a mentor
- Apply for an accelerator program
- Raise capital

The startup process of each country or private entity is quite similar, and the acquisition of patents for technology protection is a priority for tech startups based on market analysis. This is an important strategy for protecting their products from other competitors and to increase their entry barriers when their business and sales take place. In the process above, the order of finding a partner in the middle of preparing a business plan is not fixed but can be changed very flexibly in the business process.

As one of the procedures from the public sector in Korea, the ETRI startup process is illustrated in Figure 3-17. First of all, entrepreneurs who have passion and business idea using ETRI technology apply the startup incubating process.

Having the technology of ETRI and IPR can be considered secure, so they can go through simple and practical market research after creating an initial BM based on such technology. After defining customers, they figure out how many people have this problem, which they solve with a new idea. A common mistake of tech-focused startups is the idea that good technology works in the market, but that is not the case. Even if the technology is good, reviewing market adaptability is essential at the beginning of startup, so BM verification work for business ideas and market leaders is essential in the ETRI startup program.

In addition to patents, trademark registration should not be forgotten; in this process, obtaining business support and, if possible, learning basic knowledge of legal and accounting are strongly recommended before proceeding with the next step -- corporate registration.

It is recommended that mentors be involved in this process to reduce unnecessary trial and error based on their previous experience and complement them with BMs that have high chances of success at the customer's eye level. In some cases, the existing technology will be modified and customized for the target market, and these series of processes will be performed repeatedly to upgrade the idea-level BM to the actual business-type BM. Fundamental funds needed in this process are provided by ETRI during the incubation process. A group of startups in a similar situation can play an important role, since everyone is new to the experience, and they can voluntarily solve many difficulties by sharing mutual information and sharing institutions and methods for assistance.

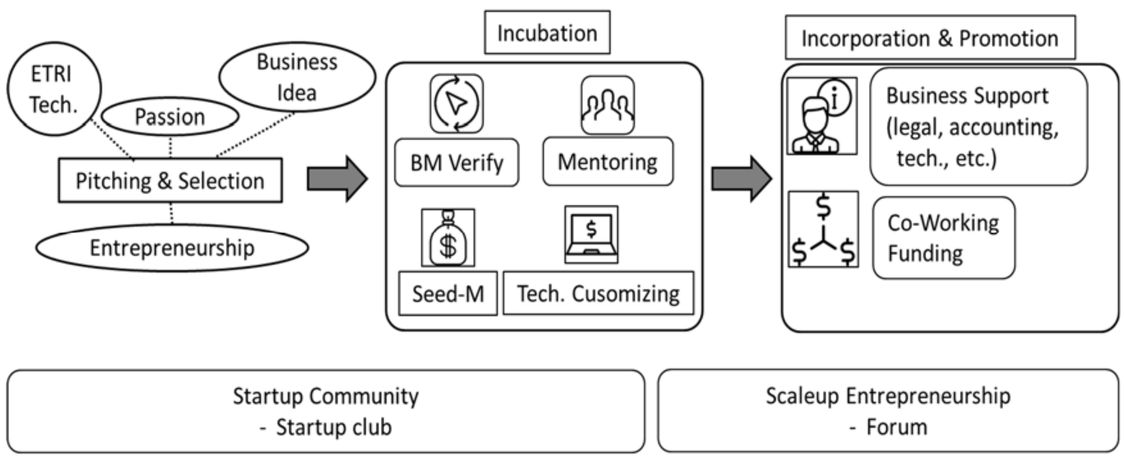
After a series of startup incubation processes such as BM establishment of ideas, market research, IPR acquisition, and mentoring, a corporation is established. When engineers or research staff start up, the process of establishing a corporation can be inconvenient, so the business support team is dedicated to eliminating unnecessary waste.

Almost all countries including Korea are supporting startups to become a 1st mover for the national economy and job creation and to offer many benefits. There is a form of corporation that can give many tax benefits to startups like venture firms. Early on, find and register a method favorable to tax and investment attraction because it gives benefits only for a limited period of time.

Workplaces or offices are also provided for prospective founders through incubating programs. The government, local governments, or universities provide places only for a certain period of time, and it is necessary to secure business space and production facilities for full-scale business in the future.

The purpose of this startup incubation in ETRI is to commercialize more developed technologies. In other words, ETRI is actively supporting startups to increase its success rate further by 20% and to commercialize small but highly explosive technologies.

[Figure 3-17] Startups and Incubation (ETRI Case)



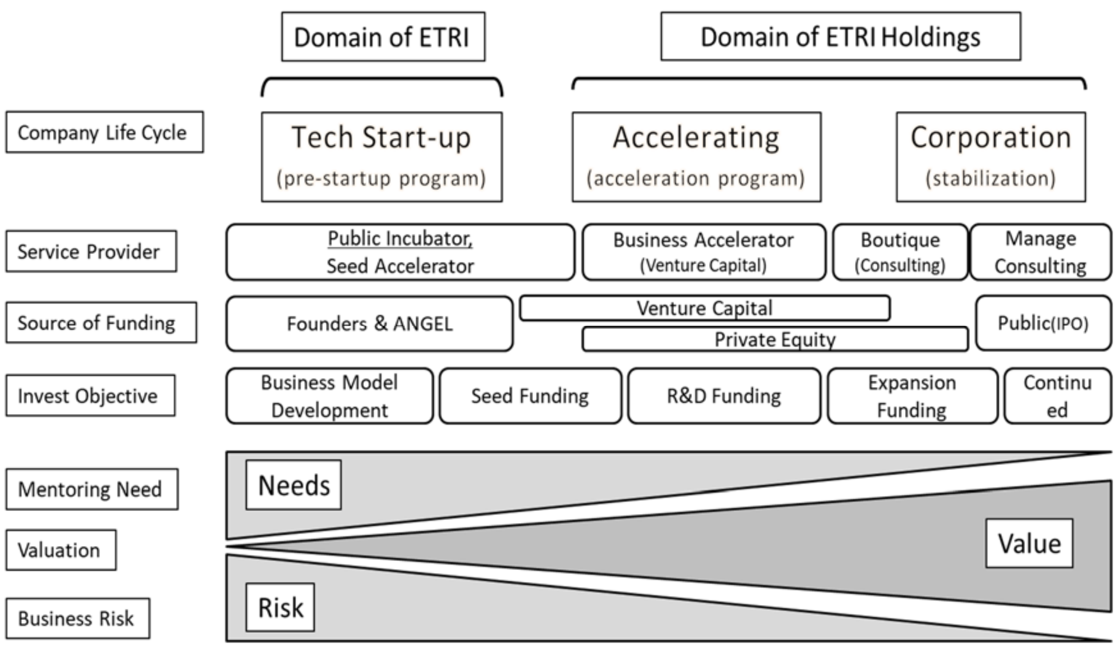
Source: ETRI (2019)

Holdings company for fund raising

When startups need more resources and expertise following incorporation, they can apply an accelerator program. An accelerator is a startup program that supports the rapid growth of public and private companies by connecting a mentor network and funding through VCs and also provides an opportunity to pitch the media and forum during demonstration day. In this case, however, the startup must divide some of the shares, and it should be very careful. Startups give equity in their company to investors or venture capitals in return for funding and, sometimes, technology transfers that should be understood before moving forward.

ETRI gets a boost in its bottom line whenever a technology is commercialized successfully and used by industries for their products. ETRI is able to support tech startups and SMEs by sharing these resources. For ETRI, there is one thing that is unique about it as shown in Figure 3-18. Since ETRI is a government-funded research institution, it cannot directly provide funding for startups. In 2010, the institute founded ETRI Holdings as a "tech-to-biz" accelerator. ETRI Holdings finds startups and matches them with the appropriate technologies of ETRI. In addition, it invests by sharing technologies with startups and offers incubation and management services (e.g., business model verification, mentoring, incubation, promotion, legal and accounting support). It even provides co-working with partners and global networks. To date, ETRI Holdings has handled a total of 60 spin-offs and joint ventures, one of which had an initial public offering.

[Figure 3-18] Role Sharing for Startups and Incubation (ETRI Case)



Source: ETRI (2019)

CHAPTER 4

Capacity building Process

Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (Ⅱ) – Policy Capacity-building Program

Chapter 4. Capacity building Process

1. Kick-off Seminar and Field-Visits

1.1 Kick-off Seminar in Uganda (Phase 1)

- Purpose: To organize the first kick-off meeting and to conduct a field study for the review of the status of industry-university linkage and incubation and commercialization of R&D outputs and conduct of interviews and meetings with stakeholders
- Period: 26 May ~ 1 June 2019 (including dates for departure and arrival), Kick-off seminar on 29 May, Site visits and meetings on 30 May
- Venue: Kampala, Uganda
- Participants

Name	Organization & Position	Area of Expertise
Dr. Deok Soon Yim	Senior Research Fellow, International Innovation Cooperation Center, STEPI	PM, Innovation Policy and System
Ms. Elly Hyanghee Lee	Researcher, STEPI	Project management, climate change policy
Prof. Jong-in Choi	Professor, Department of Management and Accounting, Hanbat National University	Industry-university linkage
Dr. Jong Heung Park	Electronics and Telecommunications Research	Incubation and commercialization of R&D

Name	Organization & Position	Area of Expertise
	Institute (ETRI)	outputs

- Main activities and agendas
  - To discuss and implement the workplan for the K-Innovation Program for Uganda 2019 such as themes, project activities, timeline (particularly on the capacity-building workshop and workshop for sharing outcomes), local consultants, and role allocation.
  - To consult with UNCST on its specific demands and expectations on policy consulting for the K-Innovation Program for Uganda 2019.
- To organize a workshop on “Sharing Korea’s policy on industry-university linkage and incubation and commercialization of R&D outputs (tentative)”
  - Local consultants are expected to give a presentation on the theme assigned to them.
- To conduct site visits and interviews with research institutions, universities, and SMEs
  - To review the status of and diagnose the industry-university linkage and incubation and commercialization of R&D outputs.
- To have meetings with the Korean Embassy and KOICA’s office in Uganda to advocate the outcomes and progress of the K-Innovation Program for Uganda 2018-2019 and to seek support for the K-Innovation Program for Uganda 2020-2022.

[Table 4-1] Schedule of Kick-off Seminar

DATE	TIME	DESCRIPTION	REMARKS
27 May (Mon.)	01:15	Departure for Uganda in Incheon via Doha	
	14:00	Arrival in Entebbe Move to the Hotel and Check-in	
28 May (Tue.)	10:00	Preparatory meeting with UNCST	UNCST
	12:30 ~ 13:30	Meeting with the Embassy of the Republic of Korea in Uganda	
	14:30	Meeting with KOICA’s Uganda Office	KOICA office

DATE	TIME	DESCRIPTION	REMARKS
	16:00	**Site visits and meetings** Makerere University	- Prof. William Kyamuhangire
	18:30	Dinner	
29 May (Wed.)	**Kick-off Seminar: Morning Session: Metropole Hotel**		
	09:00 ~ 12:00	**Opening Remarks:** Dr. Deok Soon Yim, Senior Research Fellow, STEPI **Welcome Remarks:** Dr. Peter Ndemere, Executive Secretary, Uganda National Council for Science and Technology **Congratulatory Remarks**: Mr. Obong O. David, Permanent Secretary of Ministry of Science, Technology, and Innovation **Congratulatory Remarks:** H.E. Ha Byung-Kyoo, Ambassador of the Republic of Korea to Uganda **Introduction of STEPI and K-Innovation ODA program:** Dr. Deok Soon Yim, Senior Research Fellow, STEPI **Presentations:** **Presentation 1**: Sharing Korean practices on Industry-University Linkage (30'): Prof. Jong-in Choi, Hanbat National University **Presentation 2**: Sharing Korean practices on the Incubation and Commercialization of R&D outputs (30'): Dr. Jong Heung Park, Electronics and Telecommunications Research Institute **Presentation 3**: Current status and issues of the Industry-University Linkage policy in Uganda (20'): Dr. Maxwell Otim Onapa, UNCST **Open discussion** Moderated by Dr. Deok Soon Yim, STEPI	
	12:00	Lunch	

DATE	TIME	DESCRIPTION	REMARKS
	Kick-off Seminar: Afternoon Session: Metropole Hotel		
	13:30 ~ 17:30	**Presentation 4**: Current status and issues of the policy on Incubation and Commercialization of R&D outputs in Uganda (20'): Mr. Basil Ajer, Director for Technopreneurship, MoSTI **Presentation 5**: STI Parks and Municipal Innovation Centers (20'): Mr. Geoffrey Sempiri, Head, STI Infrastructure Development, UNCST **Presentation 6:** Current status and issues with regard to the research evaluation and assessment of research quality in Uganda (20'): Mr. Bashir Kagere, Senior Science Officer, UNCST **Presentation 7:** New and emerging technologies including ICT (20'): Ms. Monica Rose Nyakaisiki, Assistant Commissioner, Science, Technology, and Innovation Infrastructure Development, MoSTI **Open discussion:** Moderated by Dr. Maxwell Otim Onapa, UNCST **Closing remarks:** Dr. Deok Soon Yim, STEPI	
	17:30	**Tea Break**	
	17:45 ~ 18:30	**Bilateral meeting for In-depth Discussion on the work plan for STEPI-UNCST cooperation (closed meeting): UNCST/STEPI research team** - i.e., project activities, timeline, expected deliverables, responsibilities of STEPI and UNCST, ToR for local consultants and so on.	
30 May (Thurs.)	09:00 ~ 12:00	**Site visits and meetings** - Research Institute	
	12:00	Lunch	
	13:00 ~ 18:00	**Site visits and meetings** - SME	
31 May (Fri.)	15:10	Departure for Incheon Via Doha	
1 Jun. (Sat.)	16:55	Arrival in Incheon	

2. Kick-off Seminar

2.1 Background

The Uganda National Council for Science and Technology (UNCST) signed a Memorandum of Understanding (MOU) with the Science and Technology Policy Institute (STEPI) of the Republic of Korea on 30 May 2016 during the Korean President's visit to Uganda. The purpose of the MOU is to establish a framework of practical cooperation between both parties to propel further collaboration in the fields of Science, Technology, and Innovation (STI). Within the framework of the agreement, the National Research Foundation of Korea (NRF) and STEPI serve as practical development practitioners in S&T Official Development Assistance (ODA) for Uganda. Specifically, STEPI, NRF, and UNCST are expected to host events jointly such as seminars on STI, exchange human resources through education and training, implement joint research projects, and conduct any other activities that may be jointly decided upon by the parties.

As part of the operationalization of the MOU above, UNCST submitted to STEPI two project proposals — (i) "Capacity Building to Enhance the Development of Science, Technology, and Innovation in Uganda" and (ii) "Strengthening Uganda's National STI System and Establishment of STI Parks" — on 17 March 2017. The project proposals were subsequently approved to run for a two-year period from February 2018 to December 2019.

The partnership has since resulted in the successful Implementation of a K-Innovation ODA program (2018) aimed at strengthening Uganda's Science, Technology, and Innovation system through capacity building. In 2018, UNCST-STEPI organized the first kick-off meeting for the K-innovation ODA Program hosted at UNCST from April 23 to 27, 2018. The purpose was (i) to identify specific areas of capacity building and (ii) discuss the scope of the 2018 program. As a result, the following thematic areas were identified as strategic for capacity building: (i) New and emerging technologies (focused on ICT); (ii) Technology transfer; (iii) Science communication; and (iv) Research, innovation, and commercialization/Roadmap for the development of science parks. Subsequently, in October 2018, STEPI hosted a knowledge-sharing seminar as well as excursions in Korea from October 28 to November 4, 2018.

The K-Innovation Program for Uganda 2019 focuses on three new themes proposed by UNCST including: Industry-University linkage; Incubation and commercialization of R&D outputs; and Research evaluation and assessment of research quality.

2.2 Objectives

The kick-off seminar will provide an opportunity for:

- Sharing Korean best practices on Industry-University linkage and Incubation and Commercialization of R&D outputs; and
- Understanding the current status and issues with regard to themes such as Industry-University Linkage, Incubation and Commercialization of R&D Outputs, Research Evaluation, STI Parks and Municipal Innovation Centers, and New and Emerging technologies including ICT.

2.3 Program

TIME	DESCRIPTION
Morning session	
09:00 ~ 12:00	**Opening Remarks:** Dr. Deok Soon Yim, Senior Research Fellow, STEPI **Welcome Remarks:** Dr. Peter Ndemere, Executive Secretary, Uganda National Council for Science and Technology **Congratulatory Remarks**: Mr. Obong O. David, Permanent Secretary of Ministry of Science, Technology, and Innovation **Congratulatory Remarks:** H.E. Ha Byung-Kyoo, Ambassador of the Republic of Korea to Uganda **Introduction of STEPI and K-Innovation ODA program:** Dr. Deok Soon Yim, Senior Research Fellow, STEPI

TIME	DESCRIPTION
Morning session	
	Presentations: **Presentation 1**: Sharing Korean practices on Industry-University Linkage (30'): Prof. Jong-in Choi, Hanbat National University **Presentation 2**: Sharing Korean practices on Incubation and Commercialization of R&D outputs (30'): Dr. Jong Heung Park, Electronics and Telecommunications Research Institute **Presentation 3**: Current status and issues of the Industry-University Linkage policy in Uganda (20'): Dr. Maxwell Otim Onapa, UNCST **Open discussion** Moderated by Dr. Deok Soon Yim, STEPI
12:00	Lunch
Afternoon session	
13:30 ~ 17:30	**Presentation 4**: Current status and issues of the policy on Incubation and Commercialization of R&D outputs in Uganda (20'): Mr. Basil Ajer, Director for Technopreneurship, MoSTI **Presentation 5**: STI Parks and Municipal Innovation Centers (20'): Mr. Geoffrey Sempiri, Head, STI Infrastructure Development, UNCST **Presentation 6:** Current status and issues with regard to research evaluation and assessment of research quality in Uganda (20'): Mr. Bashir Kagere, Senior Science Officer, UNCST **Presentation 7:** New and emerging technologies including ICT (20'): Ms. Monica Rose Nyakaisiki, Assistant Commissioner, Science, Technology, and Innovation Infrastructure Development, MoSTI **Open discussion:** Moderated by Dr. Maxwell Otim Onapa, UNCST **Closing remarks:** Dr. Deok Soon Yim, STEPI
17:30	**Tea Break**
17:45 ~ 18:30	**Bilateral meeting for In-depth Discussion on the work plan for STEPI-UNCST cooperation (closed meeting): UNCST/STEPI research team** - i.e., project activities, timeline, expected deliverables, responsibilities of STEPI and UNCST, ToR for local consultants and so on.

Group Photo of the Kick-Off Seminar

Congratulatory Remarks by H.E. Byungkoo Ha, Embassy of the Republic of Korea

Bilateral Meeting with Mr. Obong O. David, Permanent Secretary of Ministry of Science, Technology, and Innovation

Bilateral Meeting with the KOICA Office in Uganda

3. Joint Capacity-building Workshop for Uganda and Indonesia

3.1 Background

As part of the K-Innovation Program with Uganda and Indonesia, experts from the Ministry of Science, Technology, and Innovation of Uganda, Uganda National Council for Science and Technology (UNCST), and Indonesian Institute of Sciences (LIPI) are invited to participate in the joint capacity-building workshop. Through the workshop, participants are expected to deepen their understanding of the themes covered by the K-Innovation Program 2019 for Indonesia and Uganda. Major themes for the workshop include:

- Structure and history of Korean S&T policy in general
- STI policy and necessary policy measures
- Industry-University cooperation, incubation, and commercialization of R&D Outputs

The four-day workshop consists of the following: 1) lectures and group discussions; and 2) field visits. Participants are expected to participate actively in the course and present the action plan for developing the innovative science and technology policy, which could be applied to Indonesia and Uganda, based on Korea's experiences.

Theoretical and practical intensive courses are designed for Indonesian and Ugandan government officials and practitioners looking to broaden their knowledge through group discussions. Field visits to public research institutes and other related agencies are arranged.

3.2 Overview of the Capacity-building Workshop

- Dates: 8 ~ 11 July 2019
  - Arrival on 7 July and Departure on 11 July 2019 for the Ugandan team

- Arrival on 7 July and Departure on 12 July 2019 for the Indonesian team

- Venue: Shilla Stay Yeoksam, Seoul and Daejeon, Republic of Korea
- Participants: STEPI research team and 10~12 invited participants from LIPI and UNCST and Ministry of Science, Technology, and Innovation of Uganda

Organization	Name	Position/Department
Science and Technology Policy Institute (STEPI)	Dr. Deok Soon Yim	Project Manager
	Dr. Wang Dong Kim	Senior Research Fellow
	Ms. Elly Hyanghee Lee	Researcher
	Ms. Min Jung Kim	Researcher
Indonesian Institute of Sciences (LIPI)	Dr. Mego Pinandito	Deputy Chairman for Scientific Services
	Dr. Trina Fizzanty	Senior Researcher
	Mr. Qinan Maulana Binu Soesanto	Junior Researcher
	Ms. Indah Purwaningsih	Junior Researcher
Agency for the Assessment and Application of Technology (BPPT)	Mr. Boni Benyamin	Engineer (Perekayasa Madya)
Ministry of Research, Technology, and Higher Education (Ristekdikti)	Dr. Adawiah Muhammad Hasan	Deputy Director for Harmonization of Innovation Policy and Programs
Ministry of National Development Planning (BAPPENAS)	Dr. Ade Faisal	Acting Deputy Director for Research and Development in Science and Technology

Organization	Name	Position/Department
Ministry of Science, Technology, and Innovation, Uganda	Dr. Maxwell Otim Onapa	Director
Ministry of Science, Technology, and Innovation, Uganda	Ms. Monica Rose Nyakaisiki	Assistant Commissioner
Ministry of Science, Technology, and Innovation, Uganda	Mr. Morrish Ochen	Assistant Commissioner
Uganda National Council for Science and Technology	Mr. Bashir Rajab Kagere	Head, Policy Analysis, Planning and M&E
Uganda National Council for Science and Technology	Mr. Geoffrey Sempiri	Head, STI Infrastructure and Development

3.3 Program

Date	Time		Place	Key Activity
6 July (Sat.)	22:05	7:15 +1	Soekarno–Hatta International Airport (CGK)	Departure from Jakarta (Korean Air, KE0628)
	16:20		Entebbe Airport	Departure from Entebbe (EK730) Via Dubai
7 July (Sun.)	05:30		Incheon	Arrival at Incheon Airport (Korean Air, KE696)
	07:00	12:00	Shilla Stay Yeoksam	Move to Seoul & Check-in (rest)
	12:00	13:00		Lunch
	13:00	18:00		Rest & Preparation for the workshop
	16:55			Arrival at Incheon Airport (EK322)
	17:00	18:30		Move to Seoul & Check-in (rest)

Date	Time		Place	Key Activity
	19:00			Dinner
	20:00			Rest
8 July (Mon.)	9:30	10:00	Shilla Stay Yeoksam	Orientation - Opening Remarks (Dr. Chi Ung Song) - Introduction of Workshop (Dr. Deok Soon Yim)
	10:00	11:00		Lecture 1: Understanding the National Research Council of S&T by Dr. Youngjoo Ko
	11:00	11:15		Presentation 1: Status and Challenges of Science, Technology, and Innovation in Indonesia by Dr. Adawiah Muhammad Hasan
	11:15	11:30		Presentation 2: Status and Challenges of S&T Governance in Uganda by Mr. Geoffrey Sempri
	11:30	12:00		Q & A and Group Discussion
	12:00	13:30		Welcome Lunch
	14:00	15:00		Lecture 2: S&T Policies and Governance in Korea by Prof. Seong-soo Kim
	15:00	15:15		Presentation 3: Status and Challenges of STI Policy in Indonesia by Dr. Mego Pinandito
	15:15	15:30		Presentation 4: Science and Technology Innovation Policy in Uganda by Mr. Bashir Kagere
	15:30	16:00		Q & A and Group Discussion
	16:00	17:00		Lecture 3: Technology Commercialization -- “What it means to catch up” by Dr. Jong Heung Park, Executive Director, ETRI
	17:00	17:30		Q&A
	18:00			Move to Han River
	18:30	20:00		Dinner (near Han River)

Date	Time		Place	Key Activity
9 July (Tues.)	09:00	10:00	Smartwork Center for National Research Institutes in Yangjae	Lecture 4. Facts and Truth about Technology Transfer by Prof. Johng-Ihl Lee, State University of New York, Korea
	10:00	10:15		Presentation 5: Status and Challenges of Technology Commercialization in Uganda by Mr. Morrish Ochen
	10:15	10:30		Presentation 6: Model of Technology Commercialization from GRI to Private Sector in Indonesia by Mr. Boni Benyamin
	10:30	10:45		Presentation 7: Emerging Technologies with Focus on ICT: A Case for Uganda by Ms. Monica Rose Nyakaisiki
	10:45	11:15		Q & A and Group Discussion
	11:30	12:30		Lunch
	12:30			Move to KIST
	13:30	14:30	KIST	Lecture 5: How KIST develops its Technologies into Business by Dr. Yoohyung Won
	14:30	15:00	KIST	Field Visit 1: Korea Institute of Science and Technology (KIST), KIST History Hall
	15:30	18:00	Seoul	Sightseeing around the Palace and Insadong, Seoul
	18:00		Seoul	Dinner
	20:30			Move to the hotel
10 July (Wed.)	08:00	10:00	Seoul to Daejeon	Move to Daejeon
	10:00	11:00	Hanbat National University	Lecture 6: Industry-University Collaboration by Prof. Jong-in Choi, Hanbat National University (60’)
	11:00	12:00		Lecture 7: Biotechnology Industry in Korea by Prof. Byunghwan Hyun, Daejeon University (60’)
	12:00	12:15		Q & A (15’):
	12:30	13:30	Near the Univ.	Lunch
	13:30	14:00		Move to KRIBB
	14:00	15:00	Daejeon	Field Visit 2: Korea Research Institute of Bioscience and Biotechnology (KRIBB)

Date	Time		Place	Key Activity
	15:30	16:30		Field Visit 3: Daejeon Technopark (Dr. Sangho Choi)
	17:00	18:00		Field Visit 4: Advanced Nano Products Co., Ltd. (Prof. Changwoo Park, CEO)
	18:30	19:30		Dinner
	19:30		Move to Seoul	Move to the hotel
11 July (Thurs.)	09:00	10:00	Shilla Stay Yeoksam	Lecture 8: Key Factors of Korea's STI Development by Dr. Youngrak Choi, Senior Research Fellow Emeritus, STEPI
	10:00	10:15		Presentation 7: Industry-University Linkage: Issues and Current Status by Maxwell Otim Onapa
	10:15	10:30		Presentation 8: Industry-University Policy Linkage in Indonesia by Dr. Ade Faisal
	10:30	11:30		Q & A and Group Discussion - Overview of Indonesia's Science and Technology Innovation in Indonesia by Dr. Trina Fizzanty - Overview of STI Policy in Indonesia by Mr. Qinan Maulana Binu Soesanto - Overview of S&T Governance in Indonesia by Ms. Indah Purwaningsih
	12:00	13:30		Lunch
	13:30	14:30		[Wrap-up and Follow-up] Group Presentations
	14:30	15:00		Closing Ceremony
	15:00	17:00		Rest
	17:00	18:00		Dinner
	18:30			Move to the Airport (Uganda)
	23:55		Incheon Airport (ICN)	Departure from Incheon Airport (EX323)
12 July (Fri.)	11:00			Move to the Airport (Indonesia)
	15:30	20:30	Incheon Airport (ICN)	Departure from Incheon Airport (Korean Air, KE0627)

4. Lessons learnt from the workshop and future actions

4.1 Ugandan Team

Lessons learnt

First, the workshop addressed very critical aspects of MoSTI's aspirations notably in technology transfers, commercialization, and enterprise development chain.

Second, integrating STI into the national development process requires software, hardware / infrastructure, and orgware.

Third, the Korean model of reversing Brain Drain - the right approach. Modalities to do that can be further explored. In practice and not in time, Uganda can reverse to 1966 and pursue a related approach used by Korea, and there is huge possibility of success.

A reward system for researchers whose results create a new product serves as a great motivation for innovation.

4.2 Future actions

MoSTI will strengthen partnership with STEPI to enhance capacity building of the new MoSTI with contemporary skills on science and technology policy processes and practices to guide the country forward.

- Development of legal frameworks, guidelines/policy to promote technology transfer and commercialization in Uganda
- Partnership in the establishment of a Korean model of STI infrastructure
- "KIST" equivalent in Uganda, especially in those newly established universities with large land and fresh young minds to embrace these efforts
- Technology park
- "Ugandan STEPI"

Build on the capacity-building workshop to improve the innovation infrastructure. Opportunities for collaboration and capacity enhancement in bioscience and biotechnology – infrastructure will be explored.

Group photo of Capacity Building Workshop for Uganda and Indonesia

Site-visit at KIST

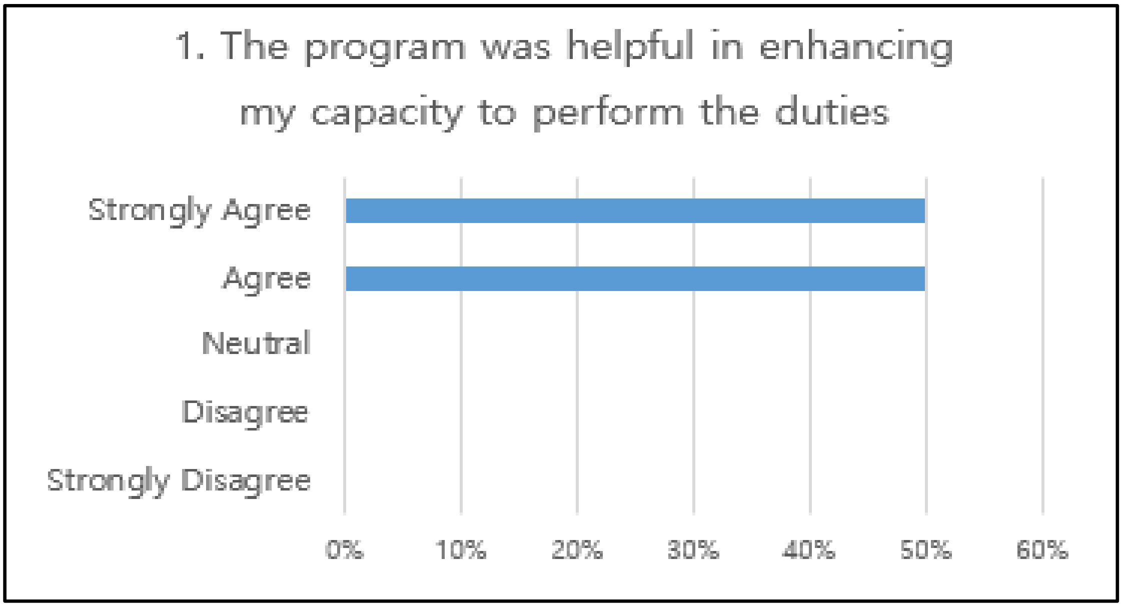
Site-visit at Korea Research Institute of Bioscience and Biotechnology

5. Evaluation Survey

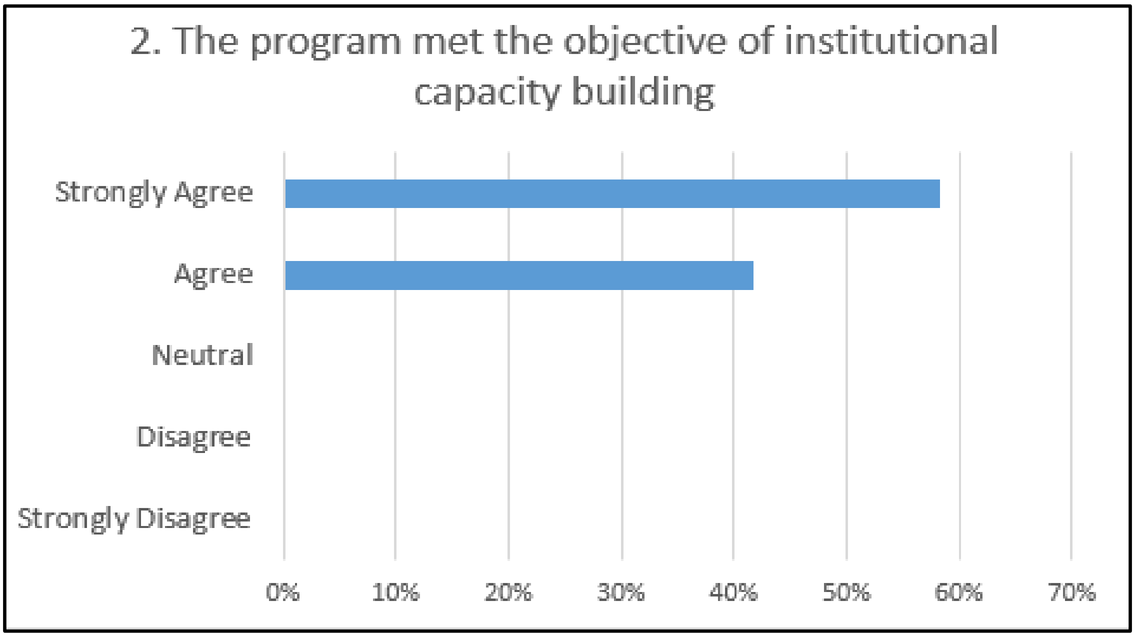
The survey indicates the participants' opinions including (1) Relevance, (2) Efficiency, (3) Effectiveness, (3) Impact, (4) Coordination, and (5) Sustainability.

The Ugandan and Indonesian participants were highly satisfied with the course and class quality. For the next invitational workshop, they would like to have more site visits and more time for discussion. They expressed special interest in the following areas: demand-side policy; public-private-people collaboration/partnership in STI; designing technology foresight for STI development, particularly for realizing industry 4.0; and R&D support mechanism.

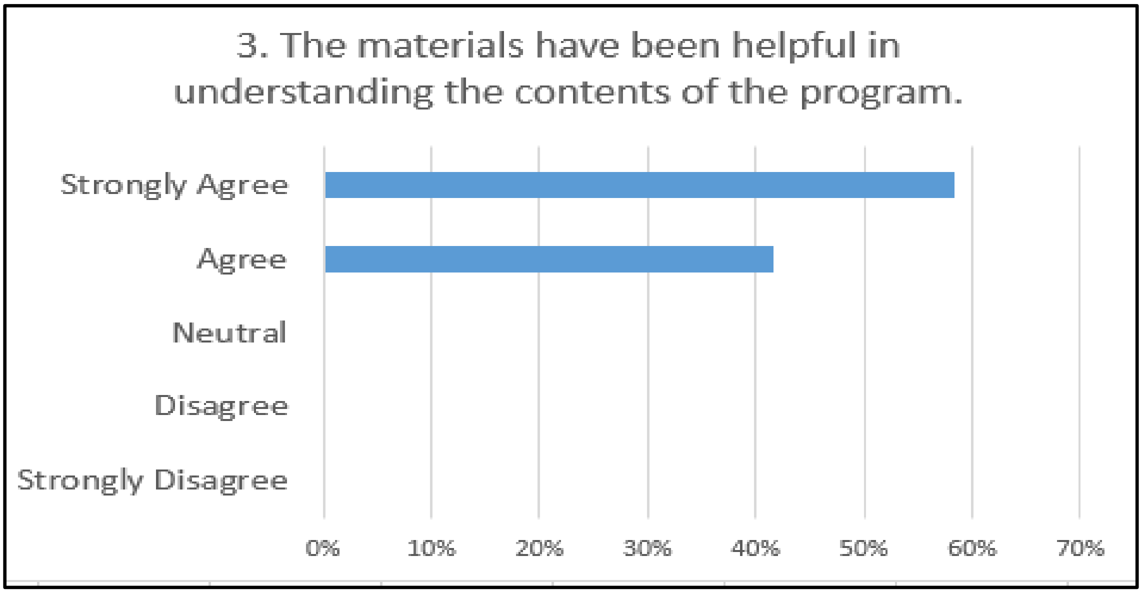
[Figure 4-1] Effectiveness of the Workshop in View of Program Arrangement



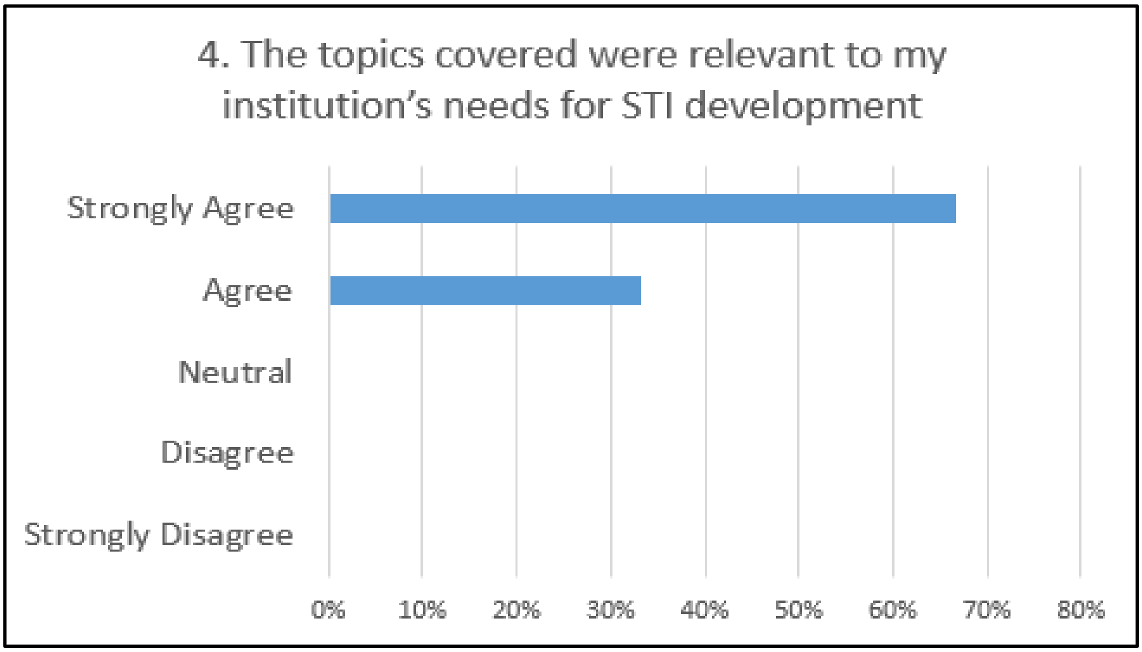
[Figure 4-2] Relevance of the Workshop in View of Program Objectives



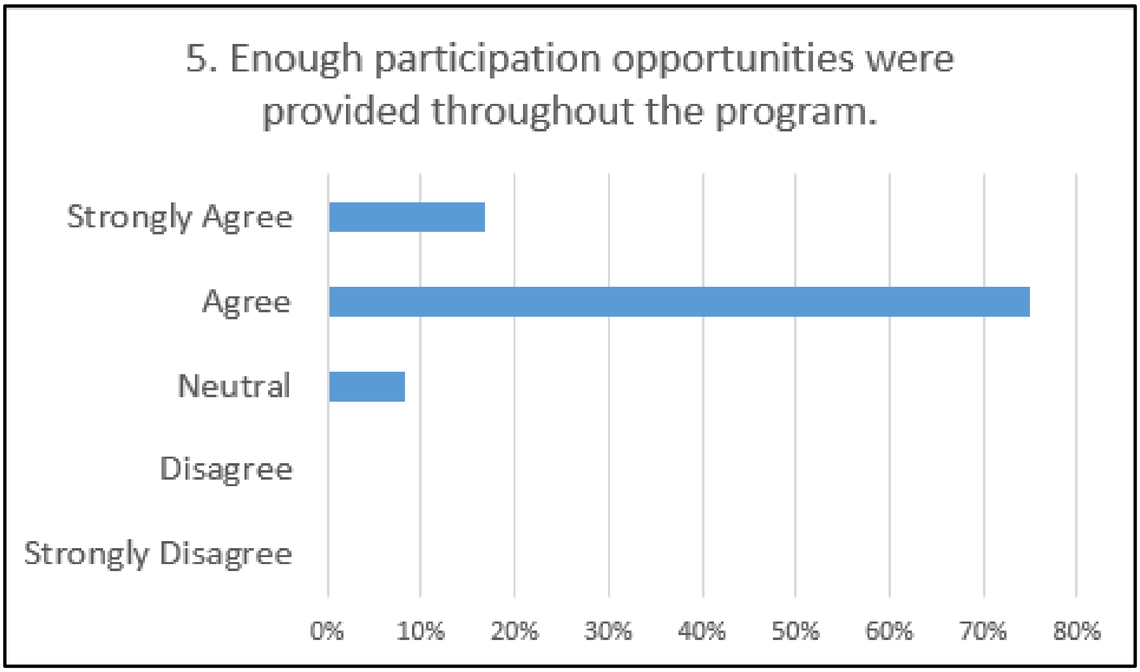
[Figure 4-3] Coordination of the Workshop in View of Educational Materials



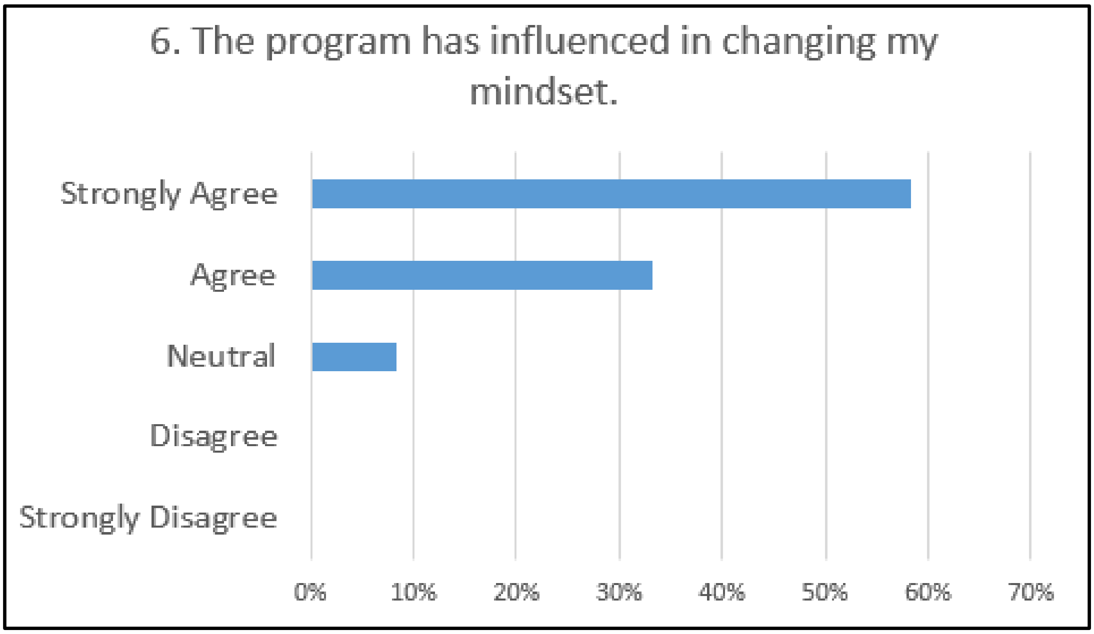
[Figure 4-4] Relevance of the Workshop in View of Educational Materials



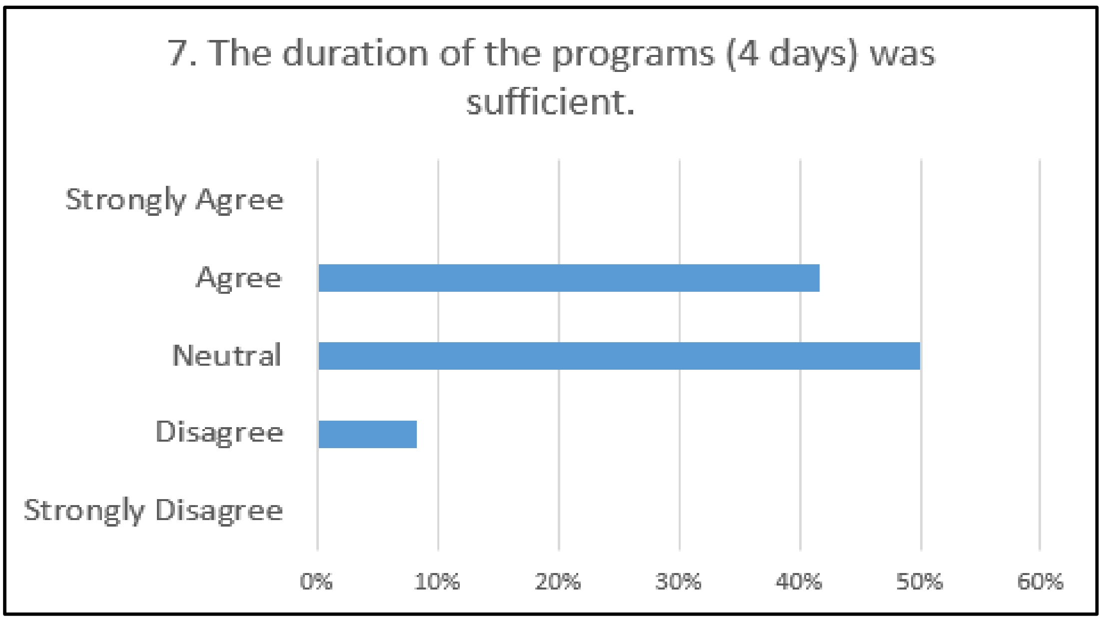
[Figure 4-5] Effectiveness of the Workshop in View of Interaction



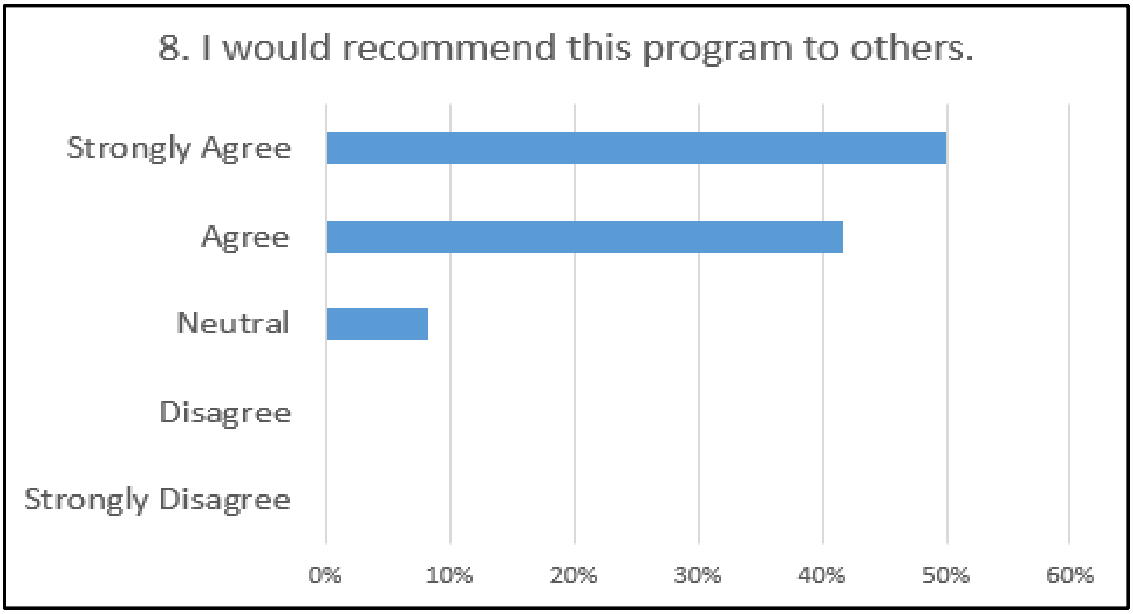
[Figure 4-6] Impact of the Workshop in View of Mindset



[Figure 4-7] Efficiency of the Workshop in View of Periods



[Figure 4-8] Sustainability of the Workshop Itself



9. What other topics/contents would you suggest we deal with in the future?

- Demand-side policy: how to promote and implement this policy
- Public-private-people collaboration/ partnership in STI
- Promoting STI for sustainable development
- Global innovation policy
- More field visits and discussion time
- Designing technology foresight for STI development, particularly for realizing industry 4.0
- Designing a national STI plan focused on the medical industry
- R&D support mechanism
- Science communication
- Technology parks management
- STI governance and policy (especially for politicians and congressmen)

- Role of the private sector in the STI ecosystem
- Planning and development of S&T infrastructure to support industry-oriented research and technology transfer
- Generation and application of STI data in STI policy planning and making

10. Are there any suggestions for improvement or requests on the overall procedure? If so, please explain in detail (Program process, contents, results, sustainability, etc.).

- Give Muslim participants time to pray
- Need feedback from experts on materials that we have already prepared
- Follow-up activities are important for sustainability through the pilot project (policy)
- Need more discussion especially in relation to the context of each participant country and scope of work
- There are some overlapping materials between experts
- Halal food can be provided for Muslim delegates for the next program
- Inviting experts from the government to stress the STI policymaking strategy
- Research collaboration in the future
- Select particular subjects and discuss deeper under the direction of the STEPI expert
- More site visits to the industry (Company)
- Need to transform the workshop into more projects that can deliver tangible values with support from KOICA
- Need to organize a workshop on governance for politicians and congressmen especially for ministers responsible for STI and members of the parliament in the STI Committee)
- Customize the workshop to benefit many officers in detail and arrange to train 3-5 officers for a period of 2-3 weeks

6. Dissemination Workshop

As the last activity of the 2019 project, STEPI, MoSTI, and UNCST co-organized the workshop to review and disseminate outcomes of the policy consulting. At the workshop, STEPI and Ugandan experts presented the outcomes of joint research along with policy recommendations proposed by Korean experts on two focused themes: Industry-University Linkage and Incubation and Commercialization of R&D outputs. This workshop gave a great opportunity for various stakeholders related to STI to publicize Korea's ongoing efforts to support Uganda's capacity building on STI through the 2019 K-Innovation ODA Program.

The dissemination workshop seminar aimed at:

- Sharing and reviewing the policy consulting outcomes of the 2019 K-Innovation ODA Program with Uganda on Industry-University Linkage and Incubation and Commercialization of R&D outputs, research evaluation, STI Parks and Municipal Innovation Centers, and new and emerging technologies focusing on ICT
- Exploring areas for further cooperation and next steps such as linking the outcomes of the 2019 program with other existing or new projects

Program

TIME	DESCRIPTION
Morning session	
09:00 ~ 12:00	• **Opening Remarks:** Dr. Deok Soon Yim, Senior Research Fellow, STEPI • **Welcome Remarks:** Dr. Maxwell Otim Onapa, Director for Research and Development, Ministry of Science, Technology, and Innovation • **Congratulatory remarks**: Mr. Obong O. David, Permanent Secretary of the Ministry of Science, Technology, and Innovation • **Presentations:** - **Presentation 1:** Major outcomes of the 2019 K-Innovation ODA program with Uganda: Dr. Deok Soon Yim, Senior Research Fellow, STEPI

TIME	DESCRIPTION
	- **Presentation 2**: Policy consulting on STI Parks by Dr. Sangho Choi, Daejeon Technopark - **Presentation 3**: Policy consulting on Emerging Technologies focusing on ICT, by Prof. Hansung Yoon of Hansung University • **Open discussion:** Moderated by Dr. Maxwell Otim Onapa, UNCST
12:00	Lunch
Afternoon session	
13:30 ~ 15:30	• **Presentations:** - **Presentation 4**: Industry-University Linkage policy in Uganda by Dr. Maxwell Otim Onapa, Director for Research and Development, Ministry of Science, Technology, and Innovation - **Presentation 5**: Policy on Incubation and Commercialization of R&D outputs in Uganda by Mr. Basil Ajer, Director for Technopreneurship, MoSTI - **Presentation 6**: New and emerging technologies including ICT by Ms. Monica Rose Nyakaisiki, Assistant Commissioner, Science, Technology, and Innovation Infrastructure Development, MoSTI - **Presentation 7:** STI Parks and Municipal Innovation Centers by Mr. Geoffrey Sempiri, Head, STI Infrastructure Development, UNCST - **Presentation 8:** Research evaluation and assessment of research quality in Uganda by Mr. Bashir Kagere, Senior Science Officer, UNCST • **Open discussion:** Moderated by Dr. Deok Soon Yim, STEPI • **Closing remarks:** Dr. Deok Soon Yim, STEPI
15:30	**Tea Break**
16:00 ~ 18:00	**Bilateral meeting on further cooperation (closed): MoSTI/UNCST/STEPI res earch team** - Review of the 2019 project and discussion on the next steps

Dr. Maxwell Otim Onapa delivered congratulatory messages on behalf of Mr. Obong O. David, Permanent Secretary of the Ministry of Science, Technology, and Innovation of the Republic of Uganda. “I would like to express appreciation on behalf of the Ministry of Science, Technology, and Innovation (MoSTI) to the Korean Government, and we indeed shared for a good collaborative relationship with the Science and Technology Policy

Institute of Korea," he said.

Followed by Dr. Deok Soon Yim's presentation, which highlighted the major outcomes of the 2019 K-Innovation ODA program with Uganda, Dr. Sangho Choi talked about how Korea's Daedeok Innopolis was established and how the Innopolis plays a role as facilitator for innovation clusters. Dr. Choi proposed policy recommendations on STP policy in 11 areas drawn from Korea's experience for Uganda's consideration.

[Table 4-2] Policy Suggestions on STP for Uganda

Areas	Actions
1. Master plan	There is a need to establish a national economic development plan and prepare a master plan to create a science and technology complex. Korea was a very poor country due to the Korean War, relying on foreign aid and US loans; in the 1960s, however, it established a five-year economic development plan and a plan for promoting national science and technology.
2. Political will	Strong political will to foster science and technology is very important. A Korean president selected the site of Daedeok Innopolis and contributed to fostering the science and technology industry with great interest. In particular, with the introduction of foreign loans, the loans were provided for both industrial development and science and technology; this investment made Korea into what it is today.
3. Governance	Governance is necessary to foster national science and technology. In the case of Korea, at the Deputy Prime Minister's level, the Ministry of Science and Technology created a science and technology infrastructure, even fostering the science and technology industry by granting authority to organize the R & D budget.
4. Laws and regulations	In order to promote and prepare science and technology, the relevant laws and regulations are required. Korea has established and operated various laws, such as the Basic Science and Technology Act, Innopolis Promotion Act, and Industrial Complex Creation Act.

Areas	Actions
5. Promotion agency	In order to create a science and technology park, a promotion agency must be designated or established. In particular, we need university (academy) infrastructure that can foster human resources along with research institutes. In Korea, GFIs in various industries led the R & D and utilized many talented graduates from universities.
6. Industrial park and Open innovation networks	It is necessary to create an industrial park suitable for science and technology. This is because technology developed at research institutes and universities must produce industrial products. It is also very important to establish Open Innovation Networks.
7. Startup infra	As part of technology commercialization, scientists' spin-off from universities and research institutes should be able to foster startups and research companies. For this one, it is necessary to build a financial infrastructure such as funds and venture capitals and various startup infrastructures such as an acceleration program.
8. Equipment infrastructure	It is necessary to establish a common equipment infrastructure for researchers from industrial companies and research institutes to conduct the necessary tests and verifications during the R&D process. In addition, research equipment DB should be set up.
9. Knowledge-based DB	Uganda will need a support program for building DB systems based on technical knowledge and acquire IPRs. A knowledge-based DB will be useful for future R&D performance management and new project development. In addition, it is necessary to build a company-related DB for each industry.
10. Business-friendly regulations	Regulations must be company-friendly to ensure that products developed through R&D can be freely placed on the market.
11. Settlement conditions	For companies entering science and technology parks, settlement conditions should be established such as tax incentives, transportation, housing, and welfare.

Noting that Uganda has great potential to pursue a platform policy for East Africa with its land-connected geographic location, abundance of natural resources, and higher education system, Prof. Hans Yoon proposed new business ideas such as cosmetics, glocal network hotel, and glocal network café. Followed by Korean experts, five Ugandan experts shared their key findings and future actions with regard to the themes for which they were responsible.

[Table 4-3] Key Findings and Policy Recommendations on Uganda's STI Policy as Suggested by Ugandan Experts

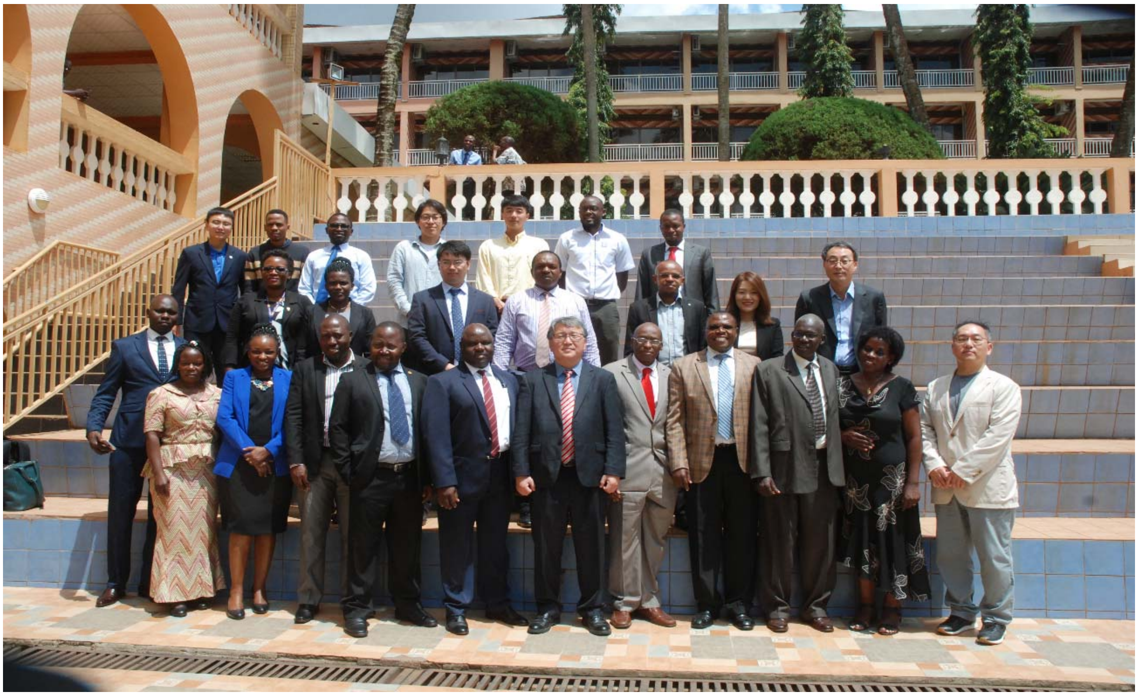
Themes	Key Findings and Future Actions
Industry-University linkage	- Gov't. still bears the biggest responsibility of financing R&I - Private sector is too weak to make any substantial investments in R&D - Venture capitalists are absent to support the commercialization of research results - Financial institutions do not appreciate their role in supporting the commercialization of university research results - Frameworks such as IP policy support linkages to support collaboration in universities and public organizations are either still weak or nonexistent. - Clear policy supporting U-I linkage is necessary
Policy on Incubation and commercialization of R&D outputs	1. Key challenge - Develop R&D for creativity and innovation to flourish, focusing on *improving local content and competitiveness* 2. Recommendations 1) Strengthen the building blocks of commercialization - Build future skills today - Nurture & develop intellectual capital - Create funding pipeline 2) Switch on enablers - Digitalize through multiple platform technologies

Themes	Key Findings and Future Actions
	- Adopt collaboration (quadruple helix – *industry, academe, society & gov't.*) - Inspire human capital through inclusiveness 3) Shoot for the stars - Nurture firms with global penetration potential - Let Ugandans compete and outperform themselves (*manufacturing scenarios that make people overachieve their limits where necessary)*
Emerging technologies	1. Refocus the National Planning Framework Priorities - Define the domain of focus for Uganda, e.g., agritech, healthcare, space, fintech, engineering, smart city, etc. Focus on areas where Uganda has competitive strength for initial consideration. - Rally the leaders to champion policies and initiatives toward emerging technologies. - Sensitize private research institutions, researchers, and private sector on the emerging technologies' paradigm shift. 2. Create a Supportive Regulatory Environment - Develop and implement a National Strategy on emerging technologies. - Review the relevant laws to address this emerging phenomenon. 3. Establish Research and Innovation Infrastructure 4. Plan and implement Human Capital Development programs - Develop a supportive curriculum as well as human resources (teachers/ instructors) and teaching methods at all levels of education. - Consider special training programs including training abroad to enable knowledge and technology transfer. - Rally regional and international collaborative partners to support the STI agenda in Uganda. 5. The government should increase investment in research (currently at 0.23%).

Themes	Key Findings and Future Actions
Research management and quality assurance	- UNCST coordinates research quality assurance nationally through regulation and works in collaboration with diverse academic and research institutions and government to strengthen capacity as well as implement the necessary policies. Nonetheless, there is a need for: - Translation of guidelines into regulations Continued training, sensitization, and mentorship Harmonization of laws Promotion of research ethics
Science and Technology Parks	1. Organization and legal framework - The proposed five regional STPs could be hosted at the regional government and universities that have large chunks of undeveloped land in their names - The STPs could be co-owned by the host Universities and the government represented by a designated competent authority and shall consequently be responsible for the management of these STPs - The proposed regional STPs in Uganda should be incorporated as a non-profit institution jointly owned by the government and host Universities - STPs should have specialized incubation and acceleration programs supporting startups - STPs should initially first focus on Biosciences (Health, Cosmetics, Agriculture), ICT, and Renewable Energy, where Uganda has comparative and competitive advantages in terms of raw materials and human capital - A clear, deliberate plan to make available qualified and skilled professionals (especially MScs & PhDs) who are able to carry out high-quality research and development within the focus areas (disciplines) of STPs to support the National Industrialization Plan - University Research, teaching faculty, and students could be encouraged to participate in STPs' activities such as joint projects and internships - Employees of tenant companies can support universities as adjacent faculty

Themes	Key Findings and Future Actions
	- Need to establish Industry-focused Government Research Institutes/or program to support the growth of local industries/manufacturing sector through applied research and, where possible, through reverse engineering - Good policy and legal framework are important in creating an enabling economic environment for startups and SMEs and attraction of large foreign technology companies into STPs - The infrastructure in STPs should be built around the concept of *"Work-Life Integration"* to create a conducive environment where today's knowledge workers can live, work, and play 2. Suggestions on STP financing 1) Initial financing for infrastructure (civil works and equipment) development: - Regional Host Universities could provide the land - Directly funded by the central government - Regional Universities could also avail themselves of their existing infrastructure such as laboratories and workshops to support these STPs - Infrastructure could be developed in phases depending on the master plan 2) STP continuous operations: The costs involved in continuing STP operations such as maintenance of buildings and other related infrastructure, management salaries, and utilities - The operations of STPs should be funded by the central government for at least 15-20 years, with a review done every 5 years to establish whether the STPs are on the right track to self-sufficiency 3) Startup capital financing for the tenant companies of STPs - The National Innovation Fund should be the first line of financing - Bank financing especially through Uganda Development Bank Limited (UDBL); the third proposed financing mechanism is through a Venture fund

Group photo of the dissemination workshop



Interview of Dr. Deoksoon Yim on 2019 K-Innovation ODA Program aired by Uganda Broadcasting Company

7. Policy Consulting Report

This policy consulting report was prepared to strengthen the national STI system and establish an STI park in Uganda through joint research and events such as kick-off seminar and capacity-building workshop. The policy consulting report on Industry-University was prepared by Professor Jong-in Choi of Hanbat National University in partnership with Maxwell Otim-Onapa, Director for Science, Research, and Innovation at the Ministry of Science, Technology, and Innovation. Dr. Jong Heung Park, who is in charge of technology commercialization for SMEs at ETRI, collaborated with Mr. Basil Ajer, who is the director for technopreneurship at MoSTI, to produce a policy consulting report on the technology commercialization of R&D outputs.

In addition to the themes being focused on, Ugandan experts wrote policy papers on Emerging Technologies focusing on ICT, research evaluation and research quality management, and STI parks. STEPI provided feedback to improve their policy papers.

CHAPTER 5

Conclusion

Policy Consulting for Strengthening Uganda's National STI System and Establishment of STI Parks (Ⅱ) – Policy Capacity-building Program

Chapter 5. Conclusion

1. Policy Implications for Uganda: Industry-University Cooperation Policy

1.1 Policies to Promote I-U Cooperation in Universities

Ultimately, specific science, technology, and innovation policies and institutional measures in HEIs ushered in the third mission from the 1990s with a thrust on innovation. Universities and HEIs, which were mainly seen as the most important source of highly skilled human resources and repositories of new research findings by public and private enterprises, came to be seen as "new frontiers of innovation," too. Every Asia-Pacific country embraced and introduced policies related to innovation in varying forms. With consultancy and collaborative links with industry as traditional forms of engagement, new policy and institutional measures in technology transfer and innovation to engage with society and business enterprises are gaining prominence. Policies for incubation, startups, and spin-offs, technology transfer offices (TTOs), and science and technology parks have gained tremendous prominence in leading Asia-Pacific universities. Two most popular concepts in higher education policies are the "triple helix" of university-government-industry collaboration and university-industry cooperation (UIC). Note, however, that only a small proportion of total universities in any given country were able to implement these policies effectively, often invoking terms such as entrepreneurial or innovation universities. Some major measures and instruments in higher education policies in Asia-Pacific countries are shown in Table 8. Before the early 1990s, universities and higher education in the Asian region were the main source of skills and human resources for industry. Universities played only a marginal role in technology transfer or innovation. Since the late 1990s, however, universities have begun to play a robust role in innovation. Let us examine some emerging innovation landscapes in Asia as shown in [Table 5-1] (Krishna, 2019).

[Table 5-1] Major Policies to Promote I-U Cooperation in Universities

Countries	Major Policy Measures to Promote Innovation
Australia	Cooperative Research Centres (CRCs), 2002; National Innovation Agenda, 2016
New Zealand	University Commercialization Offices, 2005
Japan	Law promoting technology from universities to industry 1998; Japanese version of Bayh-Dole law was enacted and in 2001 (Hiranuma Plan); National University Corporations, 2004
South Korea	Brain Korea 21 in 1999; Korean version of Bayh-Dole for TTOs, 2000; World Class University Programme, 2003; Technology Transfer and Commercialisation Promotion Act, 2006
Taiwan	Fundamental S&T Act, 1999 (for innovation)
China	Scheme to build science parks, 1980s; 973 National Basic Research Programme, 1997; 211 and 985 Programmes (1990s); Invigorating Education Towards the 21st Century
India	Software Technology Parks; Science Parks; Incubation and Technology Transfer in IITs; Renovation and Rejuvenation of Higher Education, 2009
Malaysia	Human Resource Development Fund, 1992; Malaysia Industry—Government, High Technology Program (MIGHT) 1993; National Higher Education Action Plan, 2011
Thailand	National Science and Technology Innovation Plan, 2012
Vietnam	National Fund for Technology Transfer
Philippines	Education Sector Analytical and Capacity Development Partnership
Indonesia	Ministry of Research Technology and Higher Education (UIL program); Education Sector Analytical and Capacity Development Partnership

Source: Krishna (2019)

1.2 Current Situation of Uganda

As a country in East-Central Africa, Uganda is bordered to the east by Kenya, to the north by South Sudan, to the west by the Democratic Republic of the Congo, to the southwest by Rwanda, and to the south by Tanzania. The southern part of the country includes a substantial portion of Lake Victoria, shared with Kenya and Tanzania. Uganda is in the African Great Lakes region, lying as well within the Nile basin with a varied but generally modified equatorial climate. Uganda takes its name from the Buganda kingdom, which encompasses a large portion of the south of the country including the capital Kampala. The economy of Uganda has been growing since the 1990s. Real gross domestic product (GDP) grew at an average of 6.7% annually during the period 1990–2015, whereas real GDP per capita grew at 3.3% per annum during the same period (Wikipedia, 2019).

Uganda has been a republic since it was liberated from the British Commonwealth. Since 1986, President Museveni has been in charge. Throughout these periods, the country has had stable growth; currently, it is projected to continue its growth by about 9%. Even with this growth, Uganda is still dependent on donors. On-budget donor support amounts to about USD 800 million per year, which accounts for approximately 30% of the government budget (World Bank). In addition to this, donors are supporting various separate projects around the country. The World Bank is the single largest financier with contribution of about 19% of on-budget support (Tveit and Webjørnsen, 2011).

Reinikka & Collier (2001) cite three achievements as the reasons for Uganda's success during the period 1979-1999. First, the government introduced long-term peace in a country that had been subject to violence and war. Second, the changes to the taxation system of the country made business, both export and import, more successful. Third, the government managed to maintain a stable currency that the industrial sector could trust not to devaluate in a short period of time. These achievements have helped Uganda be in a situation where it has enjoyed larger, more stable growth through the early 21st century compared to its neighboring countries. According to its African Economic Outlook report by OECD (2010), Uganda was able to maintain growth of above 7% throughout 2009 and the financial crisis compared to the average annual growth of 3.5% in Africa in the same period (Tveit and Webjørnsen, 2011).

When opportunism is high, it implies that many aspects of the government will function badly because the professional ethics normally governing conduct will have eroded (Reinikka and Collier, 2001; p. 25). Opportunism leading to corruption is indicated by the industry to be among the leading risk factors when establishing business in the country. This leads to increased cost calculation for foreign industries wishing to invest in Uganda.

In addition to corruption, the country faces major challenges in the sector of infrastructure. According to Reinikka & Collier (2001), infrastructure, combined with taxes, corruption, and cost of utilities are the top 4 factors limiting growth in Uganda. The African Economic Outlook supports the claim of infrastructure serving as a major constraint: "The major constraint to further growth remains the inadequacy of the country's infrastructure, particularly electricity and roads, due to lack of investment." OECD claims this will be a major focus for the government of Uganda in 2011 and with a lot of activity in the coming years. Uganda has recently discovered oil, and it is now working on how to benefit from this as much as possible. Because of its oil and the general economic growth, McKinsey Quarterly describes Uganda as one of the developing countries that will be the "main engines for growth" (Tveit and Webjørnsen, 2011).

Education and research institutes and infrastructure are not in good condition. Nonetheless, Makerere University (MU), the main university in the capital of Uganda, can be the main partner with various industries and companies.

1.3 Initial Suggestions for Uganda IUC

Tveit and Webjørnsen (2011) published a report titled "University-Industry Interaction and its Contribution to Economic Development in Uganda: A Study of Chosen Projects and Their Interactions with the University Sector in Kampala." The report shows that donor initiatives in developing countries may work against their intentions. This is disturbing for the donor communities and should be researched further. It is suggested that research be conducted to find ways of implementing donor work better. Some donor communities may have initiated a shift to avoid some of the problems found in this report, but further research to identify whether this shift has been effective or not is strongly suggested. This research has identified a problem with donor work and their methods in developing countries, and further research is suggested to explore this problem.

There may be differences as to the benefits of and conditions for UI-collaboration between sectors. Asian companies are not yet established in the hydropower sector in Uganda but are involved in other sectors. The impression from this study is that the Asian industry prefers using their own instead of local knowledge in foreign projects, thereby reducing the possibility of UI-collaboration. Thus, further research on cases wherein Asian involvement is present is suggested to fill this gap in the research material.

[Table 5-2] Implication Summary of Uganda: Tveit and Webjørnsen (2011)

UI Collaboration	Contents
University	- Makerere University is struggling with professors' motivation, and this is identified as the main reason for the lack of UI-collaboration. - Universities should prioritize applied research with local conditions and revise the teaching material to ensure that the students are relevant for industrial positions after graduation. - University management should make applied research part of their strategic focus and improve the university's entrepreneurial culture. Management should give the professors incentives.
Industry	- The industry should prioritize putting more pressure on the universities for increased collaboration. - The industry initiates collaboration, like what UMEME* has done, and there are observed benefits for both actors. The benefit of industrial involvement in UI-collaboration is further increased.
Government	- Establishing policy or framework for collaboration. - The government should establish policies that encourage competition between universities to increase motivation for UI-collaboration. - Applied research and curriculum development that includes communication with the industry. - Result-based salaries in universities.

Source: Umeme Limited is the largest energy distributor in Uganda, distributing 97 per cent of all electricity used in the country.

1.4 Implication for Uganda

Top Korean furniture company Hwansung Company is one of the best cases of Industry – University Collaboration in Uganda. President Kim of Hwansung Company has entrepreneurial leadership, close community engagement, employee human resource development (HRD), continuous R&D efforts, and social responsibility.

Hwansung Group

The biggest Korean investor in East Africa, with presence in Uganda, Kenya, and South Sudan.
In 1980, Mr. Sung-Hwan Kim, chairman of the Hwansung Group, tried to penetrate the African market by establishing a trading company in Kenya; in 1986, he added the necessary manufacturing facilities (bags and cookware) as well as another trading company in neighboring Uganda.

He also started a business of processing "Nile perch," a family of Latidae inhabiting Lake Victoria, after building the largest frozen warehouse in the East Africa region located in Kampala, the capital of Uganda. Exporting all the processed Nile Perch products to European countries, his business had grown to lead the market. In the process of contributing to the Ugandan economy through exporting, Mr. Kim has laid a solid business foundation as a Korean investor in East Africa.

In 1990, Vice Chairman Mr. Jeung-Bong Ahn, a maternal cousin to the chairman, has landed in Uganda to join the business. The Hwansung Group of Companies has continued its sound growth by entering the construction material industry and expanding geographically through affiliates in both Kenya and South Sudan.

Today, the **Hwansung Group** is the frontier of developing high-quality construction interior/exterior material as well as the most well-known Korean business group promulgating the "Republic of Korea" in East Africa.

[Figure 5-1] Hwansung Group in Uganda

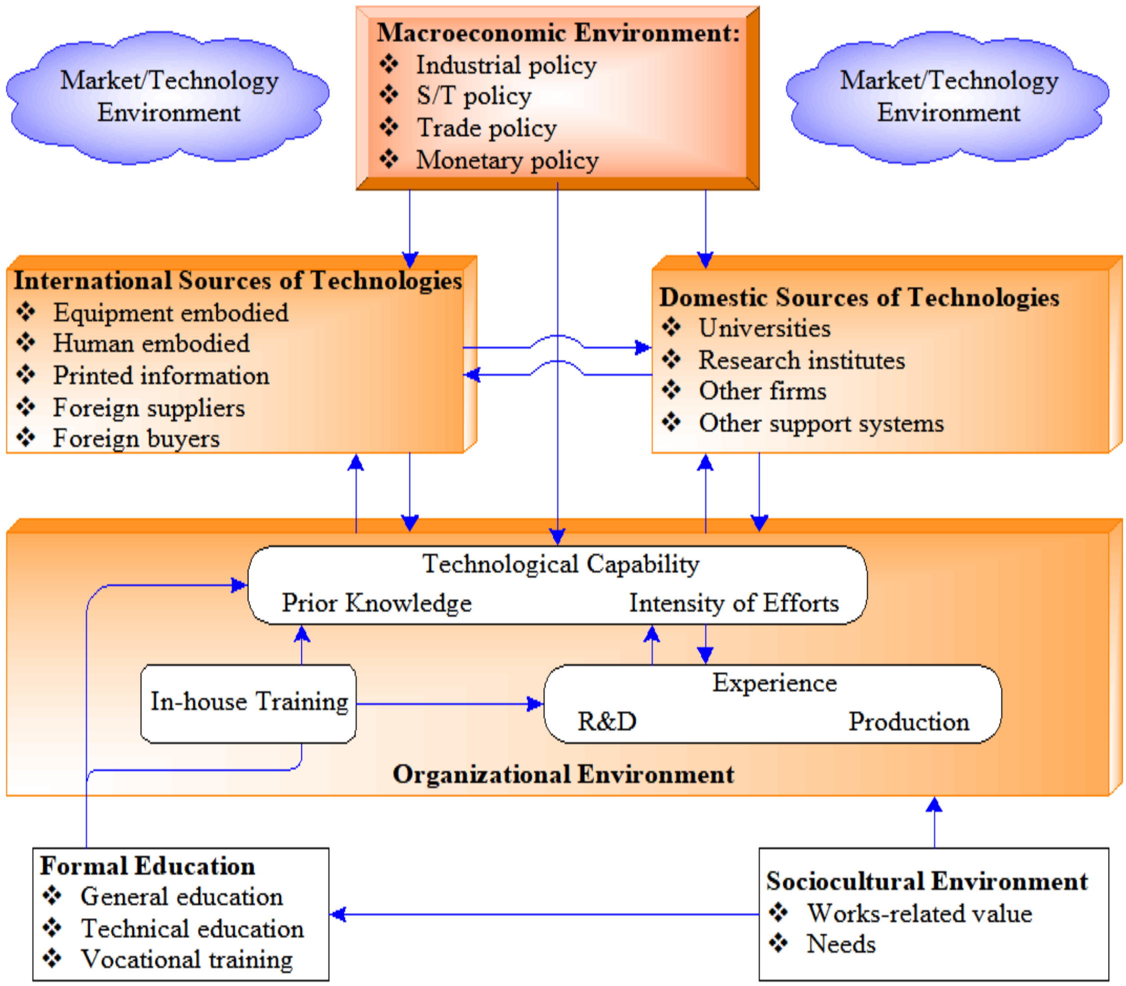
The four success components can be summarized by applying the Triple Helix model to categorize the interview and literature review results by university, industry, and government for Uganda (Choi and Markham, 2019).

Entrepreneurial Leadership and UIC Plan for Absorptive Capacity

The first suggested action is **Entrepreneurial Leadership and University-Industry Collaboration Plan.** University entrepreneurial leadership by the administration and faculty is very important for the ecosystem. The leadership of administration and faculty provides a top-down, bottom-up approach to campus-wide entrepreneurship and UIC. Entrepreneurial leadership creates a culture of industry involvement and administration support for cooperative programs. The faculty engages in programs, course creation, and charges of thought leadership and skills training. A faculty champion paves a road to UIC education and supports the UIC plan like the Korean UIC Plan for 5 years.

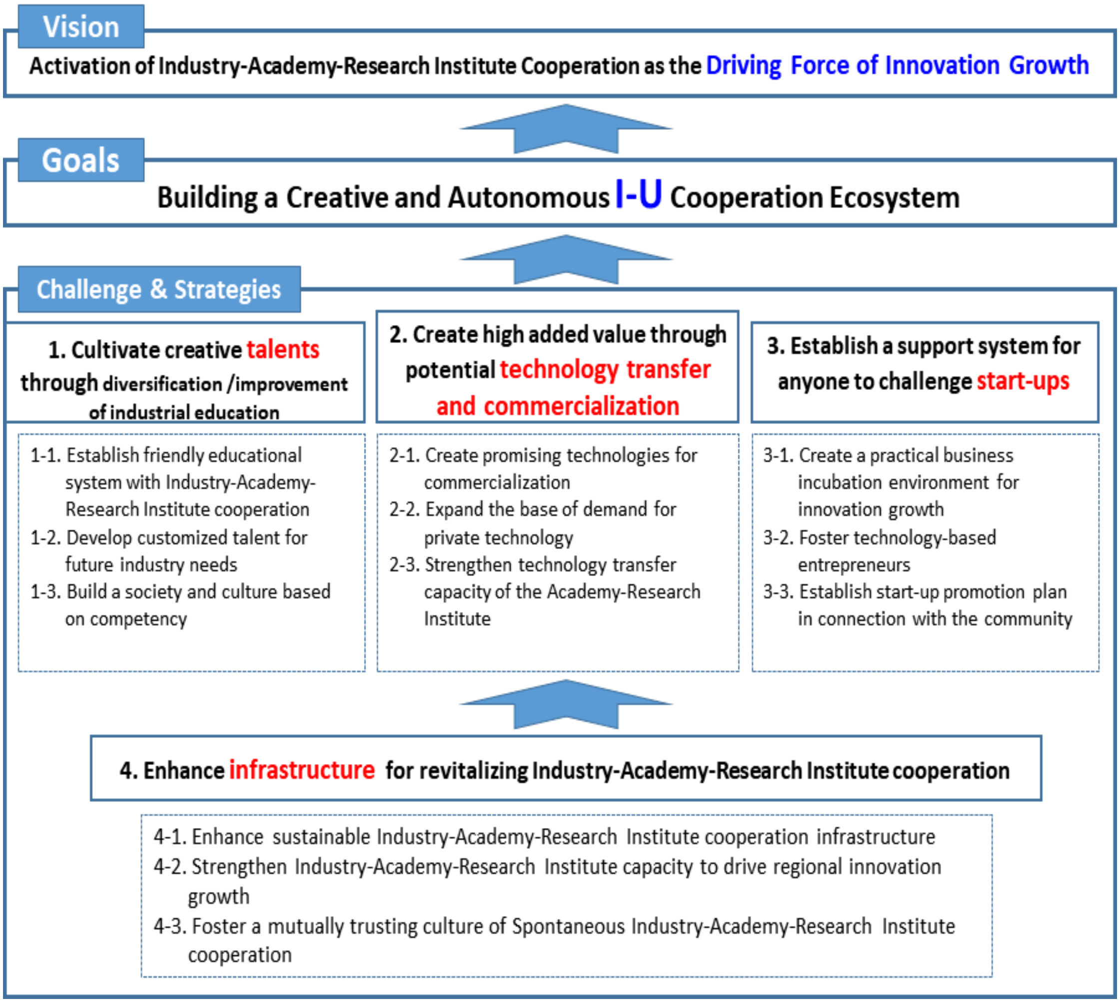
Industry leadership can lay the cornerstone for **University-Industry Collaboration** through the deep involvement and role of champions. They can lead the university and government, especially through champion proposals for university engagement with a company. Industry entrepreneurial leadership initiates the creation of UIC programs at universities, connecting leading corporate entrepreneurs with university programs, faculty, and students and developing absorptive capacity in Uganda [Figure 5-2].

[Figure 5-2] Absorptive capacity building model for Uganda



Source: Kim, Linsu (1997).

[Figure 5-3] Korean National Plan for U-I Cooperation: 2019-2023



Source: Government of Korea (2018).

Student Focus Policy in UIC

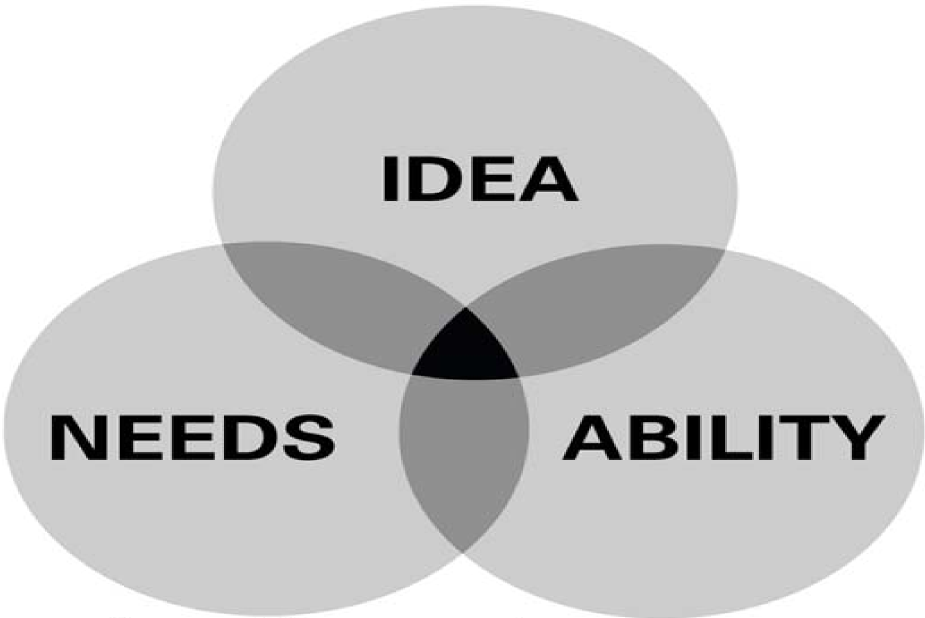
The second suggestion is "**Student Focus**." The student is a core asset of UIC, and the university has to develop and provide a UIC program to students at an adequate level for their undergraduate and graduate degrees. Universities should offer UIC education opportunities and extracurricular activities (visits, competitions). The industry should collaborate to create corporate entrepreneurship programs and host visits by university students. It should also establish hiring and promotion criteria for UIC and develop job descriptions for UIC so that students can see a new window of opportunities. This includes internship opportunities so that students can experience the real issue in the companies and the gap between the theory-based classrooms and practical large companies' activity can be narrowed down.

Faculty Champion Development

The third important suggestion is "**Faculty Champion**." University faculties are often hesitant to engage in University-Industry Collaboration because of their interests as well as time constraints, evaluation criteria, and sometimes ignorance of the industry. The faculty identify and support University-Industry Collaboration for students and corporate businesses such as INC (Idea-Needs-Capability) model. Universities can provide time and leave policies for the faculty member to participate with the industry and establish IUC-friendly evaluation criteria. They need course development support and University-Industry Collaboration funding. For instance, the Korean government provides a "faculty leave system" for UIC. Faculty should have time to develop new courses on University-Industry Collaboration.

Industries can participate in course development and assign mentors (or Executives in Residence) to courses and projects. They can provide access to company projects for courses and "faculty internships" at companies. They can create professorships in University-Industry Collaboration. The government can sponsor economic development studies and provide funding for positions to hire and retain high-profile faculty.

[Figure 5-4] INC Model for UIC



1. Idea + Ability = A product no one wants
2. Idea + Needs = A product you can't deliver
3. Needs + Ability = An unfocused product
4. **Idea + Needs + Ability = Opportunity**

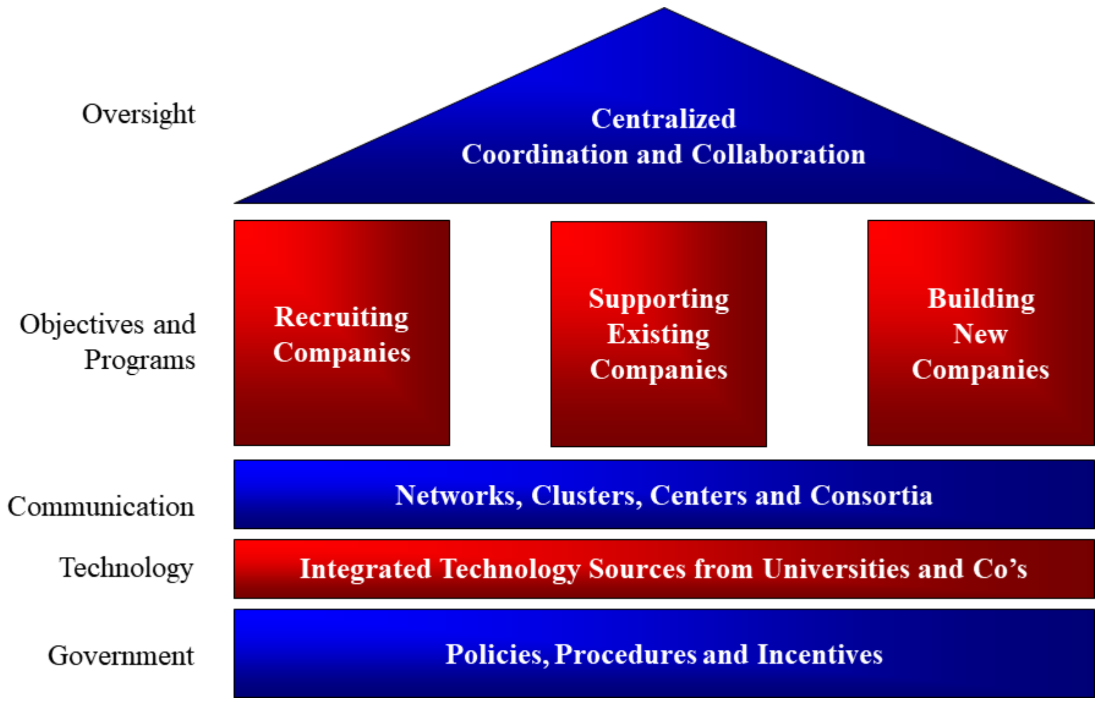
Source: Markham and Mugge (2015), Traversing the valley of death.

Community Engagement Strategy

The fourth suggestion is "**Community Engagement and Cluster**." Community engagement is the process by which community interests and individuals form a lasting, permanent relationship to apply the collective vision for the community benefit. Communities create joint programs, such as speakers, games, and competitions, with university and university extension services for corporate entrepreneurship and innovative cluster. Those programs will contribute to regional economic development and promote the formation of an innovation cluster.

The community engagement strategy supports the meetings of technical and business user groups and creates jobs with training on University-Industry Collaboration. It also supports community University-Industry Collaboration programs. Government can create and sustain collaborative relationships among business agencies including universities. Favorable tax policies can be introduced to encourage industry and university involvement and create co-investment funds, etc.

[Figure 5-5] Key Factor of Innovative UIC Clustering



Source: Zapata III (TEC, NC State)

2. Policy Implications for Uganda: Technology Commercialization of R&D Outputs

Uganda invests less than 0.2 per cent of its GDP in in R&D but is pushing for systematic technology innovation through technology commercialization. With Vision 2040, Uganda aspires for faster, more sustainable, and inclusive growth and strives to achieve its vision through the benefits of a huge pool of talent and ecosystem of science and technology engineering and innovation (STEI). It plans to focus on energy, climate change, agriculture, and health care, which have relatively high opportunities, and also hopes to create quality jobs.

Known for its high level of technology in Africa, Makerere University serves as a technology provider, and government agencies such as the Uganda Management Training and Advisory Center (MTAC) and Uganda Industrial Research Institute (UIRI) support programs held for new technology startups. The areas supported by each startup are ICT, Bio, and New Material, with small but private operators participating as well.

In view of these efforts, the systematic implementation of the aforementioned series of technology commercialization processes will result in the basic capabilities of technology commercialization to help develop the economy.

As a result of the catch-up process based on technology commercialization discussions, several suggestions are recommended for Uganda's economic growth:

- In addition to relying on Makerere University, government agencies such as the UGANDA Management Training and Advisory Center (MTAC) and Uganda Industrial Research Institute (URI) should develop new technologies themselves and act as technology providers. An attempt should be made for deficient technologies to receive technology transfers through support from advanced countries and international organizations such as the United Nations, and capacity should be enhanced through international trade, technology licensing, attraction of multinational companies, and training on human resources. In particular, it is necessary to attract transnational companies actively through various tax benefits, given that they are accompanied by capital and technology.

- Capability Building should be strengthened. In order for the aforementioned technology transfer to be successful, technology digestion to materialize the technology transferred from developed countries should be emphasized. Based on this, the government should develop detailed technologies and support existing companies or push for successful technology commercialization by fostering startups.
- R&BD strategies should be developed and implemented in order to secure the same semi-heat or better technological capability as that of developed countries and occupy the global market. The role of startup companies is paramount in the process of leading technology development, which is the last gateway to become an advanced country. Since flexible and quick action is required in the process of discovering and verifying new ideas, the startup firms can have an advantage over existing ones.
- The fact that global investors (VCs) are looking for good technologies and startups worldwide can be a good opportunity for Uganda. A startup company founded on good technology can make inroads into the global market from the beginning, and it will be able to attract investment worldwide, which is not easy in Uganda.

3. Future Directions

Implications from the policy consulting paper on Industry-University Cooperation and Technology Commercialization of R&D Outputs will be reviewed with public stakeholders of Uganda particularly with MoSTI. Close consultations on the implications and areas for further cooperation are expected.

The following action items were proposed by Korean and Ugandan experts to explore the linkage of outcomes of the project with existing national and international projects or develop pilot projects:

- Establishing a think-tank for STI policy in Uganda such as STEPI in Korea
- As for the follow-up on technology commercialization, pilot projects can be developed on the following areas:
  1) Smart Energy Pilot Project for an area with low energy supply and high climate risks
  2) Smart Agriculture Pilot Project to increase agricultural productivity
  3) e-Government Pilot Project using Open Source Software

It is recommended that a demand survey be conducted on the themes above through the Ugandan government and/or research institutes for the pilot projects to be implemented. The identified Korean partners from research institutes and companies could implement the pilot project with funding sources from the Bilateral Official Development Aid or Multilateral Development Bank such as World Bank or Africa Development Bank.

ANNEXES: SOME OF THE BUSINESS INCUBATION CENTERS IN UGANDA

Annex 1: Makerere Innovation and Incubation Center (MIIC)

As a brainchild of Makerere University, which partnered with the Ministry of Finance, Planning, and Economic Development, MIIC was established in October 2016 and launched on 05 October 2017. MIIC is a nonprofit organization overseen by a seven-member Board of Directors consisting of one member each from Makerere University, MoFPED, NPA, MOSTI, and private sector, headed by Dr. Peter Ngategize, Head of the Private Sector Development Unit (PSDU) at MoFPED as Chairperson. It is hosted at the Makerere College of Computing and Information Sciences with MoFPED, MoICT, and NG, PUM Netherlands Senior Experts, and Catalysts GmbH as partners.

Focus is on recruiting and nurturing talented youths aged 22-30 with innovative ideas in the sectors of Education, Health, Energy and Minerals, Justice, Law and Order, Financial Technology, and Agriculture, among others. These sectors are in line with the national priorities stipulated in National Development Plan II, Uganda Vision 2040, ICT Sector Strategic & Investment Plan, and National Science, Technology, and Innovation Policy.

Value Proposition

The mindset change of youth as a key element in the innovation spectrum, including the provision of a favorable ecosystem for the growth of startups into middle and upper market enterprises. This automatically leads to job creation when achieved through a structured program alongside a deal flow and supported by a corresponding organized curriculum.

Purpose

MIC seeks to achieve increased innovation, creation of new jobs, transformation of marketable ideas into sustainable businesses, and improving the youth mindset toward work and innovation.

Strategy

MIIC identifies potential entrepreneurs with the best innovative ideas that use existing or new technology in a range of market sectors such as agriculture, healthcare, and tourism. We nurture and facilitate them from ideas to sustainable companies.

Incubation

The MIIC incubation program includes both product development and business development. The number of incubatees is limited to guarantee space for all who are admitted.

Freelancing

This is an open-ended program for people and businesses looking for an innovative, vibrant working environment to run and accelerate their business startups as well as given outsourced projects by MIIC to work on.

Startup processes

Startups are managed through a stage-by-stage business development process that provides regular progress checkpoints and minimizes financial and reputational risks.

Startups to be supported

- Startups under incubation are youth 18 years and above, 60% of whom are continuing students in various Universities and the rest are recent University graduates.
- These technology-based startups are for agriculture, tourism, clean energy, and minerals, among others.

Incubation Cycle

- Organizing sourcing events wherein talented youth are identified.
- Screening applicants
- The selected are tasked to create proposals
- A team of experts picks out the best proposals
- Business Plans are then developed
- The incubatees are now trained on the sharpening of business concept, startup team formation and member roles, bookkeeping, accounting and legal essentials, customer engagement, product prototyping and design, business modeling and competitive

analysis, financial planning and modeling, creation and improvement of customer satisfaction, pitching practice, and coaching and skills development through access to experts in marketing, business development, accounting, software development, graphics and interfaces, and legal and intellectual property protection.

- They then pitch to prospective investors mainly through the Kampala Angel Investment Network (KAIN).
- Successful business plans obtain seed investment funds of up to US$2000 for Research and Development.
- Sales-ready Prototypes are developed.
- They are tested for conformity to standards.
- Early customers.
- Commercialization.

All incubated startups at MIIC are born out of R&D.

Commercialization

At least 2 out of 18 startups are ready to go commercial, for a success rate of 11%. MySchools, an online fully integrated school management system, efficiently manages the information system of schools. The other one, Seat Angel, is an airline seat optimization and ride hailing service as an airport pick-up and drop taxi venture, working with airlines like Turkish and Kenya Airlines. Airport pickup and drop costs are pre-charged by bundling them with the passenger's air fare. Upon arrival at Entebbe Airport, a Seat Angel driver picks up the passenger and drops him/her off to the designated destination.

Incubation

A favorable ecosystem for thriving is created for startups, wherein space, Internet, and other resources are provided at MIIC until the startups reach commercialization and get weaned thereafter.

Challenges

- Limited funding for operations

- Delays in receiving funds from MoFPED
- The high dropout rate, now at 50% for startups, is attributed to the need for survival, which drives them to look for job opportunities that can bring in quick returns. MIIC is partnering with an organization in the Netherlands to extend grants to the startups in order to curb the high dropout rates. The program has yet to begin, however. Additionally, MIIC has approached MoSTI for the same but has yet to get any commitment from the ministry.
- Inadequate facilitation of TOT Mentors, some of whom are asked to extent pro-bono services, which is not sustainable.
- There is limited space to accommodate all startups. There is a need for more space within or outside the University at affordable rates.
- Few startups established by females.
- Low rate of commercialization, which currently stand at 11%.
- Taxes on imported machinery for startups

Annex 2: Uganda Industrial Research Institute (UIRI)

UIRI was formally established by an Act of Parliament in 2002. and by H.E. the President pursuant to the Act on 30 July 2003. It is a progeny of the East African Industrial Research Organization (ESIRO) of the defunct East African Community (EAC)]. After the demise of the then EAC in 1977, the three member states continued with splintered Industrial Research Organizations, giving birth to:

- Kenya Industrial Research and Development Institute (KIRDI)
- Tanzania Industrial Research and Development Institute (TIRDI)
- Uganda Industrial Research Institute (UIRI)
- With Uganda's economic dislocation in the 70s and 80s, UIRI did not become fully operational until 1997 when, with a grant from the government of the People's Republic of China (GPRC), a campus was built and some technologies for pilot plants were applied. The formal handover of the modern facility to the Ugandan Government

took effect in the year 2000. UIRI's mandate is to engage in activities that will lead to the rapid industrialization of Uganda.

UIRI's Mandate and Objectives

UIRI's mandate as derived from the statute that established the institution is to:

- Undertake applied research, and
- Develop and acquire appropriate technology in order to create a strong, effective, and competitive industrial sector for the rapid industrialization of Uganda.

In order to fulfil its mandate, UIRI has established the following corporate objectives:

- To undertake Applied Industrial Research for the development of optimal production processes for Uganda's nascent industry
- To develop and/or acquire appropriate technology in order to create a strong, effective, and competitive industrial sector for the rapid industrialization of Uganda
- To provide the necessary expert input toward Government Development Initiatives such as PEAP, PMA, and MTCS

Various R&D outputs have been realized:

Majority of outputs have been born out of idea generation both at the individual entrepreneurial level and the community level. These were then shared with UIRI, which helped them develop the products, standardize, prototype, and test them for commercialization by the entrepreneur.

Food Science and Technology

Undertakes research on industrial processes and technology to add value to food products. The division is responsible for running pilot plants for processing dairy, meat, bakery, fruits, and vegetable products. Although the products from the pilot plants are available for sale to the public, the cardinal role of the pilot plants is to train entrepreneurs and others from tertiary and university institutions. The R&D products developed include:

- Sausages-fish, rabbit, mushroom, pork
- Medicinal mushrooms

- Corn bread, cassava bread
- Tomato juice
- Gooseberry juice
- Avocado Medicinal
- Yogurt
- Ice cream
- Bushera
- Wines
- Sodas

Major Brands supported with product development

- Bella Wine
- Kigezi mineral water
- Mega Milk & Mega yogurt
- Riham Cola
- Kiri Soda
- Kisubi tea

<Food Incubation Center at UIRI>

a) Ceramics

Responsible for research, design, and production of high-quality ceramic products. The department maintains a showcase at the Institute, and some of their products are available for sale to the public.

b) Training

Coordinates a spectrum of training programs from basic to advanced skill levels and from the inchoate to the fully formed entrepreneur. Most of the training is conducted in-house, and programs are run by the staff of the Institute (see Appendix (1) for the census of the Institute's training activities). UIRI offers hands-on training to boost local and community enterprise development for renewable energy technologies, food processing, laundry and cosmetics, weaving and textiles, bamboo processing, carpentry, metal fabrication, minerals, and material processing.

<Tailoring training center at UIRI>

Value Addition and Product Development: Innovative product development for rural enterprise development in order to transform local raw materials into competitive marketable products.

UIRI also engages in collaborative partnerships, Business-to-Business Networking, and participation in various platforms such as exhibitions, trade fairs, workshops, and seminars.

Intervention of the Uganda Industrial Research Institute is key to entrepreneurial development in Uganda in the drive to Middle-Income status.

c) Analytical Laboratories

Support internal research and offer services to the public in areas such as product analysis - content and context, physical and chemical properties, or any analytical service that the client might desire. Laboratories are divided into Analytical Chemistry, Microbiology, and Mineral Laboratory. Here, standardized, globally acceptable equipment can be found including other Uganda-developed equipment awaiting approval to be taken into the market after certification by the relevant authorities. This laboratory covers research on areas like DNA, genes, and hereditary traits and houses a chemistry laboratory where research on food products, aflatoxins, pharmaceuticals, cosmetics, and toiletries is done.

The following are the major R&D works done:

- Essential Oils-Cosmetics
- Portable Aflatoxin bio-sensor
- Lactic Acid formulation
- Acetic Acid and Enzymes
- Bio-fuel-Ethanol
- Zanka Gum using Cassava

Challenges

- Finding entrepreneurs who can take these prototypes to commercialization is a problem
- Brain drain of talented and trained scientists has always set back the Institute

d) Engineering and Manufacturing

Runs a maintenance engineering workshop as well as a carpentry shop. Provides all engineering and technical maintenance services to the entire Institute, coordinates any engineering/technical services that may be contracted from outside.

The following are the following R&D works done:

- Bamboo bicycle
- Bamboo furniture
- Bamboo toothpicks
- Wooden Weaving technology
- Bar Soap cutter
- Cow horn products-earrings, wall lamps, saucers

- Spinning machine
- Boilers
- Pasteurizers
- Peanut Sheller
- Mobile Kiosk
- Briquettes

Business consultancy and Advisory services in Business management, ICT, Packaging, Branding, and marketing to Entrepreneurs and to the general public through its Business Development Center.

Breakthrough in R&D Outputs

- Livara Cosmetics & Skincare-Idea-generated product development born out of individual entrepreneurial idea generation
- Vaccine production born out of R&D
- Kabale Wine from pineapples: Idea-generated standardization support born out of individual entrepreneurial idea generation
- Peanut Butter in Lira- Community Idea generation
- Prototype functionality of fabricated machines
  - Chuffer Cutter of Kyarubuga
  - Cloth material cutting machining
  - Fabrication of soap cutting machine
  - Poultry Feedmill, mixer, and pelletizer
- Equipment fabrication
  - Hatchery for poultry agro-processors
  - Poultry processing line
  - Charcoalite processing equipment
- Promote agro processing (dairy, meat, fruits, and vegetable processing)

- Prototype of Electronically Controlled Gravity Feed Kit
- Development of Mediclave – Solar-powered autoclave
- Smart Drip Irrigation System
- MUTIMA
  - Diagnostic device for Pneumonia
  - Open Source Prototyping Lab Project - remodeling PCB Lab (civil works)
  - Open Source Prototyping Lab Project - equipment purchase
  - Professional Software (subscriptions and server licenses) and purchase of server to run software applications
- Biofuel production: Production of both biodiesel and Tigernut oil. These are cheaper, clean energy products that will help in protecting the environment.
- Production of affordable cooking gas; bottled biogas from chicken droppings.
- Experimentation of solar wind hybrid system. This is an ongoing project at the Ntungamo, Energy Systems Division Pilot site
- Health safety and environment; to integrate health, safety, and environment into the core activities of UIRI.
- Design and develop a system that uses plasma technology to recycle waste. The output products include energy (from organic waste) and metal recycling and smelting
- Innovation
- Development and Commercialization of Mineral-rich Poultry feeds from Fruit Byproducts
- Development of vegetable sausage
- In-house Business incubation in the Fruits & Vegetables Sector

Mandated to engage in applied research and other activities such as value addition that will result in the rapid industrialization of Uganda, the Uganda Industrial Research Institute translates the applied research results into practical applications that lead to high-quality, efficient industrial products and processes and creates highly skilled human resources.

Typical case studies of incubated startups that have benefited from UIRI include: the prestigious Mega Milk, Lira peanut processing plant, Amagara Skincare Limited, and Mushroom Training and Resource Center (MTRC), among others.

Annex 3: CURAD Business Incubation Center

The Consortium for Enhancing University Responsiveness to Agribusiness Development Limited (CURAD Limited) is a public-private partnership institution initiated jointly by Makerere University, National Union of Coffee Agribusinesses and Farm Enterprises Limited (NUCAFE), and National Agricultural Research Organization (NARO) and whose mission is to produce young, innovative, and skillful agribusiness entrepreneurs-job makers and profitable agribusinesses through strategic partnerships that support training, mentoring, and business development.” In order to fulfil this mission, for the last two years, CURAD has been involved in the following: (i) Promoting the setup of successful agribusiness enterprises by mostly youth and women; (ii) Promoting student startups in agribusiness through its popular Earn as You Learn Program; (iii) Supporting young female and male university graduates to start and run viable agribusiness enterprises; (iv) Promoting the utilization of coffee in Uganda through coffee value chain development and value addition; and (v) Supporting farmers through agri-consultancy and enhanced market access.

CURAD is one of the first six agri-business incubators in Africa supported by the Forum for Agricultural Research in Africa (FARA). These objectives are squarely aligned with the UNCDF core aims. Over the last two years, CURAD has created and supported 64 SMEs and graduate agribusiness projects, resulting in over 1,400 new jobs. In addition, through its farmer ownership model, it has increased the household income of over 2,500 farmers. Because of these achievements, CURAD was recently recognized as the best incubator in Africa. Although CURAD has reported positive progress, it cannot scale up its services as fast as it would have liked because of limited financial and infrastructural capacity. CURAD is operating on a self-sustainable basis wherein the new successful enterprises are supposed to contribute to the continued sustenance of the institution. Since it will take some time before this is fully realized, support from government and other development partners is required. Nonetheless, the most serious challenge faced by CURAD is limited space for incubatees to pilot their business ideas and produce certifiable products that can access a wider market. Currently, CURAD is renting space for its incubatees.

To address some of these challenges, CURAD is working in partnership with the Uganda Investment Authority (UIA) to upscale its incubation activities by establishing and managing an incubation center facility for the commercialization of technologies derived from tertiary institutions, research facilities, and farmers' organizations. As a semi-autonomous agency of the government of Uganda, the Uganda Investment Authority was set up to promote, attract, and facilitate both Foreign Direct Investment (FDI) and encourage and support domestic investments based on its vision of making Uganda the preferred investment destination in Africa. UIA is developing the Kampala Industrial and Business Park (KIBP) in Namanve where industries of various sectors will be located, and KIBP is already connected with electricity, water, and access roads with access to the railway network. UIA has set aside 2.5 acres of land for CURAD to establish a Business Incubator Facility for Small and Medium Enterprises (SME) in KIBP Namanve.

The SME incubation center in Namanve shall provide full-scale business incubation services from business and innovation analysis to business plans, prototyping, and common-use production facilities and mini factories. The land earmarked for the incubation center is already planned, and the technical architectural drawing has also been approved by the Mukono District Local Government where the incubation center is to be established. A detailed feasibility study was conducted, a business plan developed, and an Environmental Impact Assessment (EIA) was done and the National Environmental Management Authority has issued a certificate of approval for the EIA and technical drawing. Therefore, this proposal is seeking funding to establish incubation facilities for CURAD in Namanve in partnership with UIA.

The CURAD-UIA MSME business incubation center in Namanve is aimed at expanding the CURAD incubation model to serve other value chains to include: i) agro-processing of fruits for juice and wine; ii) textile, art, and craft; iii) metal fabrication; iv) leather; and v) furniture.

Objectives of setting up the incubation center

The overall objective of the project is to provide an incubation services center to business startups that can run sustainably by providing premium core rentable and chargeable infrastructure facilities and services while promoting entrepreneurship and job creation along the key value chains in the country.

Custom-made, low-cost, medium-scale production facilities are crucial for SME development since local and international product certification especially for agro-products is a crucial requirement for mainstream market access, which is a key factor in enterprise sustainability and growth.

The specific objectives include, among others:

- Identify innovations and research outputs from universities, research centers, farmers, and farmer organizations and commercialize them into sustainable enterprises with focus on the youth and women
- Support existing small firms with high growth potential to develop and flourish by identifying and solving growth challenges and bottlenecks
- Provide pre-incubation services like prototype development, business plan and analysis, registration of the company, access to finance, and company management
- Provide affordable workspaces for startups for a specific period of time to kick-start viable commercial operations
- Provide business advisory, mentoring, and counselling services
- Provide technical support and product testing services
- Provision of exhibition, conference, and meeting rooms
- Transfer of technology and promotion of innovation
- Provision of a resource center
- Work with UNBS to help certify incubatee products to gain access to formal local and international markets
- Enhancing links between Universities, research institutions, business community, and market Industrial cluster development through the development of a particular industry
- Incubation services through the provision of shared production centers/ facilities for fruit and grain processing, Leather processing, mechanical workshop for fabricators

- Incubatee Marketing support through shared distribution services, exhibitions, and Expos

Summary of key Client Admission Processes

Initial contact & submission of application form (CDOs, MD)

- CURAD National Enterprise competition, calls advertised, Walk-ins, Referrals, Websites and other Media, Awareness programs, Billboards, CURAD Open days and excursions
- Submission of application form
- NDA signed where applicable
- Mail/Call the approved applicants

Review the applicant application form, business idea, or Business plan (CDOs, MD)

- Schedule a face-to-face interview.
- Use CURAD assessment tools to assess applications.
- Undertake site/enterprise visits to SMEs.
- Prepare assessment reports, select approved applicants, and send formal acceptance /rejection letters.
- Schedule presentation days for approved applicants for the TeWoCo selection committee.

Final presentations to the TeWoCo admissions committee & admission/pre-incubation or rejection by the Board

- Applicants present elevator pitchets to the TeWoCo admissions committee.
- Develop the incubation schedule with the approved applicants.
- Sign the MOA on incubation, pre-admit successful candidates after payment for membership fees.
- Present a list to the board and create and present certificates to selected CURAD incubatees.

Incubator facility management model

The Incubator Production Park and Cluster Marketing model (IPP-CM model) is aimed at incubator and incubatee sustainability. A successful incubator operational model as practiced by Timbali in South Africa (one of the most successful incubators in Africa) has been earmarked as a model of excellence for successful incubator and incubatee management. It ensures that both the incubator and incubatees are nurtured to grow into sustainable enterprises and will be employed at CURAD.

At CURAD, incubatee production marketing and distribution support is handled using the Incubation Park Cluster Marketing model (IPP-CM) whenever feasible. As mentioned before, it is modeled on the successful Timbali horticulture incubator in South Africa, ensuring steady incubatee growth toward graduation as well as incubator sustainability and financial discipline. Although employed in a horticulture value chain, this model is easily scalable to other value chains. The model targets:

- Common production facilities whenever possible for similar or related product lines, e.g., fruit juice and dairy products, baking and confectionery products charged at a user fee
- Common logistics marketing and distribution of incubatee products with the support of the incubator for a fee
- Directly manages the income from the sales of the incubatee outputs

The incubator:

- Helps with market linkages and charges a marketing and distribution fee
- Helps the incubatee in accurate bookkeeping, records, and prudent financial management

This ensures that:

- Easy management of incubatee monetary support and repayments by controlling cash inflows
- Supports the incubatee in forward planning by encouraging savings that are crucial for equipment acquisition and eventual graduation from the incubator

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